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(54) **METHOD FOR IMPROVING PLANT
GENETIC TRANSFORMATION AND GENE
EDITING EFFICIENCY**

(71) Applicants: **Institute of Genetics and
Developmental Biology, Chinese
Academy of Sciences, Beijing (CN);
Tianjin Genovo Biotechnology Co.,
Ltd., Tianjin (CN)**

(72) Inventors: **Caixia Gao, Beijing (CN); Yanpeng
Wang, Beijing (CN); Fengti Qiu,
Beijing (CN); Yidong Ran, Tianjin
(CN); Hu Xu, Tianjin (CN); Kang
Zhang, Tianjin (CN)**

(73) Assignees: **Institute of Genetics and
Developmental Biology, Chinese
Academy of Sciences, Beijing (CN);
Tianjin Genovo Biotechnology Co.,
Ltd., Tianjin (CN)**

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CPC **C12N 15/8262** (2013.01); **C12N 5/04**
(2013.01); **C12N 2310/20** (2017.05); **C12N**
2310/3519 (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Charles Logsdon
Assistant Examiner — Kelsey L McWilliams
(74) *Attorney, Agent, or Firm* — Jenkins, Taylor & Hunt, P.A.

(57) **ABSTRACT**

The present invention provides a method for improving plant genetic transformation and gene editing efficiency. Specifically, the method improves the regeneration efficiency of plant genetic transformation and/or improves the efficiency of plant gene editing by expressing genes that promote plant cell division, especially meristematic cell division.

12 Claims, 11 Drawing Sheets

Specification includes a Sequence Listing.

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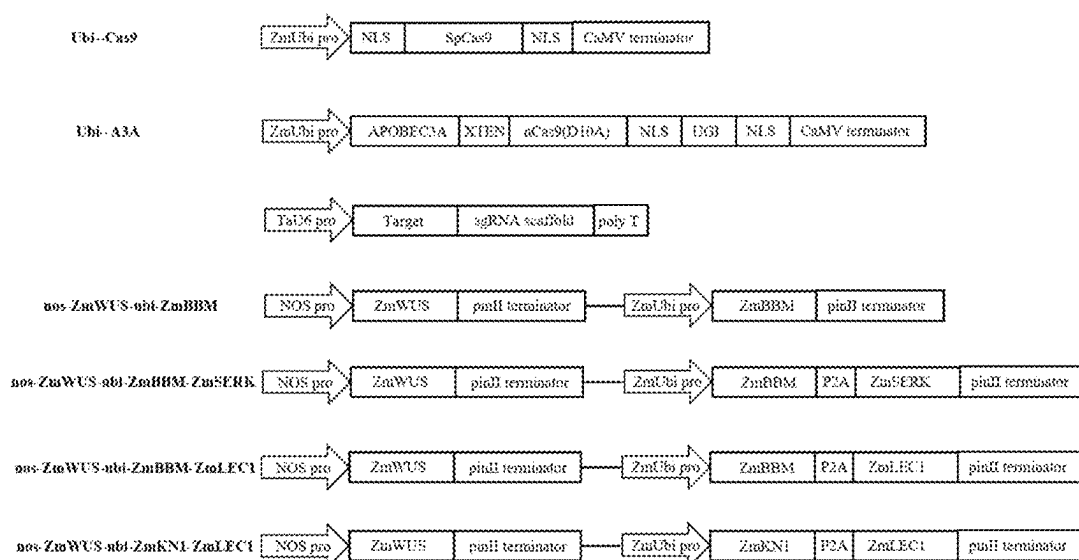


Fig. 1

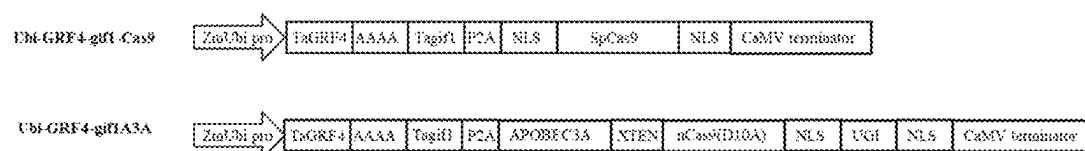


Fig. 2

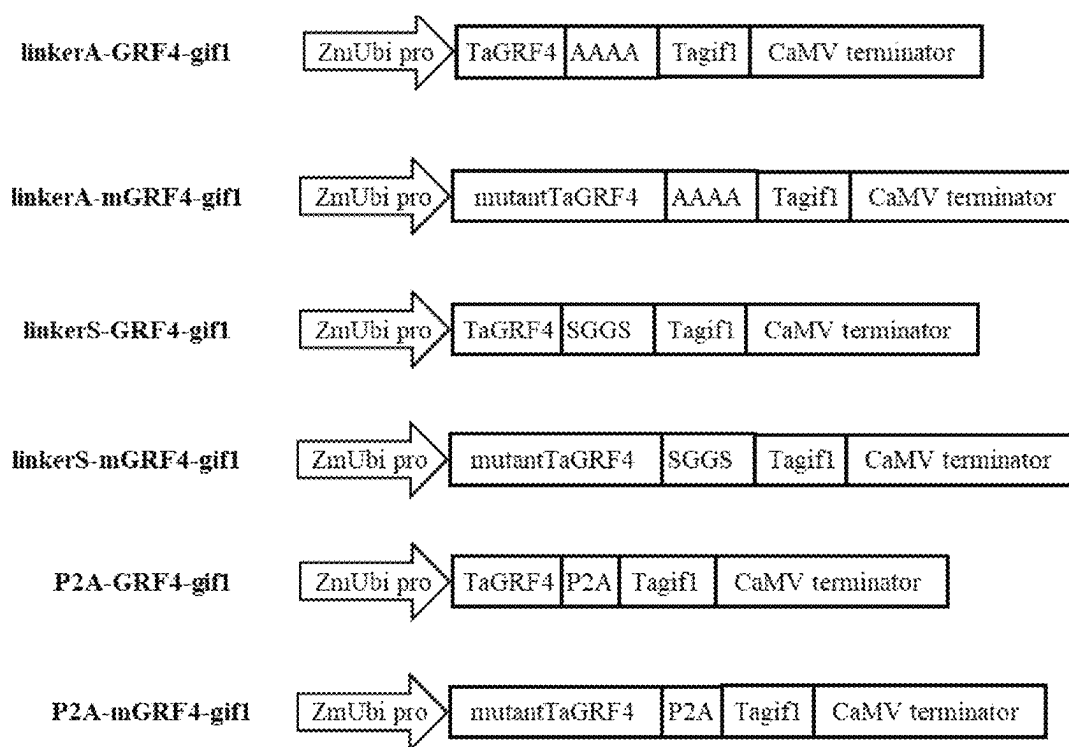


Fig. 3

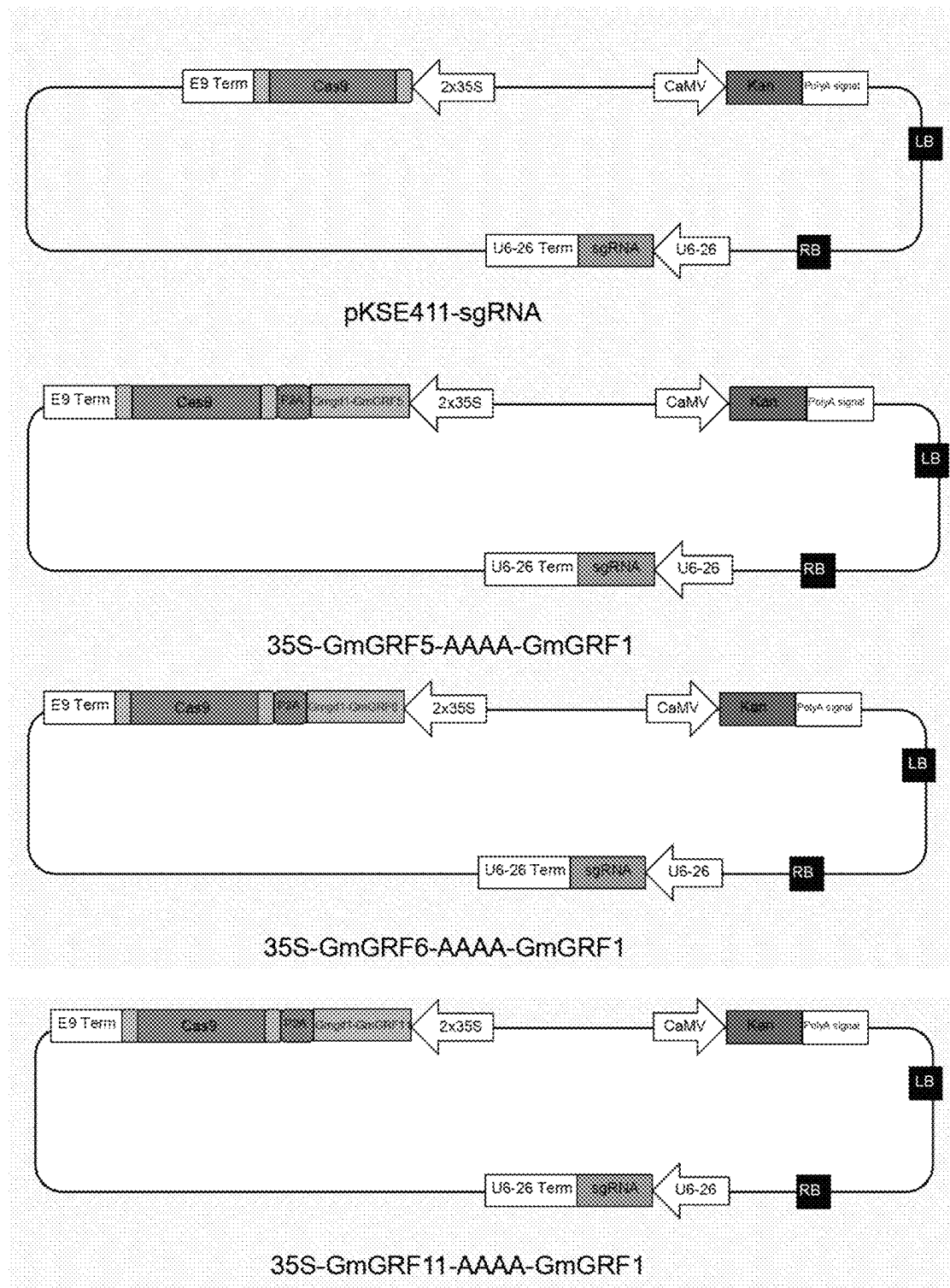


Fig. 4

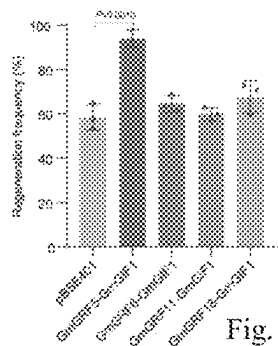
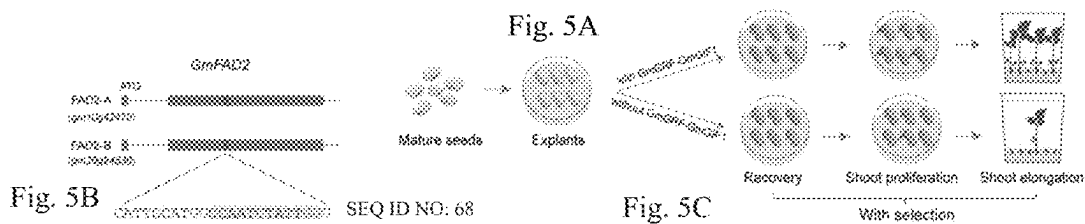
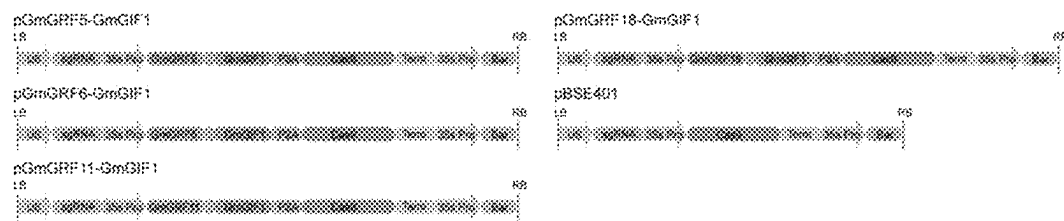


Fig. 5D

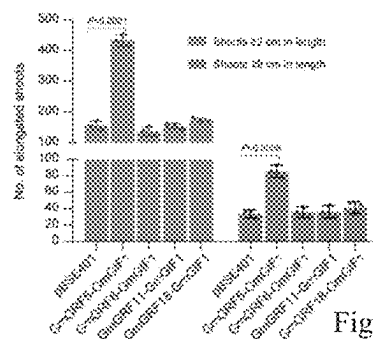


Fig. 5E

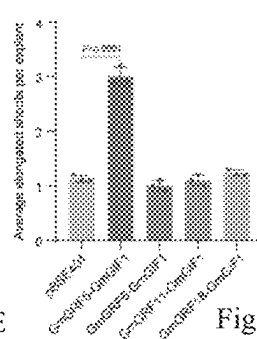


Fig. 5F

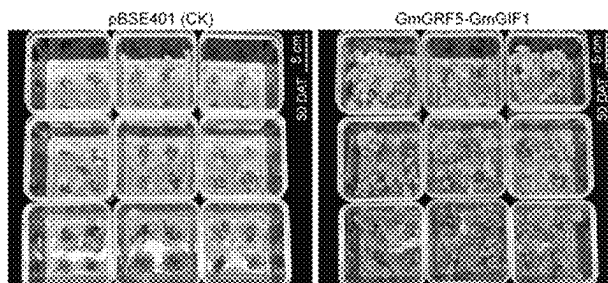


Fig. 5G

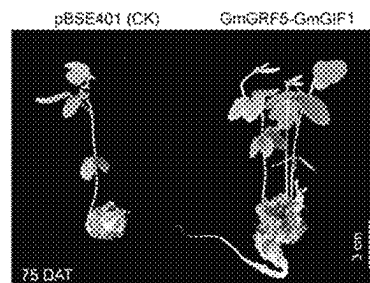


Fig. 5H

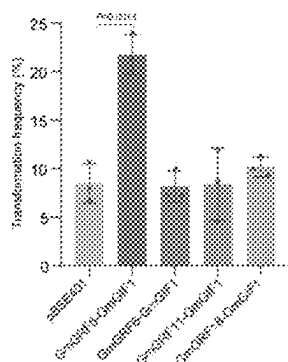


Fig. 5I

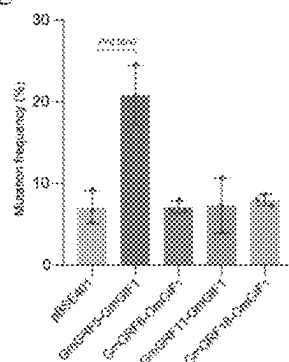


Fig. 5J

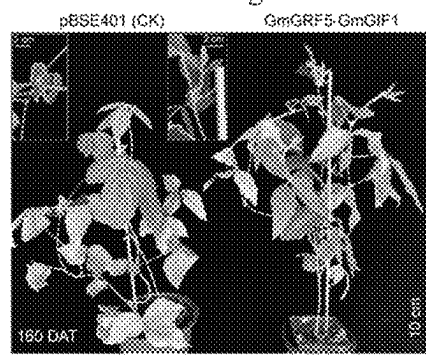
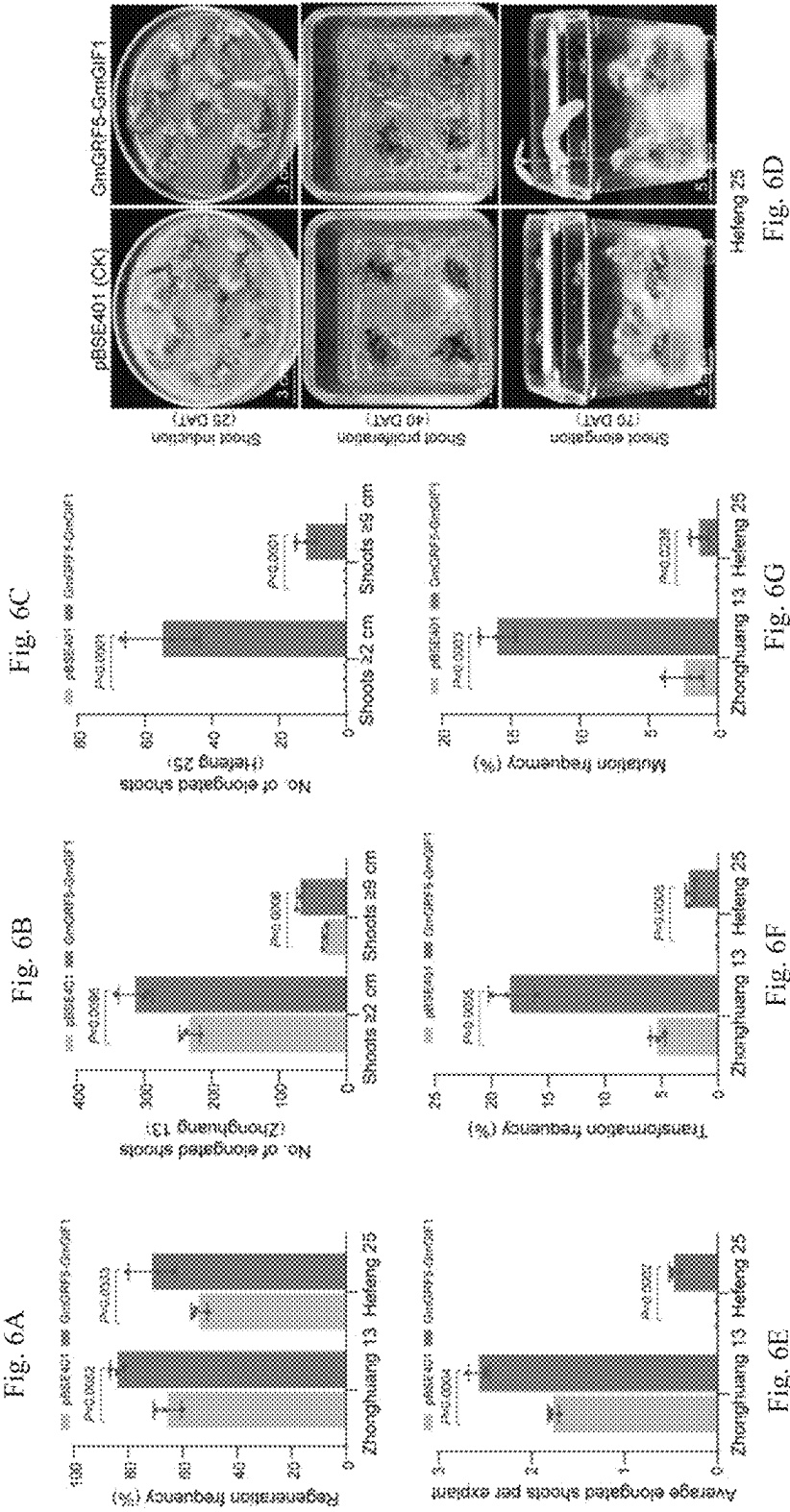
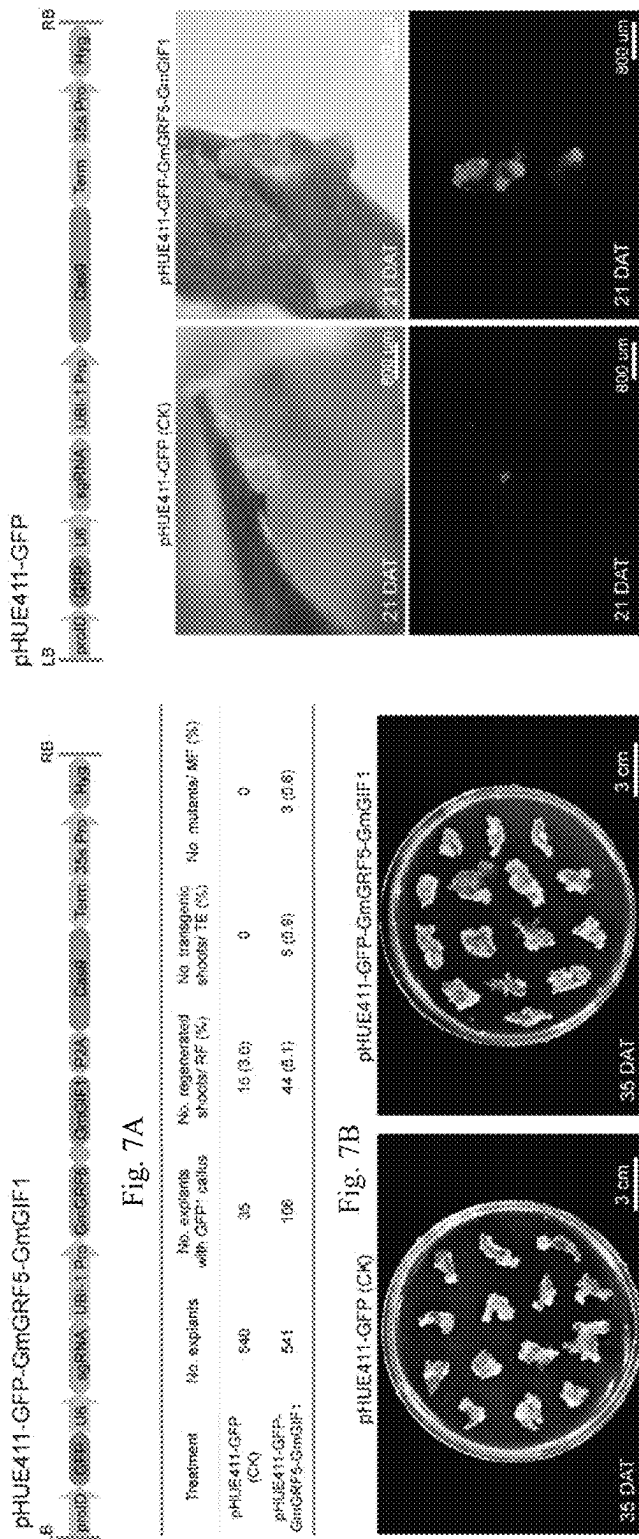
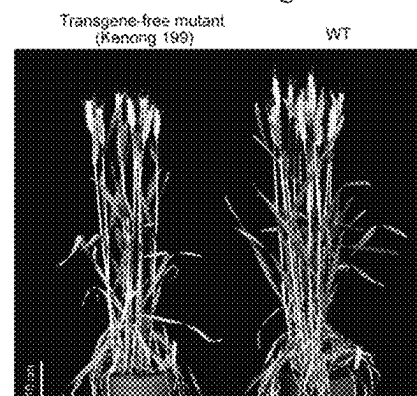
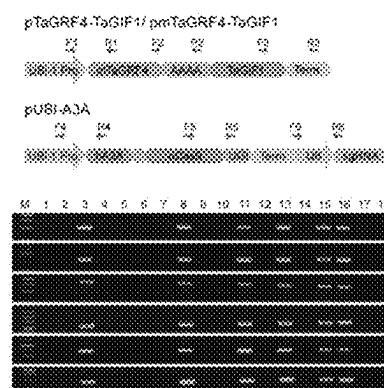
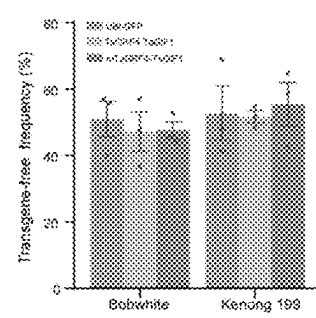
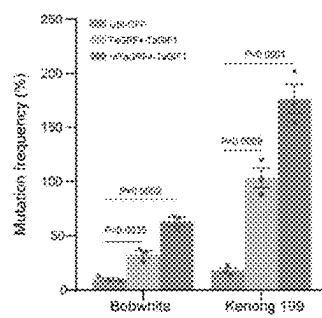
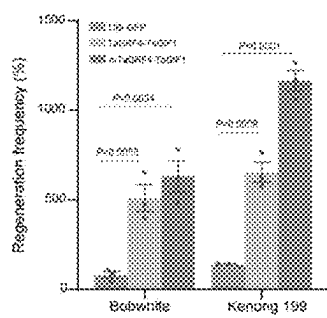
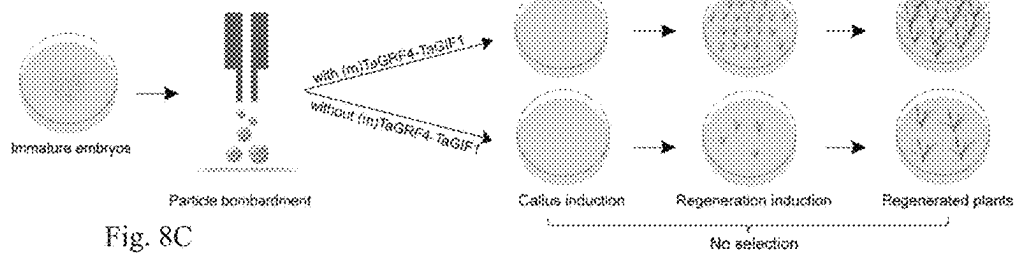
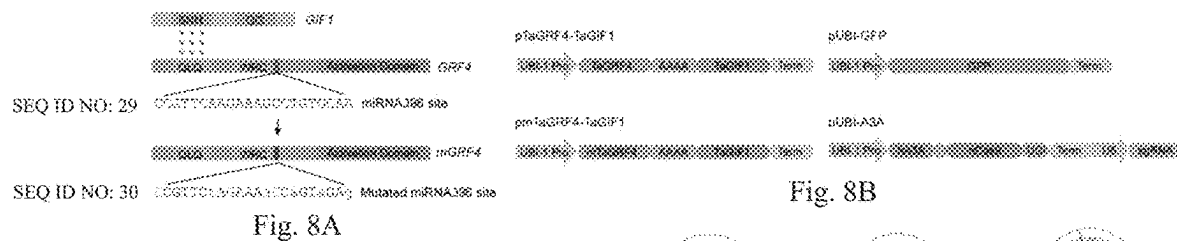
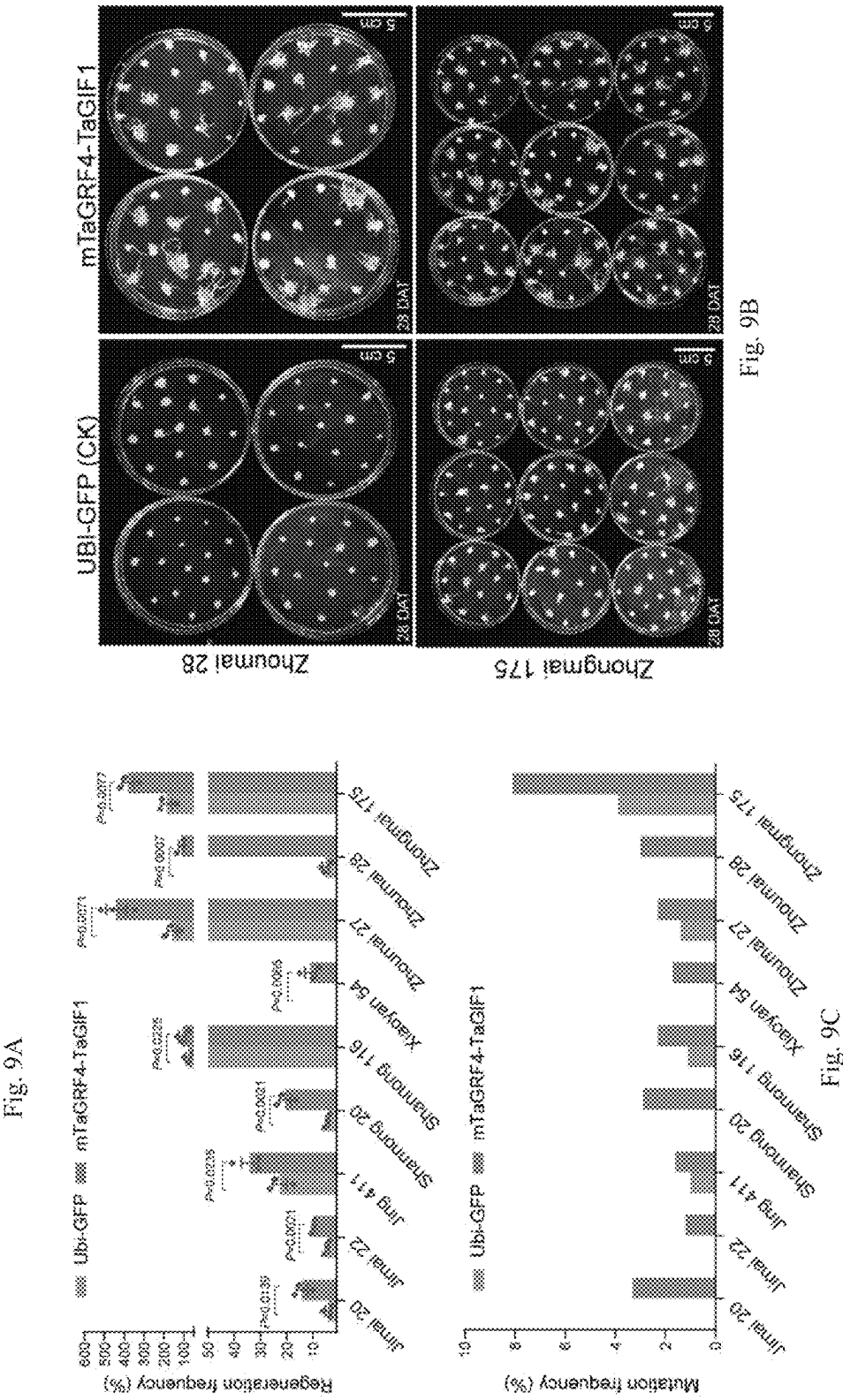


Fig. 5K









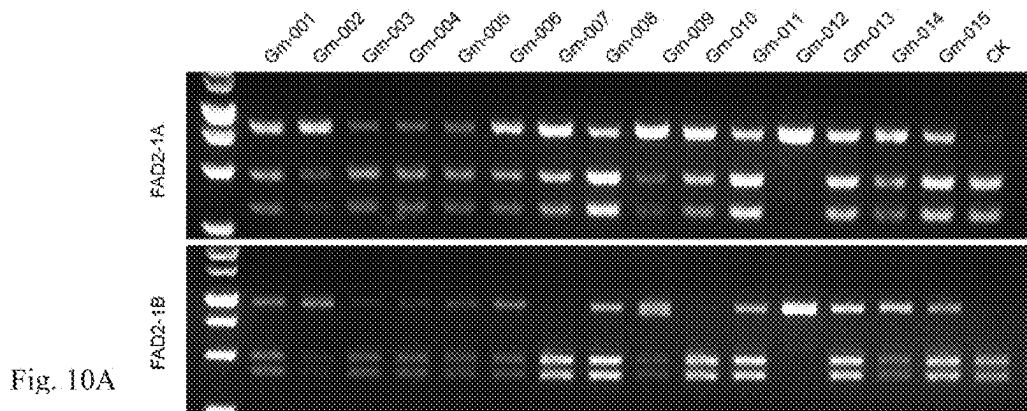
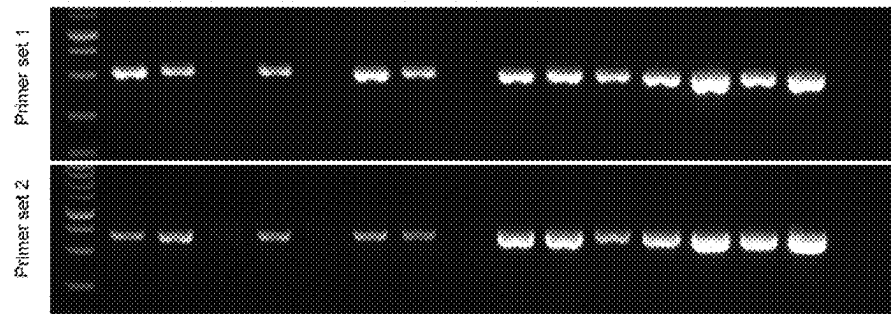


Fig. 10B



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SEQ ID NO: 33		1B: tccttctctcaaaccccttttccctcattggcatggccaatctatctgggttctctcaagggttgcccttctcaactggt	-2
SEQ ID NO: 34	Gm-002	1A: tccttctctcaaaccccttttccctcattggcatggccaatctatctgggttctctcaagggttgcccttctcaactggt	-12
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SEQ ID NO: 36	Gm-003	1A: tccttctctcaaaccccttttccctcattggcatggccaatctatctgggttctctcaagggttgcccttctcaactggt	-1
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Fig. 10C

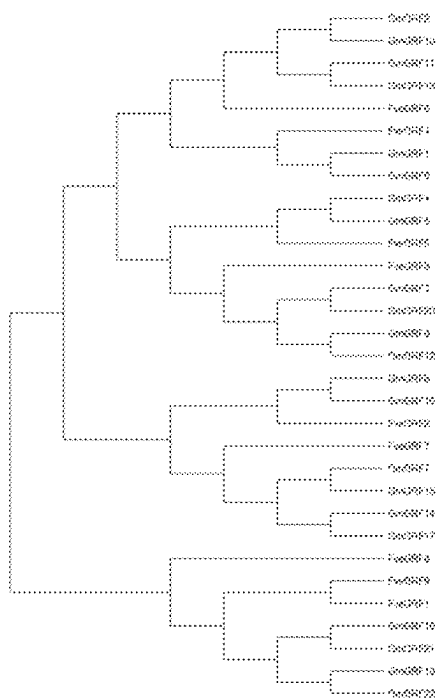


Fig. 11A

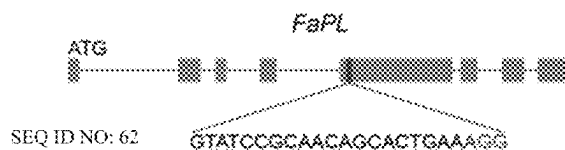


Fig. 11B

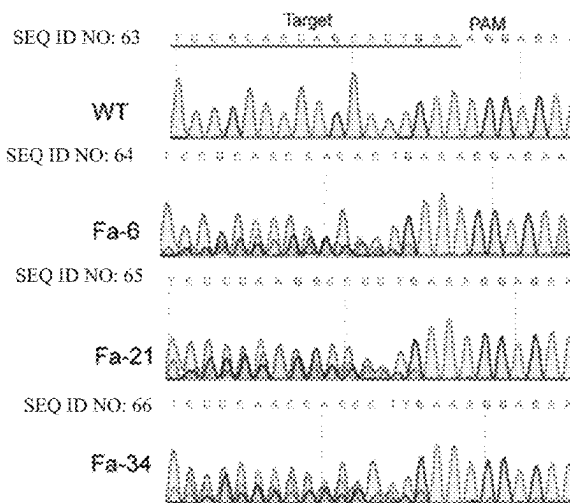


Fig. 11C

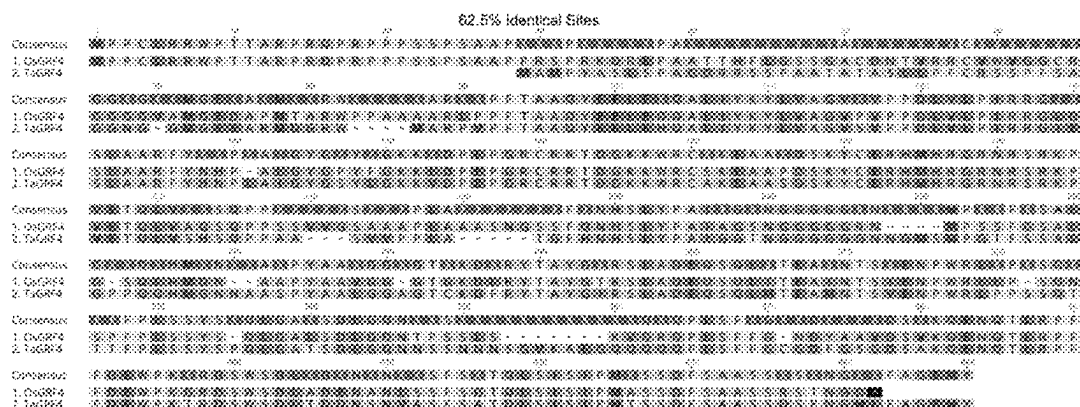


Fig. 12A.

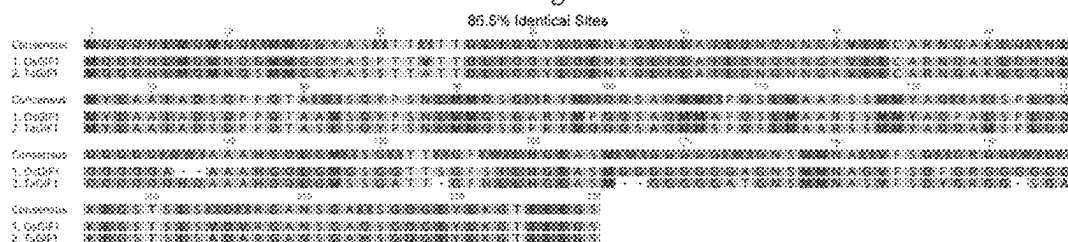


Fig. 12B

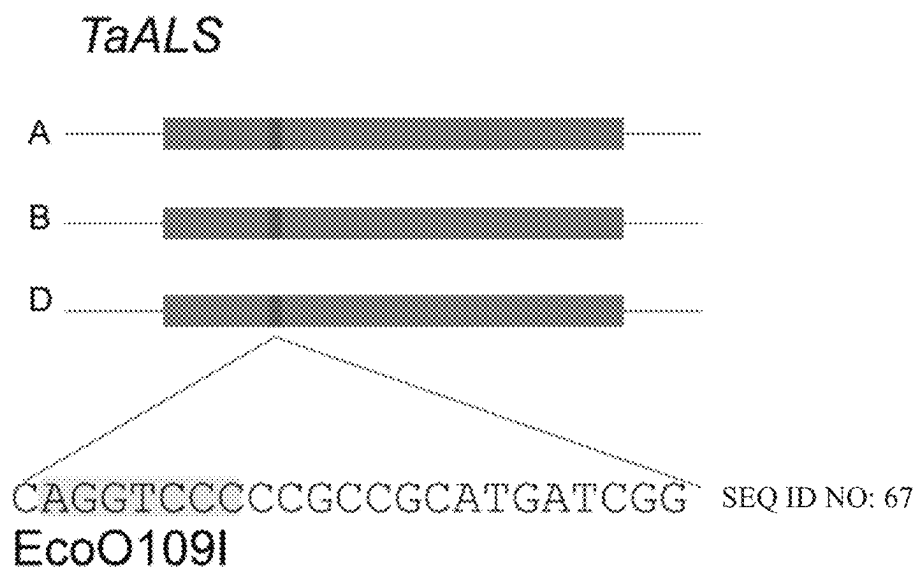


Fig. 13

1

METHOD FOR IMPROVING PLANT GENETIC TRANSFORMATION AND GENE EDITING EFFICIENCY

REFERENCE TO SEQUENCE LISTING TEXT FILE

The Sequence Listing text file associated with the instant disclosure has been electronically submitted to the United States Patent and Trademark Office via EFS as a 123.118 byte UTF-8-encoded text file created on Jul. 5, 2023 and entitled "1547_48_PCT_US_ST25.xml". The Sequence Listing submitted via EFS is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present invention belongs to the field of plant genetic engineering. Specifically, the present invention relates to a method for improving plant genetic transformation and gene editing efficiency. More specifically, the present invention relates to improving regeneration efficiency of plant genetic transformation and/or improving efficiency of plant gene editing by expression of genes which can promote division of plant cells, especially meristematic cells.

BACKGROUND OF THE INVENTION

Development of crop genetic breeding includes artificial selection breeding, cross breeding, mutation breeding and molecular marker assisted breeding using molecular technology. As the variety genetic diversity is gradually reduced, the bottleneck effect of traditional breeding is more and more obvious: it is difficult to use conventional breeding technology to obtain breakthrough new varieties to meet the requirement of human and sustainable agricultural development. The rapid development of life sciences makes it possible to enter the post-genome era from the "read" phase of the biological genetic information, and the accurate "rewriting" of the genome and the "new design" are becoming reality. This kind of biological technical means designed for creating new trait or living body shows great prospect in the field of disease treatment, medicine, manufacturing, especially agriculture and so on.

Genome editing technology is a revolutionary technical means appearing in the current life science which can realize accurate, efficient and specific rewriting of the genome and has revolutionary pushing effect to the research and exploration of life science. Gene editing refers to deleting, replacing, or inserting operation of the target gene so as to modify the genetic information to obtain a new function or phenotype, even creating a new species. Development of efficient and accurate breeding technology suitable for crops using gene editing technology will break the defect of the traditional breeding, realizing molecular design breeding of precise transformation from the genome. It has important strategic significance for the development of future agriculture.

Current gene editing technology mainly comprises ZFN, TALEN and CRISPR/Cas system. CRISPR/Cas system, due to its high efficiency and flexibility, is currently the simplest and widely used gene editing technology system. In CRISPR/Cas system, Cas protein can target any position in the genome under the guide action of the artificial designed guide RNA (guide RNA). Base editing system is a new gene editing technology developed based on CRISPR system, which can be divided into cytosine base editing system and

2

adenine base editing system, wherein the respectively deaminase and adenine deaminase are fused with Cas9 single-chain nickase. Under the targeting function of the guide RNA, Cas9 single-chain nickase generates a single-stranded DNA region, so the deaminase can efficiently respectively remove amino group of the C and A nucleotide on the single-stranded DNA of the targeting position, resulting in U base and I base, which can be repaired as T base and G base by the repair process of the cell itself. Base editing technology overcomes the defect of traditional DSB-mediated gene editing, which can efficiently realize the precise replacement of a single base. CRISPR/Cas system-mediated robust genetic engineering technology system will provide strong technical support for gene research and new plant molecular design breeding, and will accelerate the cultivation of new variety of crops and realize sustainable development of agriculture.

A key step of plant gene editing is to deliver the gene editing nuclease protein or coding nucleic acid to the plant cell to realize the editing of the target gene. The present plant genome editing delivery technology is mainly realized by genetic transformation and tissue culture technology, mainly comprising *agrobacterium* mediated method and gene gun method. Important progress has been made in plant transformation over the past few years, but transformation of many agronomically important plants (e.g., maize, soybeans, canola, wheat, indica rice, sugar cane and sorghum; and inbred lines) is still difficult and time-consuming. Generally, the only method causing the culture reaction is to optimize the culture medium components and/or explant materials and sources, which results in the success in some gene type, but many important crop plants (including excellent inbred line or variety) do not generate beneficial culture reaction and regeneration technology system. Although the transformation of a pattern genotype may be effective, the process of gradually introgressing the transgene into the product inbred line is laborious, expensive and time-consuming. Especially for monocotyledon wheat, efficiency of current gene gun and *agrobacterium* transformation method is low and greatly limited by the genotype, and a long-term tissue culture process is required. At present, the maximum bottleneck of wheat gene editing is the low efficiency of the current traditional wheat transformation system, the large technical difficulty, limitation by genotype and low throughput.

In order to facilitate the research of plant gene function and molecular design breeding more efficiently utilizing gene editing technology, establishing and digging a method to improve the plant transformation efficiency and shorten the time of tissue culture has important meaning.

SUMMARY OF THE INVENTION

In one aspect, the present invention provides a method for improving plant cell regeneration efficiency in plant transformation, the method comprising:

- (a) introducing into a cell of the plant
 - i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or
 - ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF;
- (b) regenerating an intact plant from the plant cell.

In another aspect, the present invention provides a method for improving the transformation efficiency of an exogenous nucleic acid sequence of interest in a plant or for transforming an exogenous nucleic acid sequence of interest into a plant, the method comprising:

- (a) introducing into a cell of the plant
 - i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or
 - ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF;
- (b) introducing at least one expression construct comprising at least one exogenous nucleic acid sequence of interest into the plant cell; and
- (c) regenerating an intact plant from the plant cell.

In another aspect, the present invention provides a method for improving gene editing efficiency in a plant or for gene editing in a plant, the method comprising:

- (a) introducing into a plant cell
 - i) an expression construct comprising a coding sequence of WUS, an expression construct comprising a coding sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or
 - ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a coding sequence of GIF;
- (b) introducing at least one expression construct comprising at least one exogenous sequence of interest to the plant cell, wherein the at least one exogenous sequence of interest encodes a component of a gene editing system; or introducing at least one component of a gene editing system to the plant cell; and
- (c) regenerating an intact plant from the plant cell.

The invention further provides a kit for carrying out the method of the invention, comprising at least i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF.

The present invention also provides use of i) an expression construct comprising a coding nucleic acid sequence of WUS; an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, for improving plant cell regeneration efficiency in plant transformation, for improving the transformation efficiency of exogenous nucleic acid sequence of interest in the plant or for improving the gene editing efficiency in the plant.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. Expression constructs for detecting the influence of a plurality of DR combinations on genetic transformation and gene editing efficiency.

FIG. 2 shows expression constructs for verifying the effect of GRF4 and gif1 combination on genetic transformation and gene editing efficiency in wheat.

FIG. 3 shows the expression constructs for GRF4 and gif1 combinatorial optimization experiment.

FIG. 4 shows the soybean GRF/gif expression construct.

FIGS. 5A-5K. The effect of GmGRF-GmGIF1 complexes on regeneration, transformation, and genome editing of the soybean cultivar Williams 82. (FIG. 5A) Schematic representations of T-DNA regions each containing one of the four GmGRF-GmGIF1 complexes. pBSE401 is the CRISPR/Cas9 control construct. (FIG. 5B) Schematic of the GmFAD2 sgRNA target sites in the two subgenomes. (FIG. 5C) The general procedure for *Agrobacterium*-mediated transformation of soybean. (FIG. 5D) Comparison of the effects of four GmGRF-GmGIF1 complexes on regeneration frequencies of Williams 82. RF (regeneration frequency)=no. of explants with multiple buds/explant number $\times$ 100%. (FIG. 5E) Numbers of putative glufosinate-resistant elongated shoots $\geq$ 2 cm in length (on day 50 after transformation) and shoots $\geq$ 9 cm in length (on day 75 after transformation) regenerated from explants transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5F) Average numbers of elongated shoots regenerated from single explants transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5G) Regeneration of explants transformed with pGmGRF5-GmGIF1 and pBSE401 and cultured in medium supplemented with 5.0 mg/L glufosinate on day 50 after infection. (FIG. 5H) Typical explants with regenerated shoots cultured in medium supplemented with 5.0 mg/L glufosinate on day 75 after infection with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 5I) Transformation efficiencies of Williams 82 transformed with the four GmGRF-GmGIF1 complexes and pBSE401, respectively. (FIG. 5J) GmFAD2 mutation frequencies with CRISPR/Cas9 treated by four pGmGRF-GmGIF1 constructs and pBSE401, respectively. (FIG. 5K) Performance of mature GmFAD2-edited soybean plants (160 days after transformation) transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. All values and error bars are mean values $\pm$ s.e.m. of three independent experiments.

FIGS. 6A-6G. The effect of the GmGRF5-GmGIF1 complex on regeneration, transformation, and genome editing of two soybean cultivars Zhonghuang 13 and Hefeng 25. (FIG. 6A) Regeneration frequencies of explants transformed with pGmGRF5-GmGIF1 and pBSE401 on day 50 after transformation. (FIGS. 6B and 6C) Numbers of putative glufosinate-resistant elongated shoots ($>$ 2 cm in length) on day 50 after transformation and elongated shoots ($>$ 9 cm in length) on day 75 regenerated from explants of Zhonghuang 13 (FIG. 6B) and Hefeng 25 (FIG. 6C) transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6D) Shoot induction, shoot proliferation and shoot elongation in Hefeng 25 explants transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6E) Average numbers of elongated shoots regenerated from single explants of Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6F) Transformation efficiencies of Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. (FIG. 6G) GmFAD2 mutation rates in Zhonghuang 13 and Hefeng 25 transformed with pGmGRF5-GmGIF1 and pBSE401, respectively. All values and error bars are mean values $\pm$ s.e.m. of three independent experiments.

FIG. 7A-7D. The effect of the GmGRF5-GmGIF1 complex on regeneration, transformation, and genome editing efficiencies in strawberry cultivar Benihoppe. (FIG. 7A) Schematic representation of the T-DNA region containing

the GmGRF5-GmGIF1 complex. pHUE411-GFP is the CRISPR/Cas9 control construct. (FIG. 7B) Effects of the GmGRF5-GmGIF1 complex on strawberry regeneration, transformation, and genome editing efficiencies. The data were collected on day 35 after explants were transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP. RF (regeneration frequency)=no. of regenerated shoots/total explants $\times$ 100%. TE (transformation efficiency)=no. of transgenic shoots/total explants $\times$ 100%. MF (mutation frequency)=no. of mutants/total explants $\times$ 100%. (FIG. 7C) GFP positive calli from explants transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP, on day 21 after transformation. (FIG. 7D) Calli and shoots from explants transformed with pHUE411-GFP-GmGRF5-GmGIF1 and pHUE411-GFP, respectively, on day 35 after transformation.

FIGS. 8A-8H. Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on regeneration and genome editing in two common wheat cultivars Kenong 199 and Bobwhite. (FIG. 8A) Schematic representation of common wheat GIF1, GRF4, and mutated GRF4. The dotted lines indicate the interaction between the SNH and QLQ domains. mTaGRF4 was created by introducing five point mutations in the miRNA396 target site of common wheat TaGRF4. (FIG. 8B) Schematic representations of constructs pTaGRF4-TaGIF1, pmTaGRF4-TaGIF1, and the base editor, pUBI-A3A. pUBI-GFP is the control construct. (FIG. 8C) General procedure for transgene-free genome editing in common wheat by transient expression of a cytosine base editor. (FIG. 8D) Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on regeneration frequencies of Bobwhite and Kenong199. RF (regeneration frequency)=no. of regenerated shoots/immature embryos bombarded $\times$ 100%. (FIG. 8E) Comparison of the effects of TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 on genome editing frequencies in Bobwhite and Kenong199. MF (mutation frequency)=no. of mutants/immature embryos bombarded $\times$ 100%. (FIG. 8F) Primer sets for detecting transgene-free mutants, and the outcome of tests on 22 representative TaALS mutant plants (Kenong 199). (FIG. 8G) The Transgene-free frequencies in Bobwhite and Kenong199 wheat cultivars transformed with TaGRF4-TaGIF1, mTaGRF4-TaGIF1 and pUBI-GFP (control construct), respectively. (FIG. 8H) Transgene-free mutant plants regenerated from Kenong 199 immature embryos transiently-expressing mTaGRF4-TaGIF1 and the cytosine base editor A3A-PBE, do not exhibit abnormal growth. In (FIG. 8D), (FIG. 8E) and (FIG. 8G), values and error bars are means $\pm$ s.e.m. of three independent experiments.

FIGS. 9A-9C. The effect of transient expression of the mutated TaGRF4-TaGIF1 complex on common wheat regeneration and genome editing efficiencies in nine elite wheat cultivars. (FIG. 9A) The regeneration frequencies of nine elite common wheat cultivars transformed with pmTaGRF4-TaGIF1 and pUBI-GFP (control construct). Values and error bars are means $\pm$ s.e.m. of three independent experiments. (FIG. 9B) Regenerated plants of Xiaoyan 54 and Zhongmai 175 transformed with mTaGRF4-TaGIF1 and pUBI-GFP (control construct), respectively, 28 days after transformation. (FIG. 9C) The mutation frequencies in the nine elite common wheat cultivars transformed with pmTaGRF4-TaGIF1 and pUBI-GFP (control construct). MF (mutation frequency)=total no. of mutants/total immature embryos bombarded $\times$ 100%.

FIGS. 10A-10C. Detection of mutations and transgene-free mutant plants in 15 representative Williams 82 soybean lines transformed with pGmGRF5-GmGIF1. (FIG. 10A)

Mutations in the GmFAD2 gene from the 15 representative soybean lines identified by PCR-RE assays. Lanes 1 to 15 show digests of the PCR fragments amplified from the transgenic soybean plants using BstXI. Lanes labeled CK show the digests of PCR fragments amplified from a wild-type control plant. (FIG. 10B) The outcome of tests for transgene-free mutants using two primer sets in 15 representative gmfad2 mutant plants. Lanes without a band indicate transgene-free mutants. Lanes labeled CK are the PCR fragments amplified from a WT plant. (FIG. 10C). Sanger sequencing of the GmFAD2 gene in wild type and the edited gmfad2 mutants.

FIGS. 11A-11C. Phylogenetic analysis of GRFs from *Glycine max* (Gm) and *Fragaria vesca* (Fve) and the sgRNA designed to generate mutant strawberry FaPL genes. (FIG. 11A) Clustal W was used to align 31 GRFs, including nine FveGRFs and 22 GmGRFs. MEGA 7.0 was used to construct a neighbor-joining phylogenetic tree with 1000 bootstrap replications. (FIG. 11B) Schematic of the sgRNA designed to target the FaPL gene. (FIG. 11C) Detection of mutations by Sanger sequencing in regenerated strawberry lines transformed with pHUE411-GFP-GmGRF5-GmGIF1.

FIGS. 12A and 12B. Multiple sequence alignment of the deduced GRF4 and GIF1 proteins, which are the most similar between common wheat and rice. (FIG. 12A) The GRF4 from common wheat shows 62.5% conservation with the GRF4 from rice at the amino acid level. (FIG. 12B) The GIF1 from common wheat shows 86.5% conservation with the GIF1 from rice at the amino acid level. Sequences were compared using Geneious Prime.

FIG. 13. Schematic of the TaALS sgRNA target sites in the common wheat genome. SgRNA target sites within a conserved region of common wheat TaALS homeologs were targeted by the base editing systems. The EcoO109I restriction enzyme site in the sgRNA target sequence was used for mutation detection.

DETAILED DESCRIPTION OF THE INVENTION

I. Definitions

In the present invention, unless otherwise specified, the scientific and technical terms used herein have the meaning commonly understood by those skilled in the art, and the protein and nucleic acid chemistry used herein, molecular biology, cell and tissue culture, microbiology, immunology related term and laboratory operation steps are widely used in the corresponding field and conventional steps. For example, the standard recombinant DNA and molecular cloning techniques used in the present invention are well known to those of skill in the art, and are more fully described in the following literature: Sambrook, J., Fritsch, E. F. and Maniatis, T., *Molecular Cloning: A Laboratory Manual*; Cold Spring Harbor Laboratory Press: Cold Spring Harbor, 1989 (hereinafter "Sambrook"). At the same time, for a better understanding of the present invention, the definitions and explanations of the related terms are provided below.

As used herein, the term "and/or" encompasses all combinations of items connected by the term, and each combination should be regarded as individually listed herein. For example, "A and/or B" covers "A", "A and B", and "B". For example, "A, B, and/or C" covers "A", "B", "C", "A and B", "A and C", "B and C", and "A and B and C".

When the term "comprise" is used herein to describe the sequence of a protein or nucleic acid, the protein or nucleic

acid may consist of the sequence, or may have additional amino acids or nucleotide at one or both ends of the protein or nucleic acid, but still have the activity described in this invention. In addition, those skilled in the art know that the methionine encoded by the start codon at the N-terminus of the polypeptide will be retained under certain practical conditions (for example, when expressed in a specific expression system), but does not substantially affect the function of the polypeptide. Therefore, when describing the amino acid sequence of specific polypeptide in the specification and claims of the present application, although it may not include the methionine encoded by the start codon at the N-terminus, the sequence containing the methionine is also encompassed, correspondingly, its coding nucleotide sequence may also contain a start codon; vice versa.

"Genome" as used herein encompasses not only chromosomal DNA present in the nucleus, but also organelle DNA present in the subcellular components (e.g., mitochondria, plastids) of the cell.

The term "exogenous" with respect to sequence means a sequence that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention.

"Polynucleotide", "nucleic acid sequence", "nucleotide sequence", or "nucleic acid fragment" are used interchangeably to refer to a polymer of RNA or DNA that is single- or double-stranded, optionally containing synthetic, non-natural or altered nucleotide bases. Nucleotides (usually found in their 5'-monophosphate form) are referred to by their single letter designation as follows: "A" for adenylate or deoxyadenylate (for RNA or DNA, respectively), "C" for cytidylate or deoxycytidylate, "G" for guanylate or deoxyguanylate, "U" for uridylate, "T" for deoxythymidylate, "R" for purines (A or G), "Y" for pyrimidines (C or T), "K" for G or T, "H" for A or C or T, "I" for inosine, and "N" for any nucleotide.

"Polypeptide", "peptide", "amino acid sequence" and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The terms "polypeptide", "peptide", "amino acid sequence", and "protein" are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulfation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

As used herein, an "expression construct" refers to a vector suitable for expression of a nucleotide sequence of interest in an organism, such as a recombinant vector. "Expression" refers to the production of a functional product. For example, the expression of a nucleotide sequence may refer to transcription of the nucleotide sequence (such as transcribe to produce an mRNA or a functional RNA) and/or translation of RNA into a protein precursor or a mature protein.

"Expression construct" of the invention may be a linear nucleic acid fragment (including a DNA or RNA fragment), a circular plasmid, a viral vector.

The "expression construct" of the present invention may comprise a regulatory sequence and a nucleic acid sequence of interest operably linked thereto. The regulatory sequence and the nucleic acid sequence of interest may be of different sources, or are of the same origin but are arranged in a manner different from that normally found in nature.

"Regulatory sequence" and "regulatory element" are used interchangeably and refer to a nucleotide sequence which is located upstream (5' non-coding sequences), within, or downstream (3' non-coding sequences) of a coding sequence, and influence the transcription, RNA processing or stability, or translation of the associated coding sequence. The regulatory sequence may include, but is not limited to, a promoter, a translation leader sequence, an intron and a polyadenylation recognition sequence. "Promoter" refers to a nucleic acid fragment capable of controlling transcription of another nucleic acid fragment. In some embodiments of the invention, the promoter is a promoter capable of controlling gene transcription in a cell regardless of whether it is derived from the cell. The promoter may be a constitutive promoter or a tissue specific promoter or a developmentally regulated promoter or an inducible promoter.

As used herein, the term "operably linked" refers to a regulatory element (e.g., but not limited to, a promoter sequence, a transcription termination sequence, etc.) and a nucleic acid sequence (e.g., a coding sequence or open reading frame), are linked such that transcription of the nucleotide sequence is controlled and regulated by the transcription regulatory element. Techniques for operably linking a regulatory element region to a nucleic acid molecule are known in the art.

"Introducing" a nucleic acid molecule (e.g., an expression construct) into a plant cell refers to that the nucleic acid molecule is presented to the plant cell such that the nucleic acid molecule enter the interior of the plant cell.

"Regeneration" refers to the process of growing an intact plant from one or more plant cells (e.g., plant protoplasts, callus or explants).

II. Improved Plant Transformation

In one aspect, the present invention provides a method for improving plant cell regeneration efficiency in plant transformation, the method comprising:

- (a) introducing into a cell of the plant
 - i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or
 - ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF;
- (b) regenerating an intact plant from the plant cell.

In another aspect, the present invention provides a method for improving transformation efficiency of an exogenous nucleic acid sequence of interest in a plant or for transforming an exogenous nucleic acid sequence into a plant, the method comprising:

- (a) introducing into a cell of the plant
 - i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or
 - ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF;
- (b) introducing at least one expression construct comprising at least one exogenous nucleic acid sequence of interest to the plant cell; and
- (c) regenerating an intact plant from the plant cell.

In some embodiments of this aspect, step (a) and step (b) are carried out at the same time. In some embodiments of this aspect, step (a) is performed prior to step (b). In some embodiments of this aspect, step (b) is performed prior to step (a). In some embodiments of this aspect, step (c) is performed after step (a) and step (b).

In another aspect, the present invention provides a method for improving gene editing efficiency in a plant or for gene editing in a plant, the method comprising:

- (a) introducing into a cell of the plant
 - i) an expression construct comprising a coding sequence of WUS, an expression construct comprising a coding sequence of BBM and an expression construct comprising a coding sequence of SERK; and/or
 - ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a coding sequence of GIF;
- (b) introducing at least one expression construct comprising at least one exogenous sequence of interest to the plant cell, wherein the at least one exogenous sequence of interest encodes a component of a gene editing system; or introducing at least one component of the gene editing system to the plant cell; and
- (c) regenerating an intact plant from the plant cell.

In some embodiments of this aspect, step (a) and step (b) are carried out at the same time. In some embodiments of this aspect, step (a) is performed prior to step (b). In some embodiments of this aspect, step (b) is performed prior to step (a). In some embodiments of this aspect, step (c) is performed after step (a) and step (b).

The invention further provides a kit for carrying out the method of the invention, comprising at least i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding sequence of GRF and an expression construct comprising a coding sequence of GIF.

The present invention also provides use of i) an expression construct comprising a coding nucleic acid sequence of WUS; an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK; and/or ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, for improving plant cell regeneration efficiency in plant transformation, for improving the transformation efficiency of exogenous nucleic acid sequence of interest in the plant or for improving the gene editing efficiency in the plant.

WUS (WUSCHEL), BBM (BABY BOOM) and SERK (Somatic Allogeneic receptor-like kinase) are conservative development regulatory factors (DR) widely present in plants. The inventors have surprisingly found that co-expression of the combination of WUS, BBM and SERK in the plant cell can significantly improve the efficiency of the plant cell to regenerate into intact plants, and significantly improve transformation efficiency of an exogenous nucleic acid sequence of interest into the plant. When the exogenous nucleic acid sequence of interest encodes a gene editing system, the gene editing efficiency can be significantly improved.

Examples of WUS, BBM and SERK suitable for use in the present invention include, but are not limited to, WUS,

BBM, and SERK from *Arabidopsis*, canola, strawberry, potato, rice, tomato, soybean, corn or wheat.

In some embodiments of various aspects of the invention, WUS is corn WUS (ZmWUS), BBM is the corn BBM (ZmBBM), or SERK is corn SERK (ZmSERK). In some embodiments, the ZmWUS comprises the amino acid sequence shown in SEQ ID NO: 1. In some embodiments, the ZmBBM comprises the amino acid sequence shown in SEQ ID NO: 2. In some embodiments, the ZmSERK comprises the amino acid sequence shown in SEQ ID NO: 3.

In some embodiments of various aspects of the present invention, at least two or at least three or all of the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK, and the at least one exogenous nucleic acid sequence of interest are placed in a same expression construct. In some embodiments, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK and the at least one exogenous nucleic acid sequence of interest are respectively placed in different expression constructs.

In some embodiments of various aspects of the invention, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, and the coding nucleic acid sequence of SERK are placed in the same expression construct, while the at least one exogenous nucleic acid sequence of interest is placed in another expression construct.

In some embodiments of various aspects of the present invention, the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK and/or the at least one exogenous nucleic acid sequence of interest are operatively connected to transcription regulatory elements.

Methods of expressing different proteins by the same expression construct are known in the art. For example, the different proteins can be placed in the same expression construct under the control of different transcriptional regulatory elements (e.g., different promoters). Alternatively, different proteins can be fused by self-cleaving peptide (e.g., 2A peptide, including but not limited to P2A, E2A, F2A and T2A, etc.), then expressed under the control of the same transcriptional control element (e.g., different promoter), so that separate different proteins can be generated by self-cleavage of the self-cleaving peptide after translation or translation. Alternatively, an internal ribosome entry site (IRES) can be inserted between the coding nucleic acid sequences of different proteins.

GRFs (Growth Factors) are specific transcription factors in plants, mainly controlling plant cell size, chloroplast proliferation, stamen development, osmotic stress and other plant growth and development processes. GRF transcription factors are widely existed in the plant, mainly comprising two conserved domains: QLQ and WRC. The QLQ domain of GRFs can interact with the SNH domain (SYT N-terminal domain) in GIF (GRF-interacting factor) proteins, so as to exercise transcriptional activation function. WRC domain comprises 1 functional nuclear localization signal and 1 DNA binding motif, plays a role in DNA binding. Generally, QLQ and WRC domain are located at the N terminal of the GRFs. However, some GRFs also have a second WRC domain at the C terminal.

“GIF” (GRF-interacting factor) is a protein that can form transcription co-activation factor complexes with GRF. GIFs are homologous to human transcription co-activation factor synovial sarcoma transport protein (synovial translocation protein, SYT). In *Arabidopsis thaliana*, GIF plays a role in

cell proliferation during blade development and maintains proliferation ability of the meristematic cells during flower organ development.

The inventors have further surprisingly found that co-expression of the combination of GRF and GIF in plant cell can significantly improve the regenerating efficiency of a plant cell into intact plant, and significantly improve the transformation efficiency of exogenous nucleic acid sequence of interest into the plant. When the exogenous nucleic acid sequence of interest encodes a gene editing system, the gene editing efficiency can be significantly improved.

Examples of GRFs suitable for use in the present invention include, but are not limited to, GRF from *Arabidopsis*, canola, potato, rice, tomato, soybean, corn or wheat. Examples of GIF suitable for use in the present invention include, but are not limited to, GIF from *Arabidopsis*, canola, potato, rice, tomato, soybean, corn or wheat. However, as long as it can form transcription co-activating factor complex, GRF and GIF in the present invention does not necessarily have the same origin.

In some embodiments of the invention, the GRF is wheat GRF. Suitable wheat GRFs include, but are not limited to, wheat GRF4. In some embodiments, the wheat GRF4 comprises the amino acid sequence of SEQ ID NO: 4.

In some embodiments of the invention, the GIF is wheat GIF. Suitable wheat GIFs include, but are not limited to, wheat GIF1. In some embodiments, the wheat GIF1 comprises the amino acid sequence shown in SEQ ID NO: 6.

In some embodiments of the invention, the GRF is soybean GRF. Suitable soybean GRF include, but are not limited to, soybean GRF5, soybean GRF6, soybean GRF11, or soybean GRF11. In some embodiments, the soybean GRF5 comprises the amino acid sequence of SEQ ID NO: 7. In some embodiments, the soybean GRF6 comprises the amino acid sequence of SEQ ID NO: 8. In some embodiments, the soybean GRF11 comprises the amino acid sequence of SEQ ID NO: 9. In some embodiments, the soybean GRF18 comprises the amino acid sequence of SEQ ID NO: 10.

In some embodiments of the invention, the GIF is derived from soybean GIF. Suitable soybean GIFs include, but are not limited to, soybean GIF1. In some embodiments, the soybean GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11.

Many of the transcription factors in the plant including GRF are regulated by miRNAs. For example, GRF4 is negatively controlled by miR396. The invention surprisingly found that mutating the miRNA binding site in GRF can significantly improve the effect of the GRF/GIF combination in improving efficiency of plant cell regeneration and plant genetic transformation.

Therefore, in some embodiments of the present invention, the GRF comprises a mutated miRNA binding site, so as not to be regulated by the miRNA. Examples of the miRNAs include, but are not limited to, miR396, depending on the particular GRF. In some embodiments, the GRF with the mutated miRNA binding site comprises the amino acid sequence of SEQ ID NO: 5.

In some embodiments of the present invention, at least two or all of the coding sequence of the GRF, the coding sequence of the GIF and the at least one exogenous nucleic acid sequence of interest are placed in a same expression construct.

In some embodiments of the present invention, the coding sequence of the GRF and the coding sequence of GIF are

placed in a same expression construct, while the at least one exogenous nucleic acid sequence of interest is placed in another expression construct.

In some embodiments of the present invention, the coding sequence of the GRF, the coding sequence of the GIF and the at least one exogenous nucleic acid sequence are operatively linked to transcription control sequences.

In some embodiments of the invention, the GRF is fused to the GIF. In some embodiments, the GRF is fused to the N terminal of the GIF. In some embodiments, the GRF is fused to the GIF through a linker. An exemplary linker comprises the sequence AAAA (SEQ ID NO: 12) or SGGS (SEQ ID NO: 13). Preferably, the linker is AAAA.

In some embodiments of the present invention, “an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF” encompasses an expression construct comprising a coding sequence of the fusion protein of GRF and GIF.

In some embodiments of the present invention, the fusion protein of the GRF and GIF comprises the amino acid sequence encoded by any one of SEQ ID NO: 17-22. In some embodiments of the present invention, the fusion protein of the GRF and GIF is encoded by any one of SEQ ID NO: 17-22. In some embodiments of the present invention, the fusion protein of GRF and GIF comprises the amino acid sequence encoded by any one of SEQ ID NO: 23-28.

The “at least one exogenous nucleic acid sequence of interest” may be any nucleic acid sequence to be transformed into the plant. For example, the exogenous nucleic acid sequence of interest can encode an agronomic trait, insect resistance, disease resistance, herbicide resistance, sterility, grain characteristics, and a trait important for the commercial product. The nucleic acid sequence of interest may also include those nucleic acid sequences involved in oil, starch, carbohydrate or nutrient metabolism, and those nucleic acid sequences affecting seed size, sucrose content, and the like.

In some preferred embodiments of the present invention, the at least one exogenous nucleic acid sequence encodes a component of a gene editing system, so as to carry out gene editing in the plant.

“Gene editing”, also known as genome editing, uses a sequence-specific nuclease or a derivative thereof for nucleotide insertion, deletion or substitution in the genome of an organism. Gene editing generally results in site-specific double-strand break (DSB) at the desired position in the genome, and then introducing desired DNA insertion, deletion or substitution by the process of repairing DSB. However, gene editing can also cover base editing technology, transcriptional activation or inhibition, epigenetic modification technology, which does not relate to DSB, as long as it has sequence specificity.

The gene editing system used is not particularly limited in the present invention. For example, a gene editing system suitable for use in the present invention includes but is not limited to zinc finger nuclease (ZFN), meganuclease (MGN); transcription activating factor-like effector nuclease (TALEN) and CRISPR (Clustered regularly interspaced short palindromic repeats) system. “Zinc finger nucleases” are artificial restriction enzymes prepared by fusing a zinc finger DNA binding domain to a DNA cleavage domain. The zinc finger DNA binding domain of a single ZFN typically contains 3-6 individual zinc finger repeats, each of which can identify a sequence of, for example, 3 bp. By combining different zinc finger repeating sequences, different genome sequences can be targeted.

13

Meganucleases generally refer to homing endonucleases capable of identifying a nucleic acid sequence of 14-40 base in length. Long recognition sequence allows the meganucleases to have strong specificity so as to reduce the off-target effect.

“Transcription activator-like effector nucleases” are restriction enzymes that can be engineered to cleave specific DNA sequences, and that are typically prepared by fusing the DNA-binding domain of a transcription activator-like effector (TALE) to a DNA cleavage domain. TALE can be engineered to bind almost any desired DNA sequence.

“CRISPR system” generally comprises two components capable of forming complexes with sequence specificity: CRISPR nuclease or a variant thereof, and a corresponding guide RNA. Therefore, for the CRISPR system, the at least one exogenous nucleic acid sequence of interest of the invention can comprise a nucleic acid sequence encoding a CRISPR nuclease or a variant thereof, and/or a coding nucleic acid sequence of a corresponding guide RNA. Alternatively, at least one component of the gene editing system introduced into the plant cell may include a CRISPR nuclease or a variant thereof, and/or a corresponding guide RNA.

In some preferred embodiments of the present invention, the gene editing system is a CRISPR system. A large number of different CRISPR gene editing systems are known in the art, which can be applied to the invention. For example, suitable CRISPR gene editing system can be found at the Addgene company website (Addgene, Watertown, Massachusetts). CRISPR gene editing systems cover the systems capable of changing genomic sequence, but also comprise the systems for transcription control but not changing the genomic sequence.

As used herein, the term “CRISPR nuclease” generally refers to a nuclease present in the naturally occurring CRISPR system. The CRISPR nuclease variant comprises a modified form of natural CRISPR nuclease, artificial mutant (including the nickase mutant), catalytic active fragment, or a fusion with other functional protein/polypeptide and so on. A variety of artificial functional variants of CRISPR nucleases are known in the art, such as high specific variant or nickase variant, or a cytidine deaminase or adenosine deaminase fusion protein and so on. CRISPR nuclease or a variant thereof can interact with the corresponding guide RNA for recognizing, binding and/or cutting target nucleic acid structure. Those skilled in the art know how to select a suitable CRISPR nuclease or a variant thereof to achieve the purpose of the present invention.

CRISPR nuclease or a variant thereof used in the CRISPR gene editing system of the invention can be selected from Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3; GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, Csf1, Cas9, Csn2, Cas4, Cpf1 (Cas12a), C2c1, C2c3 or C2c2 protein, or functional variants of these nucleases.

In some embodiments of the present invention, the CRISPR nuclease or a variant thereof comprises a Cas9 nuclease or a variant thereof. CRISPR gene editing system based on Cas9 nuclease or variant thereof is also referred to herein as CRISPR-Cas9 gene editing system. The Cas9 nuclease can be a Cas9 nuclease from different species, such as spCas9 (having an amino acid sequence of SEQ ID NO: 15) from *Streptococcus pyogenes* (*S. pyogenes*).

Cas9 nuclease variant can include Cas9 nickase (nCas9), wherein one of two sub-domain (HNH nucleic acid enzyme sub-domain and RuvC sub-domain) of the Cas9 nuclease DNA cutting domain is inactivated to form a nickase. In some embodiments, combination of a Cas9 nickase and two

14

gRNAs targeting upstream and downstream of the sequence to be edited can be used to realize deletion of the sequence to be edited, or to realize the replacement of the sequence to be edited in the presence of a donor sequence.

In some embodiments of the present invention, the CRISPR nuclease or variant thereof may also comprises Cpf1 (Cas12a) nuclease or a variant thereof such as a high specific variant. The Cpf1 nuclease can be Cpf1 nuclease from different species, such as from *Francisella novicida* U112; *Acidaminococcus* sp.BV3L6 and *Lachnospiraceae* bacterium ND2006. CRISPR gene editing system based on Cpf1 nuclease or variant thereof is also referred to herein as CRISPR-Cpf1 system.

In some embodiments of the present invention, the CRISPR nuclease variant further comprises a base editor. The base editor is typically a fusion protein comprising a deaminase and a CRISPR nuclease variant lack of DNA cleavage activity.

As used in the present invention, the CRISPR nuclease variant lack of DNA cleavage activity comprises but not limited to Cas9 nickase (nCas9), nuclease-dead Cas9 nuclease (dCas9) or nuclease-dead Cpf1 nuclease (dCpf1). Nuclease-dead Cas9 nuclease (dCas9) or nuclease-dead Cpf1 nuclease (dCpf1) completely lacks DNA cutting activity. A plurality of CRISPR nuclease variants lack of DNA cleavage activity are known in the art.

As used in the present invention, “deaminase” refers to an enzyme that catalyzes the deamination reaction. In some embodiments of the present invention, the deaminase is a cytosine deaminase, which is capable of receiving single-stranded DNA as a substrate and capable of catalyzing cytidine or deoxycytidine respectively deaminated as uracil or deoxyuracil. In some embodiments of the present invention, the deaminase is adenosine deaminase, which is capable of receiving single-stranded DNA as a substrate and capable of catalyzing adenosine or deoxyadenosine (A) to form inosine (I). A variety of suitable cytosine deaminases or adenine deaminases with single-stranded DNA as substrate are known in the art. Suitable cytosine deaminases include, but are not limited to, APOBEC1 deaminase, activation-induced cytidine deaminase (AID), APOBEC3G, CDA1, human APOBEC3A deaminase. In some preferred embodiments, the cytosine deaminase is human APOBEC3A. Examples of suitable adenine deaminases include, but are not limited to, the DNA-dependent adenine deaminase disclosed by Nicloe M. Gaudelli et al. (doi: 10.1038/nature24644, 2017).

By using a fusion of a CRISPR nuclease variant lack of DNA cleavage activity and a deaminase (forming a so-called “base editor”), base editing in the target nucleotide sequence, such as conversion from C to T or conversion from A to G, can be achieved. A variety of base editors are known in the art, and those skilled in the art will know how to select a suitable base editor to achieve the object of the present invention. The base editor-based CRISPR gene editing system is also referred to as a base editing system.

In some preferred embodiments of the present invention, the CRISPR system is a base editing system. Preferably, the base editing system comprises a base editor having the amino acid sequence shown in SEQ ID NO: 14.

As used herein, “guide RNA” and “gRNA” can be interchangeably used, which refers to a RNA molecule that can form a complex with the CRISPR nuclease or its functional variant and is capable of targeting the complex to a target sequence because it has a certain identity to the target sequence. The guide RNA targets the target sequence through base pairing between the guide RNA and the

15

complementary strand of the target sequence. For example, gRNA used by Cas9 nuclease or its functional mutant is often composed of crRNA and tracrRNA molecules that are partially complemented to form the complex, wherein crRNA contains a guide sequence (referred to as seed sequence) that has sufficient identity to the target sequence so as to be hybridized with the complementary strand of the target sequence and directs a CRISPR complex (Cas9+crRNA+tracrRNA) to specifically bind to the target sequence. However, it has been known in the art that single guide RNA (sgRNA) can be designed, which simultaneously contains the features of crRNA and tracrRNA, gRNA used by Cpf1 nuclease or its functional variant is often only composed of matured crRNA molecules, which is also referred to as sgRNA. Designing suitable gRNA based on the CRISPR effector protein as used and the target sequence to be edited is within the skill of those skilled person in the art.

In some specific embodiments of the invention, the guide RNA is a sgRNA. For example, the sgRNA comprises a scaffold shown in SEQ ID NO: 16.

The sequence specific nuclease for gene editing in the invention, such as zinc finger nuclease, transcription activating factor-like effector nuclease or CRISPR nuclease or a variant thereof, may further comprise a sub-cellular localization signal (such as a nuclear localization signal), a peptide linker, a detectable label and other elements. For example, the base editor in the CRISPR base editing system generally comprises one or more nuclear localization signal (NLS) for entering the cell nucleus to realize the editing of the chromosomal DNA.

The expression construct of the invention can be introduced into the plant cell by a variety of methods known in the art, the methods comprise but are not limited to gene gun method, PEG-mediated protoplast transformation and *Agrobacterium*-mediated transformation.

In some embodiments of the present invention, the plant cell of the invention is a cell suitable for regenerating an intact plant cell by tissue culture. Examples of suitable plant cells include, but are not limited to, protoplast cells, callus cells, immature embryonic cells, and explants cells.

Methods for regenerating a transformed intact plant by culturing the transformed protoplast, callus, immature embryo or explant is known in the art. In the regeneration process, the transformant can also be selected based on the selectable marker carried on the introduced expression construct. In some embodiments, the regeneration is carried out in the absence of a selection pressure. In some embodiments, the transformant can be selected under a moderately stringent selection condition. The moderately stringent condition refers to a condition that does not completely inhibit the growth of the non-transformants. For example, moderately stringent selection does not inhibit the growth of the transformants but partially inhibit the growth of the non-transformants. For example, under moderately stringent selection, the non-transformants can grow but slower or weaker than the transformant. The moderately stringent selection can be determined by those skilled in the art for specific plants and specific selectable markers.

In some embodiments of the invention, the expression construct of the invention is transiently transformed into the plant cell. Transient transformation refers to introducing the construct into the cell, allowing it to exert the function but not integrated into the cell genome. This is particularly useful for gene editing, because transgene-free modified plants can be produced. Another surprising discovery of the present invention is that even transient expression of the combination of WUS, BBM and SERK, or a combination of

16

GRF and GIF, can improve the efficiency of regeneration, transformation and/or gene editing of the plant.

Plants suitable for transformation or gene editing using the methods of the present invention may be monocotyledonous plants or dicotyledonous plants. For example, examples of the plants include, but are not limited to, wheat, strawberry, rice, corn, soybean, sunflower, sorghum, canola, alfalfa, cotton, barley, millet, sugar cane, tomato, tobacco, cassava and potato.

The method of the invention is particularly suitable for genetic transformation or gene editing in a plant variety or genotype that previously is difficult to be transformed. In some specific embodiments, the plant is wheat, for example, the wheat is Jimai 20, Jimai 22, Beijing 411, Shannong 20, Shannon 116, Xiaoyan 54, Zhoumai 27, Zhoumai 28 and Zhongmai 175. In some specific embodiments, the plant is soybean, for example, the soybean is Williams 82, zhonghuang 13 and Hefeng 25. In some specific embodiments, the plant is strawberry, such as strawberry Benihoppe.

In order to obtain effective expression in the plant, in some embodiments of the invention, the coding nucleic acid sequence or the nucleic acid sequence of interest is codon optimized against the plant species of which the genome is to be modified.

Codon optimization refers to a process of modifying a nucleic acid sequence for enhanced expression in the host cells of interest by replacing at least one codon (e.g. about or more than about 1, 2, 3, 4, 5, 10, 15, 20, 25, 50, or more codons) of the native sequence with codons that are more frequently or most frequently used in the genes of that host cell while maintaining the native amino acid sequence. Various species exhibit particular bias for certain codons of a particular amino acid. Codon bias (differences in codon usage between organisms) often correlates with the efficiency of translation of messenger RNA (mRNA), which is in turn believed to be dependent on, among other things, the properties of the codons being translated and the availability of particular transfer RNA (tRNA) molecules. The predominance of selected tRNAs in a cell is generally a reflection of the codons used most frequently in peptide synthesis. Accordingly, genes can be tailored for optimal gene expression in a given organism based on codon optimization. Codon usage tables are readily available, for example, at the "Codon Usage Database" available at <<www>>.<<kazusa>>.<<or>><<jp>><</codon>> and these tables can be adapted in a number of ways. See Nakamura, Y., et al. "Codon usage tabulated from the international DNA sequence databases: status for the year 2000" Nucl. Acids Res. 28:292 (2000).

In one aspect, the invention provides a plant obtained by the method of the invention and progeny thereof.

EXAMPLES

Further understanding of the present invention can be obtained by reference to some specific examples given herein, which examples are only used to illustrate the invention, and are not intended to limit the scope of the invention. Obviously, various modifications and changes may be made to the present invention without departing from the spirit of the present invention, and therefore these changes and variations are also within the scope of the present application.

Example 1. Development Regulatory Factor WUS, SERK, LEC1, KN1, BBM Improving Wheat Genetic Transformation and Gene Editing Efficiency

Recently, using the totipotency of plant cell, ectopic expression of specific DR combination (such as WUS, STM,

MP, BBM and so on) in somatic cell has the potential for inducing meristematic tissues. Co-expression of DRs and gene editing system can greatly shorten the plant tissue culture process to obtain the genome edited plant individuals. In addition, researches showed that by expressing some genes associated with the cell division of meristematic tissues can also achieve plant genome editing without tissue culture. The development of new plant cell (especially apical meristematic tissue cells) division-promoting and development-related genes, is of important significance especially for plant such as wheat which is difficult to transform.

This example is to select candidate genes which may promote plant cell division and plant regeneration. These candidate genes include WUS, BBM, SERK, KN1 and LEC.

1.1. Vector Construction

Corn WUS, SERK, LEC, BBM and KN1 were selected to test. The following plant expression vectors were constructed:

1. CRISPR/Cas9 knockout vector: UBI-Cas9
2. The single base editing vector: UBI-A3A
3. sgRNA expression vector: TaU6-sgRNA
4. ZmWUS/BBM combination: nos-ZmWUS-ubi-ZmBBM
5. ZmWUS/BBM/SERK combination: nos-ZmWUS-ubi-ZmBBM-ZmSERK
6. ZmWUS/BBM/LEC1 combination: nos-ZmWUS-ubi-ZmBBM-ZmLEC1
7. ZmWUS/KN1/LEC combination: nos-ZmWUS-ubi-ZmKN1-ZmLEC1

The structures of the plant expression vectors are shown in FIG. 1. The vectors 4-7 were also referred to as promoting (booster) vectors.

1.2 Transforming Wheat Immature Embryo with Gene Gun and Analyzing Transformation and Editing Efficiency

UBI-Cas9 or UBI-A3A respectively combined with TaU6-sgRNA and four promoting (booster) expression vectors were co-transformed. Wheat ALS gene was selected as the editing site. The established wheat gene gun transformation technology was used to transform. Plants were obtained after tissue culture and regeneration, and mutants were detected by PCR/RE method. The results of UBI-A3A are shown in Table I below.

TABLE 1

booster plasmid	No. of bombardment	Selective agent	No. of mutants
nos-ZmWUS-ubi-ZmBBM	5	F7.5	10
Ubi-A3A	5	F7.5	9
UBI-A3A	5	F7.5	4
nos-ZmWUS-ubi-ZmBBM-ZmSERK	5	F7.5	17
nos-ZmWUS-ubi-ZmBBM-ZmLEC1	5	F7.5	4
nos-ZmWUS-ubi-ZmKN1-ZmLEC1	5	F7.5	7

The above results showed that: compared with the control UBI-A3A without booster, the efficiency of nos-ZmWUS-ubi-ZmBBM-ZmSERK has significant improving effect for base editing system UBI-A3A.

Example 2. Plant Growth Control Factor GRF/Gif Improve Wheat Genetic Transformation and Gene Editing Efficiency

GRF transcription factors are specific transcription factors in plants, mainly control plant cell size, chloroplast proliferation,

gynogenesis, osmotic stress and other plant growth and development processes. GRF and GIF can form transcription co-activation factor complex. GIFs are homologous to human transcription co-activation factor synovial sarcoma transport protein (synovial translocation protein, SYT). In plants, the GRF4 and gif1 plays a role in cell proliferation and maintains the proliferation ability of the meristematic cell during the flower organ development. The inventors selected wheat GRF4 and wheat gif to detect whether the genetic transformation and gene editing of the wheat can be improved.

2.1. Vector Construction

Wheat GRF4 and gif1 were fused by linker AAAA; the ZmUBI promoter was used for driving expression. In addition, to reduce the number of vectors, the GRF4-AAAA-gif1 fusion protein and Cas9 or A3A-nCas9 were constructed on the same plant expression vector, and the P2A was used for separate expression. The structure of the constructed vector is shown in FIG. 2.

2.2. Transforming Wheat Immature Embryo with Gene Gun and Analyzing Transformation and Editing Efficiency

The established wheat gene gun transformation technology was used to co-transform the combination of the booster vector and TaU6-sgRNA constructed in 2.1; combination of UBI-Cas9 or UBI-A3A and TaU6-sgRNA was co-transformed as a control. Wheat ALS gene was selected as the editing site. Plants were obtained after tissue culture and regeneration, and mutants were detected by PCR/RE method. The results are shown in the following table.

TABLE 2

Plasmid bombarded	No. of bombardment	No. of seedling	No. of mutants
Ubi-GRF4-gih1-Cas9	5	158	12
Ubi-Cas9	5	91	0
UBI-GRF4-gif1-A3A	5	F7.5	21
Ubi-A3A	5	F7.5	6

The above results show that: as compared with the control UBI-Cas9 and UBI-A3A. UBI-GRF4-gif-Cas9 and UBI-GRF4-gif1-A3A with wheat GRF4/gif1 added showed significantly increased number of mutants, which was increased by a factor of about 3-12. The plant growth control factors can be used as important regulatory proteins to promote plant genetic transformation and improve efficiency of genome editing.

Example 3. Optimizing GRF4/Gif1 to Improve the Wheat Gene Editing Efficiency

In order to further improve the efficiency of wheat genome editing, the linker of the GRF4 and gif1 fusion protein was optimized. The linkerA (AAAA) was replaced by linkerS (SGGS). In addition, the miR396/GRF control pattern is very conservative in plants, miR396 has down-regulation effect to GRF4. The miR396 binding site of GRF4 was mutated, which may increase expression of GRF4, so as to improve the efficiency of wheat genome editing. The booster vectors shown in FIG. 3 were constructed.

The vectors respectively combined with the wheat ALS gene targeting A3A base editing system in the above Examples were transformed through gene gun into wheat immature embryo. The results shown in the following table.

19

TABLE 3

Booster plasmid	Transformation time	No. of bombardment	No. of seedling	No. of mutants
linkerA-GRF4-gifl	2019 Sep. 24	3	275	71
linkerA-mGRF4-gifl	2019 Sep. 24	3	403	125
linkerS-GRF4-gifl	2019 Sep. 24	2	154	12
linkerS-mGRF4-gifl	2019 Sep. 24	2	182	28
P2A-GRF4-gifl	2019 Sep. 26	4	191	4
P2A-mGRF4-gifl	2019 Sep. 26	3	165	1

As can be seen from the above results, editing efficiency of linkerA was higher than LinkerS. Most importantly, the mutation of the binding site of the miR396 of GRF4 increased the regeneration efficiency of the wheat with large amplitude, and also increased the mutation efficiency in large amplitude.

Example 4. GRF4/Gifl Improves the Editing Efficiency in Wheat of Different Genotype

In order to investigate whether GRF4/gifl can improve the editing efficiency of other wheat varieties, namely breaking the limitation of wheat genotype on genetic transformation, several varieties with low transformation efficiency by traditional methods were selected for testing. The results are as follows:

TABLE 4

No.	Booster plasmid	Transformation time	No. of bombardment	Selective agent	No. of mutants
Xiaoyan 54					
Pm20.25	linkerA-GRF4-gifl	2019 May 16	5	F7.5	12
Pm20.27	/	2019 May 16 Jimai 22	5	F7.5	0
Pm20.26	linkerA-GRF4-gifl	2019 May 16	5	F7.5	1
Pm20.28	/	2019 May 16	5	F7.5	0

The results show that the wheat growth transcription factor GFR4/gifl can significantly realize editing of a plurality of varieties, breaking the restriction of genotypes.

Example 5. GRF4/Gifl Improves Wheat Transformation Efficiency and Gene Editing Efficiency Mediated by *agrobacterium*

In order to further investigate whether GRF4/gifl can improve the efficiency of wheat transformation efficiency and gene editing mediated by *agrobacterium*, the GRF4/

20

gifl was constructed on the *agrobacterium* gene editing vector, transfecting wheat immature embryo by *agrobacterium*. The results are shown in the following table.

TABLE 5

transformed <i>agrobacterium</i>	Recipient material	No. of young embryo	mutant
ubi-GRF4-A3A-ALS174-Hyg	Kenong 199	519	12
A3A-ALS174-Hyg	Kenong 199	199	0

It can be seen that GRF/gif can significantly improve the editing efficiency mediated by *agrobacterium* transformation, increasing the gene editing efficiency from zero to 2.3%.

Example 6. GRF/Gif Improves Soybean Transformation Efficiency and Gene Editing Efficiency Mediated by *Agrobacterium*

6.1. Soybean GRF/Gif Vector Construction

The result of the above Examples shows that the plant growth factor GRF/gif combination can improve the regeneration efficiency of tissue culture of monocotyledon wheat and efficiency of gene editing. In order to investigate whether GRF can increase tissue culture plant regeneration efficiency of dicotyledonous plants, endogenous GRF genes of dicotyledon soybean were selected to test whether they can improve the plant regeneration efficiency of soybean. GFR5, GRF6, GRF11, GRF18 of soybean were selected for test, and connected with gifl of soybean by AAAA linker. The fusion proteins and Cas9 were separated and linked by P2A; the expression was driven by 35s promoter; the sgRNA was initiated by the U6-26 promoter; and the sgRNA was constructed on the same carrier. The constructed expression vector is shown in FIG. 4.

6.2 Soybean GRF/Gif Improve the Regeneration Efficiency of Soybean Callus

The above constructed vectors were transformed into soybean by *agrobacterium*; the result of the soybean regeneration efficiency is shown in the following table.

TABLE 6

project starting time	RECO	SBA	EBA	ETC	20200303 elongated seedlings/explants	elongation rate
35S-GmGFR5-AAAA-Gmgifl	187	147	178	144	35/124	0.282258
35S-GmGFR6-AAAA-Gmgifl	195	100	64	56	3/56	0.053571
35S-GmGFR18-AAAA-Gmgifl	188	161	156	124	9/124	0.072581
35S-GmGFR11-AAAA-Gmgifl	194	168	160	157	14/124	0.112903
pKSE411-shRNA	178	107	72	64	4/60	0.066667

RECO: representing soybean infection, recovery culture stage; SBA: represents the soybean entering selection stage; EBA: represents the soybean enters the elongation stage. ETC: represents soybean elongation culture stage, not selection added.

The result shows that the soybean endogenous GFR5 in combination with gifl can significantly improve the regeneration efficiency of soybean callus. In addition, the gene editing efficiency of the endogenous gene of the regenera-

tion plant is detected, showing that the gene editing efficiency was also significantly improved.

Example 7. The GmGRF5-GmGIF1 Boosts Soybean Regeneration

Of the 22 GRF genes predicted in the soybean genome, GmGRF5, GmGRF6, GmGRF11, and GmGRF18, which are specifically (or preferentially) expressed in flower and shoot apical meristems (Chen et al., 2019), were chosen to form fusion proteins with soybean GmGIF1, whose homolog gene in *Arabidopsis* and rice strongly expresses in flowers and shoot apical meristems to control plant growth (Kim, 2019). Each GmGRF-GmGIF1 fusion protein was co-expressed with a CRISPR/Cas9 expression cassette, generating four constructs (pGmGRF5-GmGIF1, pGmGRF6-GmGIF1, pGmGRF11-GmGIF1, and pGmGRF18-GmGIF1) (FIG. 5A), targeting two copies of the soybean fatty acid desaturase 2 (GmFAD2) gene (FIG. 5B), whose product catalyzes the conversion of oleic acid to linoleic acid and lowers the quality of soybean oil quality (Haun et al., 2014). The transformation efficiencies and frequencies of editing of GmFAD2 by the GmGRF-GmGIF1 complexes were evaluated.

As shown in FIG. 5C, each of the four pGmGRF-GmGIF1 constructs was transformed into soybean cultivar Williams 82, one of the most popular transformable genotypes, by *Agrobacterium*-mediated transformation (Jia et al., 2015). Construct pBSE401 (Xing et al., 2014), which contains the GmFAD2-targeting CRISPR/Cas9 expression cassette but no GmGRF-GmGIF1 complex, was used as control (FIG. 5A). The number of explants recovered after introduction of each of the four GmGRF-GmGIF1-containing constructs was similar to that obtained with pBSE401 (Tables 7 and 8). However, after two weeks of bud induction and two weeks of shoot proliferation with glufosinate selection, the explants transformed with pGmGRF5-GmGIF1 produced large numbers of buds: 93.8% of the explants in each transformation event produced multiple buds, compared with 58.4% in the control and similar proportions in the explants infected with other GmGRF-GIF1 constructs (FIG. 5D, Tables 7 and 8). On elongation medium with moderate stringency of glufosinate selection, the average number of elongated shoots (≥ 2 cm) increased 2.8-fold with pGmGRF5-GmGIF1 (432.3) relative to the control (156.0) and more than 2.5-fold relative to the other constructs (FIGS. 5E and 1G, Tables 7 and 8). Similarly, average numbers of elongated large shoots (≥ 9 cm) were 2.5-fold greater with pGmGRF5-GmGIF1 (85.3) than with pBSE401 (33.7) (FIGS. 5E and 5H, Tables 7 and 8). The inventors also found that the explants transformed with pGmGRF5-GmGIF1 produced significantly more putative glufosinate resistant shoots per explant (3.0) than the control (1.1) (FIGS. 5F, 5G and 5H, Tables 7 and 8). Also, the time from the initial transformation to when the first elongated shoot reached 9 cm was 58 days for pGmGRF5-GmGIF1, compared with 70 for the control (Tables 7 and 8). There were no significant differences in shoot induction, shoot elongation, and time from the initial transformation to first 9 cm shoot between explants infected with pBSE401 and the three other GmGRF-GmGIF1 complexes (Tables 7 and 8). Collectively, these data show that the GmGRF5-GmGIF1 complex stimulates regeneration of soybean explants.

Example 8. The GmGRF5-GmGIF1 Complex Enhances Genome Editing in Soybean

Efficient regeneration is a prerequisite for successful plant genetic transformation and genome editing. Consistent with

the above regeneration frequencies, the transformation efficiency (TE) of explants when pGmGRF5-GmGIF1 was used was 21.8%, significantly higher than the 8.5% with pBSE401, while the other complexes all yielded frequencies of no more than 10.2% (FIG. 5I, Tables 7 and 8). The inventors used PCR-RE assays and Sanger sequencing to detect mutations of GmFAD2 in the regenerated lines. Again, in agreement with the TEs, explants transformed with pGmGRF5-GmGIF1 contained more edits and significantly higher mutation frequencies (MF) than those transformed with pBSE401 (FIG. 5J, Tables 7 and 8). Also, a large number of the transformants had mutations in both FAD2-1A and FAD2-1B, indicating that both genes were efficiently mutated (FIG. 10, Table 9). Intriguingly, among the 94 mutants from transformation with pGmGRF5-GmGIF1, 15 (16.0%) were transgene-free. (FIG. 10, Tables 7 and 8).

To see whether the constitutive expression of GmGRF-GmGIF1 complexes caused phenotypic abnormalities, the inventors grew GmFAD2-edited Williams 82 plants transformed with pGmGRF5-GmGIF1 and pBSE401 in a greenhouse. Throughout the whole growth period, Williams 82 plants transformed with pGmGRF5-GmGIF1 were fertile and no morphologic differences were observed between the two sets of lines (FIG. 5K), suggesting that that GmGRF5-GmGIF1 greatly stimulates genome-editing of soybean without producing undesirable side-effects.

Example 9. The GmGRF5-GmGIF1 Complex Enables Genome Editing of Marginally Transformable Soybean Cultivars

Because the pGmGRF5-GmGIF1 construct yielded the highest rates of regeneration, genetics transformation, and mutation among the four GmGRF-GmGIF1 complexes, it was used in an attempt to increase the transformation and genome-editing efficiencies in two marginally transformable but commercially important soybean cultivars, Zhonghuang 13 and Hefeng 25. The GmGRF5-GmGIF1 complex was transformed into Zhonghuang 13 and Hefeng 25 by *Agrobacterium*-mediated transformation, in combination with a GmFAD2-targeting CRISPR/Cas9 expression cassette, with pBSE401 as control. In Zhonghuang 13, GmGRF5-GmGIF1 gave rise to substantially higher frequencies of regeneration (84.0%), elongated putative glufosinate-resistant shoots (312.6, ≥ 2 cm in length), and elongated putative glufosinate-resistant shoots per explant (2.6), as well as higher TE (18.2%) and MF (16.0%), than did pBSE401 (65.6%, 233.0, 1.7, 5.3% and 2.4%, respectively) (FIGS. 6A, 6B, 6E and 6G, Tables 7 and 8). In Hefeng 25, about 71.1% of explants transformed with pGmGRF5-GmGIF1 produced at least one elongated shoot ≥ 2 cm, whereas none of the explants transformed with pBSE401 produced shoots (FIG. 6A, 6C-6E, Tables 7 and 8). The transformation efficiency of the explants transformed with pGmGRF5-GmGIF1 was 2.5%, and five transgenic plants had mutations in the GmFAD2 target sites (FIGS. 6F and 6G, Tables 7, and 4). One transgene-free editing event was also identified (Tables 7 and 8). Evidently the GmGRF5-GmGIF1 complex greatly stimulates regeneration and hence genetic transformation and genome editing in marginally transformable soybean cultivars. Also, the GmGRF5-GmGIF1 complex is able to generate transgene-free mutants due to transient expression of the CRISPR DNA and the fact that expression of the

booster genes stimulated much more extensive cell proliferation, while the selection pressure was relatively low.

Example 10. The Soybean GmGRF5-GmGIF1 Complex Promotes Regeneration of Strawberry and Genome Editing

The regeneration of another dicot, strawberry (*Fragaria ananassa*), is genotype-dependent. Transformation of one of the most widely cultivated octoploid strawberry cultivars, Benihoppe, is extremely difficult, and no genome editing events have yet been obtained in this cultivar (Folta and Dhingra, 2006).

The inventors examined the regeneration and genome editing of Benihoppe using the soybean GmGRF5-GmGIF1 complex because strawberry GRF proteins are close to soybean GRFs (FIG. 11A). The fusion protein was co-expressed with Cas9 in a single expression cassette, in vector pHUE411-GFP-GmGRF5-GmGIF1 (FIG. 7A). The pectate lyase gene, FaPL, which contributes to loss of strawberry fruit firmness (Jimenez-Bermudez et al., 2002), was targeted by Cas9 (FIG. 11B) and green fluorescent protein (GFP) was used as a visual reporter that allowed continuous monitoring of transgenic events. The control vector (pHUE411-GFP) contained Cas9, the sgRNA targeting FaPL, and the GFP gene. A total of 541 and 540 strawberry leaf explants were infected by the *Agrobacterium* contained pHUE411-GFP-GmGRF5-GmGIF1 and the control vector, respectively (FIG. 7B). Calli from leaf explants induced on callus induction medium were screened for GFP expression three weeks after infection. 19.6% (106/541) of the explants transformed with GmGRF5-GmGIF1 produced at least one GFP positive callus, compared with 6.5% (35/540) in the control (FIG. 7B).

Moreover, the leaf explants transformed with GmGRF5-GmGIF1 tended to form multiple and large calli with strong GFP fluorescence, whereas, only a few small GFP-positive spots were observed on callus masses in the control (FIG. 7C). During growth on shoot induction medium the calli on explants transformed with GmGRF5-GmGIF1 grow more vigorously than those on the control (FIG. 7D), and shoot initiation occurred on 8.1% of the former calli from explants infected with GmGRF5-GmGIF1 five weeks after infection compared with only 3.0% of the latter (FIG. 7D). The transformation efficiency of the explants transformed with GmGRF5-GmGIF1 was 0.9%, and mutations in the FaPL target sites were detected in 0.6% of the explants (FIG. 7B, FIG. 11C), whereas neither transgenic plants nor mutant plants were detected in the control (FIG. 7B). These data demonstrate that the GmGRF5-GmGIF1 complex could promote the regeneration, genetic transformation and genome editing of a transformation-recalcitrant strawberry cultivar, showing that GmGRF5-GmGIF1 stimulates genetic transformation and genome editing in other dicot plants.

Example 11. Comparison of the Effects of Wild Type TaGRF4-TaGIF1 and Mutated TaGRF4-TaGIF1 on Regeneration and Genome Editing of Common Wheat

To broaden the application of GRF-GIF1 complexes, the inventors examined their effect in the monocot, common wheat, in which transformation is limited to a narrow range of genotypes (He et al., 2015; Jones et al., 2005). Phylogenetic tree analysis suggested that the soybean GRF family, including GmGRF5, is distantly related to the monocot GRF family of rice (Chen et al., 2019). Previous work has also

shown that overexpression of OsGRF4 and its coactivator OsGIF1 stimulates the cell proliferation and seed size in rice (Sun et al., 2016; Hu et al., 2015). Therefore, the inventors tested the effects of the rice homologs, TaGRF4, and TaGIF1, in common wheat (FIG. 12) while adding two modifications.

First, TaGRF4 was fused with TaGIF1 to form a TaGRF4-TaGIF1 complex, and the miR396 target site in TaGRF4 was inactivated by five point mutations (mTaGRF4-TaGIF1) to increase transcription of TaGRF4 (FIGS. 8A and 8B). Second, a transient expression strategy (Liang et al., 2017; Zhang et al., 2016; Zhang et al., 2018), which avoids any selection pressure throughout the whole tissue culture procedure, was used to generate transgene-free mutants in the TO generation (FIG. 8C). The wild type complex TaGRF4-TaGIF1, and mutant complex, mTaGRF4-TaGIF1, were separately transformed into the efficiently transformable common wheat cultivars Bobwhite and Kenong 199 (FIGS. 8B and 8C), together with the cytosine base editor pUBI-A3A vector containing an sgRNA for the common wheat acetolactate synthase gene (TaALS) (FIG. 13), in which C-to-T substitutions at Pro197 confers resistance to nicosulfuron (Zhang et al., 2019; Zong et al., 2018). The combination of pUBI-A3A and pUBI-GFP served as a control. After six weeks on non-selective media, numbers of regenerated plants and TaALS mutations were analyzed.

The inventors found that immature embryos of both Bobwhite and Kenong 199 transformed with TaGRF4-TaGIF1 produced more regenerated plants (508.0% and 654.5%, respectively) than the control groups (81.0% and 136.9%, respectively) (FIG. 8D, Table 11). Moreover, mTaGRF4-TaGIF1 gave even higher regeneration ratios of 630.1% and 1165.4%, in Bobwhite and Kenong 199, respectively, 1.2- and 1.7-fold higher than the regeneration frequency with the wild type TaGRF4-TaGIF1 complex, and 7.8- and 8.5-fold higher than with the control construct (FIG. 8D, Tables 11 and 12). The inventors also examined genome editing at the TaALS target site using PCR-RE assays and Sanger sequencing. Consistent with the regeneration ratios, the Bobwhite and Kenong 199 plants transformed with the TaGRF4-TaGIF1 complex had higher mutation frequencies (32.7% and 103.4%, respectively) than plants transformed with the control construct (9.9% and 17.7%, respectively) (FIGS. 8E and 8D, Tables 11 and 12). Among the plants generated by the mTaGRF4-TaGIF1 complex the inventors identified 216 mutants in 343 embryos (63.3%) and 577 mutants in 328 embryos (176.1%) in Bobwhite and Kenong 199, respectively, 1.7-1.9-fold higher than in the plants generated by the TaGRF4-TaGIF1 complex, and 6.4-9.9-fold higher than in those in the control groups (FIG. 8E, Tables 11 and 12).

Since the plasmids were delivered using a transient expression approach (FIG. 8C), there seemed a high probability that the TaGRF4-TaGIF1 and base editor DNA constructs had not been integrated into the genome of the mutant plants. To test for the presence of plasmid DNAs in the regenerated TO mutants the inventors used a total of six primer sets (three for pUBI-A3A, three for pTaGRF4-TaGIF1 or pmTaGRF-TaGIF1) to amplify different regions of the TaGRF4-TaGIF1 and the base editor constructs, together covering almost the entire constructs (FIG. 8F). Based on the PCR analysis, the two vectors (pUBI-A3A and pTaGRF4-TaGIF1/pmTaGRF4-TaGIF1) were absent from 47.1%-55.4% of the mutants of Bobwhite and Kenong 199 treated with TaGRF4-TaGIF1, mTaGRF4-TaGIF1 and the control construct (FIG. 8G, Tables 11 and 12). The total numbers of transgene-free mutants in the mTaGRF4-Ta-

GIF1-treated groups were 6.0-11.6-fold higher than in the control groups (Tables 11 and 12). Moreover, when the inventors grew the transgene-free mutants of Kenong 199 derived from mTaGRF4-TaGIF1 in a greenhouse the inventors found that they were fertile and displayed no obvious phenotypic differences from WT plants throughout all of development (FIG. 8H).

In summary, the TaGRF4-TaGIF1 complex increased both regeneration frequency and genome-editing efficiency in common wheat and disruption of the miR396 target site further enhanced its efficiency. Furthermore, transient expression of mTaGRF4-TaGIF1 had no ill-effect on phenotype.

Example 12. The Mutated TaGRF4-TaGIF1
Complex Extends Genome Editing in Broad
Common Wheat Cultivars

The inventors tested whether the mTaGRF4-TaGIF1 complex improved regeneration rates and genome editing efficiencies in various common wheat cultivars that are widely grown in China, pUBI-A3A vector, containing an sgRNA for TaALS, was delivered with mTaGRF4-TaGIF1 into immature embryos of nine major Chinese common wheat cultivars Jimai 20, Jimai 22, Jing 411, Shannong 20, Shannong 116, Xiaoyan 54, Zhoumai 27, Zhoumai 28 and Zhongmai 175 according to the procedure in FIG. 8C. The combination of pUBI-A3A and pUBI-GFP served as a control.

After six weeks on non-selective medium, the inventors found that mTaGRF4-TaGIF1 stimulated regeneration of these common wheat cultivars, and regeneration frequencies, which ranged from 9.9%-440.8%, were significantly higher than in the corresponding controls (0%-187.3%) (FIGS. 9A and 9B, Tables 13 and 14). In particular, mTaGRF4-TaGIF1 provoked substantial the regeneration and genome editing of Xiaoyan 54, Zhoumai 28, Jimai 20, Jimai 22 and Shannong 20, which is normally very difficult (FIGS. 9A and 9B, Table 13). Mutations in the TaALS sites were detected by PCR-RE assays and Sanger sequencing in all of these nine common wheat cultivars transformed with mTaGRF4-TaGIF1. Mutation frequencies ranged from 1.2-8.1%, compared with 0-3.9% in the control groups, five of which contained no mutants (FIG. 5C, Table 8). Moreover, homozygous mutant frequencies (HMF) in the nine cultivars transformed with mTaGRF4-TaGIF1 ranged from 10.0-55.6%, which was much higher than in the controls (0-15.4%) (Tables 13 and 15). The inventors also identified 22.2-66.7% transgene-free mutants among the nine cultivars transformed with the mTaGRF4-TaGIF1 complex (Table 13). In contrast, transgene-free mutants were only found in Shannong 116, Zhoumai 27 and Zhongmai 175 transformed with the control plasmid (Table 13). Thus, the mTaGRF4-TaGIF1 complex stimulates regeneration and genome editing in all nine common wheat cultivars.

Soybean is one of the most important sources of edible oils and proteins, but its transformation rates remain very low despite the many available genetic transformation methods (Christou, 1992; Trick and Finer, 1998; Yang et al., 2016). Thus, the development of an improved transformation system for soybean is urgently needed. Plant GRF and GIF genes are highly expressed in meristematic tissues, including leaf and floral organ primordia and shoot apical meristems (Kim, 2019; Omidbakhshfar et al., 2015; Zhang et al., 2018), and might potentially be developed as boosters to stimulate plant regeneration so enhance genetic transformation and genome editing rates. The inventors screened and tested four GmGRF-GmGIF1 complexes in Williams

82, and found a transformation rate of 21.8% using the GmGRF5-GmGIF1 complex (FIG. 5I, Table 7), which is much higher than the transformation rate obtained using traditional transformation methods (Finer and McMullen, 1991; Hinchey et al., 1988). It should be noted that GmGRF5 is predicated as GmGRF7 in GenBank (XM_003526701.4), since it is most similar to *Arabidopsis* AtGRF7.

The genotype limitation depends on the susceptibility to *Agrobacterium* infection of the donor genotype and the regeneration ability of the explant (Jia et al., 2015). Fortunately, the GmGRF5-GmGIF1 complex improved the transformation efficiencies of Zhonghuang 13 (18.2%) and Hefeng 25 (2.5%), which are the most widely grown cultivars in China but are difficult to engineer genetically by current methods. This indicates that the GmGRF5-GmGIF1 complex can overcome the bottleneck of genotype-dependence in soybean transformation. GmGRF5-GmGIF1 enhanced recovery of transgene-free FAD2-edited events due to large number of regenerated shoots and use of a moderate stringency selection which promotes regenerated shoots as well as transgenic plant to elongate. This simulation of shoot regeneration by the GmGRF5-GmGIF1 complex and the resulting improved transformation and genome editing efficiencies in soybean and strawberry were further confirmed by the ability of soybean GROWTH-REGULATING FACTOR 5 (GmGRF5) and its cofactor soybean GIF1 complex to enhance regeneration, transformation and genome editing in diverse dicotyledonous plants.

Considering their extremely wide distributions in all land plants, and their preferential expression in meristematic tissue (Kim, 2019; Omidbakhshfar et al., 2015; Shimano et al., 2018), native GRF-GIF1 complexes could be effective in promoting regeneration in most plants. The regeneration of monocotyledonous common wheat remains genotype-dependent (He et al., 2015; Jones et al., 2005), but the inventors demonstrated that transient expression of wheat mTaGRF4-TaGIF1 dramatically improved regeneration and genome editing, and also greatly increased numbers of transgene-free mutants in nine recalcitrant Chinese commercial elite cultivars, most of which had never been regenerated and gene-edited before. The improved TaGRF4-TaGIF1 complex (mTaGRF4-TaGIF1) thus overcomes the genotype limitations to regeneration in common wheat and boosts its genome editing.

An ideal booster should not influence the morphology of regenerated plants. In the present work, no morphological changes were observed in the transgenic soybean plants, even when that the GmGRF5-GmGIF1 complex was constitutively expressed, probably due to the moderate level of its expression, which is regulated at the post-transcriptional level (Kim and Tsukaya, 2015; Liu et al., 2009; Rodriguez et al., 2010). In common wheat, the inventors found that the mTaGRF4-TaGIF1 complex performed better in improving the regeneration ability and genome editing frequency than the original TaGRF4-TaGIF1 complex. Although mTaGRF4-TaGIF1 was not repressed by miR396, the transient expression system prevented prolonged presence of this complex in the plant cells, this not only minimized morphological side effects but also generated a large number of transgene-free mutants.

In summary, the GmGRF5-GmGIF1 complex enhances the regeneration of dicotyledonous soybean and strawberry and hence stimulates genetic transformation and genome editing. Similarly, transient expression of the improved TaGRF4-TaGIF1 complex containing the inactivated

Given that members of the GRF gene family exists in many plant species, the GRF-GIF1 complexes described here holds great promise for improving genome-editing efficiency in a wide range of crop plants.

Effect of GmGRF-GmGIF1 fusions on soybean transformation and genome editing efficiency.

RF (regeneration frequency) = no. of explants with multiple buds/total explant number $\times$ 100%.
Growth rates were measured from the date of the initial transformation to the date when the first elongated shoot reached 9 cm in length.
TE (transformation efficiency) = no. of transgenic plants/total explants $\times$ 100%.
Elongated shoots (≥ 9 cm in length) were collected for transgenic assay only once on day 75 after *Agrobacterium*-mediated transformation.
MF (mutation frequency) = no. of mutants/total transgenic plants $\times$ 100%.
TFF (transgene-free frequency) = no. of transgene-free mutants/total mutants $\times$ 100%.
The numbers and means for each treatment were calculated from data collected from three replicates of the experiment.
Data are mean $\pm$ SD (n = 3).
^{a,b,c}indicate significant differences compared with the CK group
(two-sided Student's t-test. ^aindicates P < 0.05; ^bindicates P < 0.01; ^cindicates P < 0.001).

Effect of the GmGRF-GmGIF1 fusions on soybean transformation and genome editing efficiency (Williams 82, Hefeng 25 and Zhonghuang 13, raw data).

Targeted gene	Soybean cultivar	Repeat	Treatment	No. explants	No. explants recovered	No. explants with multiple buds/RF (%)	No. elongated putative glufosinate-resistant shoots (≥2 cm in length)	No. elongated putative glufosinate-resistant shoots (≥9 cm in length)
GmFAD2 (Knockout)	Williams 82	I	pBSE401 (CK)	150	138	97 (64.7)	155	36
			GmGRF5-GmGIF1	150	147	147 (98.0)	432	85
			GmGRF6-GmGIF1	150	145	92 (61.3)	138	36
			GmGRF11-GmGIF1	150	141	85 (56.7)	145	35
			GmGRF18-GmGIF1	150	136	106 (70.7)	180	44
		II	pBSE401 (CK)	150	140	86 (57.3)	171	37
			GmGRF5-GmGIF1	150	142	135 (90.0)	451	93
			GmGRF6-GmGIF1	150	144	103 (68.7)	123	42
			GmGRF11-GmGIF1	150	143	90 (60.0)	159	44
			GmGRF18-GmGIF1	150	141	89 (59.3)	170	47
		III	pBSE401 (CK)	150	134	80 (53.3)	142	28
			GmGRF5-GmGIF1	150	143	140 (93.3)	414	78
			GmGRF6-GmGIF1	150	135	95 (63.3)	152	29
			GmGRF11-GmGIF1	150	138	94 (62.7)	159	29
			GmGRF18-GmGIF1	150	142	109 (72.7)	170	33

TABLE 8-continued

Effect of the GmGRF-GmGIF1 fusions on soybean transformation and genome editing efficiency (Williams 82, Hefeng 25 and Zhonghuang 13, raw data).							
Zhonghuang 13	I	pBSE401 (CK)	150	145	90 (60.0)	216	35
		GmGRF5-GmGIF1	150	147	126 (84.0)	300	68
	II	pBSE401 (CK)	150	141	105 (70.0)	245	29
		GmGRF5-GmGIF1	150	149	122 (81.3)	342	63
	III	pBSE401 (CK)	150	149	100 (66.7)	238	33
		GmGRF5-GmGIF1	150	144	130 (86.7)	296	74
Hefeng 25	I	pBSE401 (CK)	150	121	83 (55.3)	0	0
		GmGRF5-GmGIF1	150	132	121 (80.7)	45	9
	II	pBSE401 (CK)	150	128	84 (56.0)	0	0
		GmGRF5-GmGIF1	150	123	95 (63.3)	52	15
	III	pBSE401 (CK)		123	76 (50.7)	0	0
		GmGRF5-GmGIF1		117	104 (69.3)	67	12

Targeted gene	Growth rate (days)	Average elongated shoots per explant	No. transgenic plants/TE (%)	No. mutants/MF (%)	No. transgene-free mutant/TEF (%)
GmFAD2 (Knockout)	70	1.1	12 (8.0)	10 (6.7)	0
	55	2.9	30 (20.0)	26 (17.3)	3
	68	1.0	11 (7.3)	10 (6.7)	0
	69	1.0	13 (8.7)	11 (7.3)	0
	69	1.3	17 (11.3)	13 (8.7)	0
	66	1.2	16 (10.7)	14 (9.3)	0
	58	3.2	36 (24.0)	37 (24.6)	7
	71	0.9	15 (10.0)	12 (8.0)	0
	71	1.1	18 (12.0)	16 (10.7)	0
	69	1.2	14 (9.3)	12 (8.0)	0
	74	1.1	10 (6.7)	8 (5.3)	0
	60	2.9	32 (21.3)	31 (20.6)	5
	72	1.1	11 (7.3)	10 (6.7)	0
	70	1.2	7 (4.6)	6 (4.0)	0
	70	1.2	15 (10)	11 (7.3)	0
	89	1.7	7 (4.7)	3 (2.0)	0
	87	2.5	24 (16.0)	22 (14.7)	2
	89	1.8	9 (6.0)	2 (1.3)	0
	87	2.7	28 (18.7)	24 (16.0)	4
	90	1.8	8 (5.3)	6 (4.0)	0
	88	2.5	30 (20.0)	26 (17.3)	3
	1	0.0	0	0	0
	70	0.4	4 (2.7)	1 (0.7)	0
	—	0.0	0	0	0
	76	0.5	4 (2.7)	2 (1.3)	1
	1	0.0	0	0	0
	75	0.5	3 (2.0)	3 (2.0)	0

RF (regeneration frequency) = no. of explants with multiple buds/total explant number $\times$ 100%.

Growth rates were measured from the date of the initial transformation to the date when the first elongated shoot reached 9 cm in length.

TE (transformation efficiency) = no. of transgenic plants/total explants $\times$ 100%.

Elongated shoots (≥ 9 cm in length) were collected for transgenic assay only once on day 75 after *Agrobacterium*-mediated transformation.

MF (mutation frequency) = no. of mutants/total transgenic plants $\times$ 100%. T

FF (transgene-free frequency) = no. of transgene-free mutants/total mutants $\times$ 100%.

TABLE 9

Mutation identification using PCR-RE assays for mutants regenerated from Williams 82 explants transformed with pGmGRF5-GmGIF1, pGmGRF6-GmGIF1, pGmGRF11-GmGIF1, pGmGRF18-GmGIF1, and pBSE401, respectively.						
Vector	Mutant ID	Genotype	Mutants ID	Genotype	Mutants ID	Genotype
pGmGRF5-GmGIF1	Gm-001	AaBb	Gm-033	AaBb	Gm-065	AaBB
	Gm-002	AaBb	Gm-034	AaBB	Gm-066	AaBb
	Gm-003	AaBb	Gm-035	AaBb	Gm-067	AaBb
	Gm-004	AaBb	Gm-036	AaBB	Gm-068	AaBB
	Gm-005	AaBb	Gm-037	aabb	Gm-069	AaBb
	Gm-006	AaBb	Gm-038	AaBb	Gm-070	aabb
	Gm-007	AaBB	Gm-039	AaBb	Gm-071	AaBb
	Gm-008	AaBb	Gm-040	AaBb	Gm-072	AaBb
	Gm-009	AaBb	Gm-041	AaBb	Gm-073	AaBb
	Gm-010	AaBB	Gm-042	AaBb	Gm-074	AaBb
	Gm-011	AaBb	Gm-043	AaBb	Gm-075	AaBb
	Gm-012	aabb	Gm-044	AaBb	Gm-076	AaBb
	Gm-013	AaBb	Gm-045	AaBB	Gm-077	AaBb
	Gm-014	AaBb	Gm-046	AaBb	Gm-078	AaBB
	Gm-015	AaBb	Gm-047	AaBB	Gm-079	AaBb

TABLE 9-continued

Mutation identification using PCR-RE assays for mutants regenerated from Williams 82 explants transformed with pGmGRF5-GmGIF1, pGmGRF6-GmGIF1, pGmGRF11-GmGIF1, pGmGRF18-GmGIF1, and pBSE401, respectively.						
Vector	Mutant ID	Genotype	Mutants ID	Genotype	Mutants ID	Genotype
pGmGRF6-GmGIF1	Gm-016	AaBb	Gm-048	AABb	Gm-080	AaBB
	Gm-017	AaBb	Gm-049	AaBb	Gm-081	aabb
	Gm-018	AaBb	Gm-050	AaBb	Gm-082	AaBB
	Gm-019	AaBb	Gm-051	AaBb	Gm-083	AABb
	Gm-020	AaBB	Gm-052	AaBb	Gm-084	AaBb
	Gm-021	AaBb	Gm-053	AaBb	Gm-085	AaBb
	Gm-022	AaBB	Gm-054	AaBB	Gm-086	AaBb
	Gm-023	aabb	Gm-055	AaBb	Gm-087	AaBb
	Gm-024	AaBb	Gm-056	AaBB	Gm-088	AaBb
	Gm-025	AaBb	Gm-057	aabb	Gm-089	AaBB
	Gm-026	AaBb	Gm-058	AaBb	Gm-090	AaBb
	Gm-027	AaBb	Gm-059	AaBb	Gm-091	AaBb
	Gm-028	AaBb	Gm-060	AaBb	Gm-092	AaBB
	Gm-029	AaBB	Gm-061	AaBb	Gm-093	AaBb
	Gm-030	AaBb	Gm-062	AaBb	Gm-094	AaBb
	Gm-031	AaBB	Gm-063	AaBb		
	Gm-032	aabb	Gm-064	AaBB		
	Gm-101	AaBb	Gm-112	AaBb	Gm-123	AaBB
	Gm-102	AaBB	Gm-113	AaBb	Gm-124	aabb
	Gm-103	AaBB	Gm-114	AaBB	Gm-125	AaBb
	Gm-104	AaBb	Gm-115	AaBb	Gm-126	AaBb
	Gm-105	AaBb	Gm-116	AaBb	Gm-127	AaBb
	Gm-106	AaBB	Gm-117	AaBB	Gm-128	AaBb
	Gm-107	AaBB	Gm-118	AaBB	Gm-129	AaBb
	Gm-108	AaBb	Gm-119	AaBb	Gm-130	AaBb
	Gm-109	AaBb	Gm-120	AaBb	Gm-131	AaBb
	Gm-110	AaBb	Gm-121	AaBb	Gm-132	AaBb
	Gm-111	AaBB	Gm-122	AaBb		
pGmGRF11-GmGIF1	Gm-201	AaBb	Gm-212	AaBb	Gm-223	AaBB
	Gm-202	AaBB	Gm-213	AaBB	Gm-224	AaBb
	Gm-203	AaBb	Gm-214	AaBb	Gm-225	AaBB
	Gm-204	AaBB	Gm-215	AaBB	Gm-226	AaBb
	Gm-205	AaBB	Gm-216	AaBb	Gm-227	AaBb
	Gm-206	AaBb	Gm-217	AaBb	Gm-228	AaBb
	Gm-207	AaBb	Gm-218	AaBb	Gm-229	AaBb
	Gm-208	AaBb	Gm-219	AaBb	Gm-230	AaBb
	Gm-209	AaBb	Gm-220	AaBb	Gm-231	AaBb
	Gm-210	AaBb	Gm-221	AaBb	Gm-232	AaBb
	Gm-211	AaBB	Gm-222	AaBb	Gm-233	AaBb
pGmGRF18-GmGIF1	Gm-301	AaBb	Gm-313	AaBb	Gm-325	AaBb
	Gm-302	AaBb	Gm-314	AaBb	Gm-326	AaBB
	Gm-303	AaBB	Gm-315	AaBb	Gm-327	AaBb
	Gm-304	AaBb	Gm-316	AaBb	Gm-328	AaBb
	Gm-305	AaBB	Gm-317	AaBB	Gm-329	AaBb
	Gm-306	AaBB	Gm-318	AaBb	Gm-330	AaBb
	Gm-307	AaBb	Gm-319	AaBB	Gm-331	AaBb
	Gm-308	AaBb	Gm-320	AaBB	Gm-332	AaBB
	Gm-309	AaBb	Gm-321	AaBB	Gm-333	AaBb
	Gm-310	AaBb	Gm-322	AaBb	Gm-334	AaBB
	Gm-311	AaBb	Gm-323	AaBB	Gm-335	AaBb
pBSE401 (CK)	Gm-312	AaBb	Gm-324	AaBB	Gm-336	AaBb
	Gm-401	AaBb	Gm-413	AaBB	Gm-425	AaBb
	Gm-402	AaBb	Gm-414	AaBb	Gm-426	AaBb
	Gm-403	AaBb	Gm-415	AaBB	Gm-427	AaBb
	Gm-404	AaBb	Gm-416	AaBB	Gm-428	AaBb
	Gm-405	AaBb	Gm-417	AaBb	Gm-429	AaBb
	Gm-406	AaBb	Gm-418	AaBb	Gm-430	AaBB
	Gm-407	AaBb	Gm-419	AaBb	Gm-431	AaBb
	Gm-408	AaBb	Gm-420	AaBb	Gm-432	AaBB
	Gm-409	AaBb	Gm-421	AaBb		
	Gm-410	AaBb	Gm-422	AaBb		
	Gm-411	AaBB	Gm-423	AaBB		
	Gm-412	AaBb	Gm-424	AaBb		

Aa/aa and Bb/bb represent heterozygous/homozygous mutations in FAD2-1A and FAD2-1B, respectively.

TABLE 10

Mutation identification using PCR-RE assays for mutants regenerated from explants (Zhonghuang 13 and Hefeng 25) transformed with pGmGRF5-GmGIF and pBSE401.							
Soybean cultivar	Vector	Mutant ID	Genotype	Mutant ID	Genotype	Mutant ID	Genotype
Hefeng 25	pGmGRF5- GmGIF	Gm-501	AaBb	Gm-503	AaBB	Gm-505	AaBB
		Gm-502	AaBB	Gm-504	AaBb	Gm-506	AaBb
Zhonghuang 13	pGmGRF5- GmGIF1	Gm-601	AaBB	Gm-625	AaBb	Gm-649	AaBb
		Gm-602	AaBb	Gm-626	AaBb	Gm-650	AaBB
		Gm-603	AaBb	Gm-627	AaBb	Gm-651	AaBb
		Gm-604	AaBb	Gm-628	AaBB	Gm-652	AaBB
		Gm-605	AaBB	Gm-629	AaBb	Gm-653	AaBB
		Gm-606	AaBb	Gm-630	AaBB	Gm-654	AaBb
		Gm-607	AaBB	Gm-631	AaBB	Gm-655	AaBb
		Gm-608	AaBB	Gm-632	AaBb	Gm-656	AaBb
		Gm-609	AaBb	Gm-633	AaBb	Gm-657	AaBb
		Gm-610	AaBb	Gm-634	AaBb	Gm-658	AaBb
		Gm-611	AaBb	Gm-635	AaBb	Gm-659	AaBB
		Gm-612	AaBb	Gm-636	AaBb	Gm-660	AaBb
		Gm-613	AaBb	Gm-637	AaBB	Gm-661	AaBB
		Gm-614	AaBB	Gm-638	AaBb	Gm-662	AaBB
		Gm-615	AaBb	Gm-639	AaBB	Gm-663	AaBb
		Gm-616	AaBB	Gm-640	AaBB	Gm-664	AaBb
		Gm-617	AaBB	Gm-641	AaBb	Gm-665	AaBb
		Gm-618	AaBb	Gm-642	AaBb	Gm-666	AaBb
		Gm-619	AaBb	Gm-643	AaBb	Gm-667	AaBb
		Gm-620	AaBb	Gm-644	AaBb	Gm-668	AaBB
pBSE401		Gm-621	AaBb	Gm-645	AaBb	Gm-669	AaBb
		Gm-622	AaBb	Gm-646	AaBB	Gm-670	AaBB
		Gm-623	AaBB	Gm-647	AaBb	Gm-671	AaBB
		Gm-624	AaBb	Gm-648	AaBB	Gm-672	AaBb
		Gm701	AaBb	Gm705	AaBb	Gm709	AaBb
		Gm702	AaBb	Gm706	AaBb	Gm710	AaBB
		Gm703	AaBB	Gm707	AaBb	Gm711	AaBb
		Gm704	AaBb	Gm708	AaBb		

Aa/aa and Bb/bb represent heterozygous/homozygous mutations in FAD2-1A and FAD2-1B, respectively.

TABLE 11

Effect of the TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 complexes on regeneration and genome editing efficiencies of common wheat cultivars Bobwhite and Kenong 199.						
Targeted gene	Common wheat cultivar	Treatment	No. bombarded immature embryos	No. regenerated plants/ RF (%)	No. mutants/ MF (%)	No. transgene- free mutants/ TFF (%)
TaALS (C to T)	Bobwhite	pUBI-GFP (CK)	333	266 (81.0 ± 19.8)	33 (9.9 ± 1.3)	17 (50.9 ± 5.5)
		TaGRF4-TaGIF1	277	1507 (508.0 ± 75.9)	100 (32.7 ± 3.45)	47 (47.1 ± 5.9)
		mTaGRF4-TaGIF1	343	2080 (630.1 ± 86.2)	216 (63.3 ± 4.0)	103 (47.5 ± 2.6)
		pUBI-GFP (CK)	307	419 (136.9 ± 5.7)	55 (17.7 ± 2.6)	28 (52.8 ± 8.4)
	Kenong 199	TaGRF4-TaGIF1	325	2095 (654.5 ± 55.5)	331 (103.4 ± 9.3)	170 (51.6 ± 1.8)
		mTaGRF4-TaGIF1	328	3796 (1165.4 ± 58.8)	577 (176.1 ± 13.9)	324 (55.4 ± 6.6)

RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%.

MF (mutation frequency) = no. of mutants/total immature embryos × 100%.

The numbers and means for each treatment were calculated from data collected from three replicates of the experiment.

Data are means ± s.e.m (n = 3).

TABLE 12

Effect of the TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 complexes on regeneration and genome editing efficiencies of common wheat cultivars Bobwhite and Kenong 199 (raw data).							
Targeted gene	Common wheat cultivar	Treatment	Repeat	No. bombarded immature embryos	No. regenerated plants/RF (%)	No. mutants/MF (%)	No. transgene- free mutants/ TFF (%)
TaALS (C to T)	Bobwhite	pUBI-GFP (CK)	1	119	50 (42.0)	10 (8.4)	4 (40.0)
			2	112	119 (106.2)	14 (12.5)	8 (57.1)
			3	102	97 (95.0)	9 (8.8)	5 (55.6)

TABLE 12-continued

Effect of the TaGRF4-TaGIF1 and mTaGRF4-TaGIF1 complexes on regeneration and genome editing efficiencies of common wheat cultivars Bobwhite and Kenong 199 (raw data).							
Targeted gene	Common wheat cultivar	Treatment	Repeat	No. bombarded immature embryos	No. regenerated plants/RF (%)	No. mutants/MF (%)	No. transgene-free mutants/TFF (%)
	Kenong 199	TaGRF4-TaGIF1	1	116	547 (472.6)	42 (25.9)	20 (47.6)
			2	86	562 (653.5)	30 (34.9)	11 (36.7)
			3	75	398 (530.7)	28 (37.3)	16 (57.1)
		mTaGRF4-TaGIF1	1	103	498 (483.5)	70 (68.0)	32 (45.7)
			2	117	915 (782.1)	78 (66.7)	41 (52.6)
			3	123	767 (624.6)	68 (55.3)	30 (44.2)
		pUBI-GFP (CK)	1	102	146 (143.1)	17 (16.7)	7(41.2)
			2	110	138 (125.5)	25 (22.7)	12 (48.0)
			3	95	135 (142.1)	13 (13.7)	9 (69.2)
		TaGRF4-TaGIF1	1	132	763 (578.0)	116 (87.9)	56 (48.3)
			2	100	623 (623.0)	120 (120)	62 (51.7)
			3	93	709 (762.4)	95 (102.2)	52 (54.7)
		mTaGRF4-TaGIF1	1	114	1329 (1165.8)	230 (201.8)	135 (58.7)
			2	120	1276 (1063.3)	185 (154.2)	120 (64.9)
			3	94	1191 (1267.0)	162 (172.3)	69 (42.6)

RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%.

MF (mutation frequency) = no. of mutants/total immature embryos × 100%.

TABLE 13

Effect of mTaGRF4-TaGIF1 on regeneration and genome editing efficiencies of nine elite common wheat cultivars.							
Targeted gene	Common wheat cultivar	Treatment	No. bombarded embryos	No. regenerated plants/ RF (%)	No. mutants/ MF (%)	No. homozygous mutants/HMF (%)	No. transgene-free mutants/TFF (%)
TaALS (C-to-T)	Jimai 20	UBI-GFP (CK)	335	14 (4.2 ± 1.1)	0	0	0
	Jimai 22	mTaGRF4-TaGIF1	270	37 (14.2 ± 2.1)	9 (3.3)	5 (55.6)	2 (22.2)
	Jing 411	UBI-GFP (CK)	272	13 (4.7 ± 0.6)	0	0	0
	Shannong 20	mTaGRF4-TaGIF1	344	34 (9.9 ± 0.4)	4 (1.2)	0	1 (25.0)
	Shannong 116	UBI-GFP (CK)	203	65 (18.8 ± 2.7)	2 (1.0)	0	0
	Xiaoyan 54	mTaGRF4-TaGIF1	320	135 (34.0 ± 3.2)	5 (1.6)	1 (25.0)	2 (50.0)
	Zhoumai 27	UBI-GFP (CK)	335	15 (4.5 ± 0.6)	0	0	0
	Zhoumai 28	pmTaGRF4-TaGIF1	344	71 (20.7 ± 2.2)	10 (2.9)	1 (10.0)	5 (50.0)
	Zhongmai 175	UBI-GFP (CK)	285	128 (66.3 ± 8.2)	3 (1.1)	0	1 (33.3)
		mTaGRF4-TaGIF1	308	342 (101.7 ± 18.4)	7 (2.3)	1 (14.3)	2 (28.6)
		UBI-GFP (CK)	320	0	0	0	0
		mTaGRF4-TaGIF1	343	36 (11.0 ± 2.1)	6 (1.7)	1 (16.7)	3 (50.0)
		UBI-GFP (CK)	286	441 (157.6 ± 24.1)	4 (1.4)	0 (0.0)	1 (25.0)
		mTaGRF4-TaGIF1	343	1578 (440.8 ± 50.3)	8 (2.3)	1 (12.5)	5 (62.5)
		UBI-GFP (CK)	308	16 (5.2 ± 1.3)	0	0	0
		mTaGRF4-TaGIF1	331	370 (112.3 ± 11.4)	10 (3.0)	2 (20.0)	4 (40.0)
		UBI-GFP (CK)	335	614 (187. ± 26.1)	13 (3.9)	2 (15.4)	8 (61.6)
		mTaGRF4-TaGIF1	332	1265 (380.6 ± 29.0)	27 (8.1)	10 (37.0)	18 (66.7)

RF (regeneration frequency) = no. of regenerated plants/total embryos × 100%.

MF (mutation frequency) = no. of mutants/total immature embryos × 100%.

HMF (homozygous mutant frequency) = no. of homozygous mutants/total mutants × 100%.

TFF (transgene-free frequency) = no. of transgene-free mutants/total mutants × 100%.

The numbers and means for each treatment were calculated from data collected from three replicates of the experiment.

Data are means ± s.e.m (n = 3).

TABLE 14

Effect of expression of the mutated common wheat TaGRF4-TaGIF1 on the regeneration and genome editing efficiencies of nine elite common wheat cultivars (raw data).				
Common wheat cultivar	Treatment	Repeat	No. bombarded immature embryos	No. regenerated plants/RF (%)
Jimai 20	pUBI-GFP (CK)	1	115	4 (3.5)
		2	110	7 (6.4)
		3	110	3 (2.7)
	mTaGRF4-TaGIF1	1	109	11 (10.1)
		2	91	14 (15.4)
		3	70	12 (17.1)

TABLE 14-continued

Effect of expression of the mutated common wheat TaGRF4-TaGIF1 on the regeneration and genome editing efficiencies of nine elite common wheat cultivars (raw data).				
Common wheat cultivar	Treatment	Repeat	No. bombarded immature embryos	No. regenerated plants/RF (%)
Jimai 22	pUBI-GFP (CK)	1	102	6 (5.9)
		2	66	3 (4.5)
		3	104	4 (3.8)
	mTaGRF4-TaGIF1	1	118	11 (9.3)
		2	123	13 (10.6)
		3	103	10 (9.7)
Jing 411	pUBI-GFP (CK)	1	115	20 (17.4)
		2	116	28 (24.1)
		3	113	17 (15.0)
	mTaGRF4-TaGIF1	1	130	40 (30.8)
		2	140	43 (30.7)
		3	128	52 (40.6)
Shangnong 20	pUBI-GFP (CK)	1	114	5 (4.4)
		2	115	4 (3.5)
		3	106	6 (5.7)
	mTaGRF4-TaGIF1	1	115	19 (16.5)
		2	116	25 (21.6)
		3	113	27 (23.9)
Shangnong 116	pUBI-GFP (CK)	1	60	49 (81.7)
		2	75	40 (53.3)
		3	61	39 (63.9)
	mTaGRF4-TaGIF1	1	133	124 (93.2)
		2	90	110 (122.2)
		3	85	108 (127.1)
Xiaoyan 54	pUBI-GFP (CK)	1	98	0
		2	108	0
		3	114	0
	mTaGRF4-TaGIF1	1	98	10 (10.2)
		2	100	15 (15.0)
		3	141	11 (7.8)
Zhoumai 27	pUBI-GFP (CK)	1	81	160 (197.5)
		2	105	120 (114.3)
		3	100	161 (161.0)
	mTaGRF4-TaGIF1	1	109	500 (458.7)
		2	132	457 (346.2)
		3	120	621 (517.5)
Zhoumai 28	pUBI-GFP (CK)	1	115	8 (7.0)
		2	86	5 (5.9)
		3	107	3 (2.8)
	mTaGRF4-TaGIF1	1	129	130 (100.8)
		2	105	142 (135.2)
		3	97	98 (101.0)
Zhongmai 175	pUBI-GFP (CK)	1	95	198 (208.2)
		2	110	240 (218.2)
		3	130	176 (135.4)
	mTaGRF4-TaGIF1	1	120	389 (324.2)
		2	107	450 (420.6)
		3	105	417 (397.1)

RF (regeneration frequency) = no. of regenerated plants/total immature embryos × 100%.

TABLE 15

Genotypes of the T0 taals mutants regenerated from immature embryos of nine elite common wheat cultivars treated with mTaGRF4-TaGIF1 and UBI-GFP (CK), respectively.							
Wheat cultivar	Vectors delivered	Mutant ID	Genotype	A subgenome	B subgenome	D subgenome	Transgene-free
Zhongmai 175	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-001	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-002	aabbdd	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
		Ta-003	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-004	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	Del CCCC GCCC; Del CCCC	YES
		Ta-005	aabbdd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈ ; C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-006	AaBbDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	WT	YES
		Ta-007	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ > T ₆ T ₇	YES
		Ta-008	AaBbDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	WT	WT	NO
		Ta-009	aabbdd	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈		YES
		Ta-010	aabbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
		Ta-011	AaBbDD	AGGTCCCCGCCGC > GCCCC	ATCTGTTTCGGCCAGGT	C ₆ C ₇ > T ₆ T ₇	NO
		Ta-012	AaBbDD	Del AGGTCCCC; WT	CCCC > TGTTC-----;	WT	YES
		Ta-013	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-014	AaBbDd	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	YES
		Ta-015	AaBbDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	WT	NO
		Ta-016	AaBbDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	WT	WT	NO
		Ta-017	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-018	aabbdd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₇ C ₉ C ₁₀ > T ₇ T ₉ T ₁₀	YES
		Ta-019	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₇ C ₉ C ₁₀ > T ₇ T ₉ T ₁₀	YES
		Ta-020	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈ ;	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-021	aabbDd	Del CCCC GCCC AATC; WT	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	YES
		Ta-022	aabbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	WT	Del CCCC GCCC	YES
		Ta-023	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-024	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-025	AaBbDd	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-026	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-027	aabbDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	NO
		Ta-051	AaBbDd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈ ;	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
	pUBI-GFP + pUBI-A3A	Ta-052	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀ ;	NO
						C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	NO

Genotypes of the T0 taals mutants regenerated from immature embryos of nine elite common wheat cultivars treated with mTaGRF4-TaGIF1 and UBI-GFP (CK), respectively.

[illegible]

TABLE 15-continued

Genotypes of the T0 taals mutants regenerated from immature embryos of nine elite common wheat cultivars treated with mTaGRF4-TaGIF1 and UBI-GFP (CK), respectively.							
Wheat cultivar	Vectors delivered	Mutant ID	Genotype	A subgenome	B subgenome	D subgenome	Transgene-free
Jimai 22	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-158	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	NO
		Ta-159	aaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7 > T_6T_7$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	NO
		Ta-160	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	NO
		Ta-171	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-172	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
Shannong 116	pUBI-GFP + pUBI-A3A	Ta-173	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	WT	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-174	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	WT	NO
		Ta-211	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-212	AaBbDd	WT	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
		Ta-213	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	NO
	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-201	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
		Ta-202	aaBbDd	CCCCGCCGCAT > TTTTGCCGCTC	CCCCGCCGCAT > TTTTGCCGCTC	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	YES
		Ta-203	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	NO
		Ta-204	AaBbDd	WT	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	NO
		Ta-205	AaBbDd	WT	Del GTCCGCCGCCGCAT	WT	NO
Jing 411	pUBI-GFP + pUBI-A3A	Ta-206	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	GAICGGACGG	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-207	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	CAATGAT	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-301	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-302	AaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8 > T_6T_7T_8$	NO
		Ta-401	aaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-402	aaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
		Ta-403	aaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$A_7C_6C_7C_8C_9 > G_7T_6T_7T_9$	NO
		Ta-404	aaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_9 > G_9$	$C_6C_7 > T_6T_7$	NO
		Ta-405	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-431	AaBbDd	$C_6C_7 > T_6T_7$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	NO
Zhoumai 27	pUBI-GFP + pUBI-A3A	Ta-432	AaBbDd	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-433	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	YES
		Ta-434	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	NO
		Ta-450	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	NO
		Ta-451	aaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8 > T_6T_7T_8$	YES
	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-452	AaBbDd	$C_{5-5} > T_{5-5}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
		Ta-453	aaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	YES
		Ta-454	aaBbDd	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-455	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	$C_6C_7C_8C_9C_{10} > T_6T_7T_8T_9T_{10}$	NO
		Ta-456	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7C_8 > T_6T_7T_8$	YES
		Ta-457	AaBbDd	$C_6C_7C_8 > T_6T_7T_8$	$C_6C_7C_8C_9 > T_6T_7T_8T_9$	$C_6C_7 > T_6T_7$	YES

TABLE 15-continued

Genotypes of the T0 taals mutants regenerated from immature embryos of nine elite common wheat cultivars treated with mTaGRF4-TaGIF1 and UBI-GFP (CK), respectively.							
Wheat cultivar	Vectors delivered	Mutant ID	Genotype	A subgenome	B subgenome	D subgenome	Transgene-free
Zhoumai 28	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-411	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
		Ta-412	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-413	AaBbDd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-414	aaBBDD	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	WT	WT	NO
		Ta-415	aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-416	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	NO
Xiaoyan 54	pmTaGRF4-TaGIF1 + pUBI-A3A	Ta-417	AaBBDD	C ₆ C ₇ > T ₆ T ₇	WT	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
		Ta-418	AabbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-419	AaBbDd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	NO
		Ta-420	AabbDd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-501	AaBbDd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	YES
		Ta-502	aaBBDD	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉	YES
		Ta-503	Aabbdd	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉ ; C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	NO
		Ta-504	aabbdd	Del CCCCCGCCGCA	C ₆ > T ₆	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	NO
		Ta-505	aabbDd	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	C ₆ > T ₆	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈	YES
		Ta-506	AabbDd	C ₆ C ₇ C ₈ C ₉ > T ₆ T ₇ T ₈ T ₉	C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ T ₁₀	C ₆ C ₇ C ₈ > T ₆ T ₇ T ₈ C ₆ C ₇ C ₈ C ₉ C ₁₀ > T ₆ T ₇ T ₈ T ₉ A ₁₀	NO

Aa/aa, Bb/bb, and Dd/dd represent heterozygous/homozygous mutations in TaALS-1A, TaALS-1B, and TaALS-1D, respectively.

Sequences

SEQ ID NO: 1

ZmWUS amino acid sequence

MAANAGGGGAGGGSSGSAAPAVCRPSGSRWPTPEQIRMLKELYYGCGIRSPSSSEQIQRITAMLRQHGKIE
GKNVFWFQNHKARERQKRRLTSLDVNVPAAGAADATTSQLGVLSLSSPPPSGAAPPSTLGFYAAGNGGGS
AVLLDTSSDWGSSGAAMATETCFLQDYMVGVTDTGSSSQWPRFSSSDTMAAAAAARAATRAPETLPLFPPTCG
DDGGSGSSSYLFPWGAASTTAGATSSVAIQQHQHQLQEYVSFYNSNSNTQLAGTGNQDVSATAAAAAALELSL
SSWCSPPYPAAGSM

SEQ ID NO: 2

ZmBBM amino acid sequence

MATVNNWLAFLSLSPQELPPSQTTDSTLISAATADHVSGDVCFNIPQDWSMRGSELSALVAEPKLEDFLGGISFS
EQHHKANCNMIPSTSTSTVCYASSGASTGYHHQLYHQPTSSALHFADSVMVASSAGVHDGGAMLSAAAANG
VAGAASANGGIGLSMIKNWLRSQPAPMQPRVAAEQAQGLSLSMNMAGTTQGAAGMPLLAGERARAPES
VSTSAQGGAVVVTAPKEDSGSGSVAGALVAVSTDTGSGSGASADNTARKTVDTFGQRTSIYRGVTRHRWTG
RYEAHLWDNSCRREGQTRKGRQVYLGGYDKEEKAARAYDLAALKYWGATTTTNFPVSNYEKELEDMKHMTR
QEFVASLRKKSSGFSRGASIRGVTRHHQHGRWQARIGRVAGNKDLYLGTFTSTQEEAAEAYDIAAIFKFRGLNAV
TNFDMSRDYDKSILDSALPIGSAAKRLKEAEEAASQHHHAGVVSYDVGRIASQLGDGGALAAAYGAHYHG
AAWPTIAFQPGAASGLYHPYAQQPMRGGGWCKQEQDHAVIAAAHSLQDLHHLNLGAAGAHDFPSAGQQ
AAAAAMHGLGSDTSASLEHSTGSNSVVYNGGVGDSNGASAVGGSGGYMMPMSAAGATTTSAMVSHEQV
HARAYDEAKQAQMGYESYLVAENNGGGRMSAWGTVVSAAAAAASNDNMAADVGHGAQLFSVW
NDT

SEQ ID NO: 3 ZmSERK amino acid sequence

MAAAEARRRSVAVWALLPLLRLHHPAALVLANTEGDALHSLRTNLNDPNNVLQSWDPTLVNPFCTWFHVTN
NNSVIRVLDGNAALSGTLVLPQLGQLKQLYLELYSNNISGTIPSELGNLTNLVSLDLVUNNFTGPIDSLGKLLK
LRFLRLNNSLSGSIKSLTAITALQVLDLSNNNLSGEVPTSGTSGSLFTPIISFGNNPNLCGPGTTKPCPGAPPFSP
PYNPTTPVQSPGSSSSTGATAGVAAAGALLFAIPAIISPAYWRRRKQEHFFDVPAEEDPEVHLGQLKRFSLR
ELQVATDGFSSNKNIILGRGGFGKVKYGRSLADGSLVAVKRLKEERTPGGELQFQTEVEMI SMAVHRNLRLRGFC
MTPTERLLVYPYMANGSVASRLRDRPPAEPPLDWQTRQRIALGSARGLSYLHDHCDPKI IHRDVKAANI LLD
FEAVVGDFGLAKLMDYKDTHTVTTAVRGTIHIAPEYLSGTGKSEKTDVFGYGI TLELITGQRAFDLARLANDDD
VMLLDWVNCATISSVQIVPQVFMPLWLPHYCAGAGHLFLCNC

SEQ ID NO: 4 TaGRF4 amino acid sequence

MAMPYASLSPAGDRRSSPAATATASLLPFCRSSPFSAGNGMGMEEARMDGRWMARVPVFTAAQYEELEHQ
ALIYKYLAVGVSPDPDLVLPPIRRGIESLAARFYHNPLAIGYGSYLGGKVDPEPGRCCRDTGKKWRCACEAASDSKY
CERHMRHGRNRSRKPVEATQLVSHSQPPAASVVPPLATGFHNHSLYPAIGGTNGGGGGNNGMSMPGTFFS
ALGPPQQHMGNNAASPYAALGGAGTCKDFRYTAYGIRSLADEQSOLMTEAMNTSVENPWRLPSPSQTTTFFP
LSSYSPLGATSDLGQNNSSNNNSGVKAEGQQQQQLSPFGCGDFGSGDSAKQENQTLRPFDEWPKTRDS
WSDLTDDNSNVASFATQLSISIPMTSSDFSASSQSPNGMLFAGEMY

SEQ ID NO: 5 miR396 binding site mutated TaGRF4 amino acid sequence

MAMPYASLSPAGDRRSSPAATATASLLPFCRSSPFSAGNGMGMEEARMDGRWMARVPVFTAAQYEELEHQ
ALIYKYLAVGVSPDPDLVLPPIRRGIESLAARFYHNPLAIGYGSYLGGKVDPEPGRCCRDTGKKWRCACEAASDSKY
CERHMRHGRNRSRKPVEATQLVSHSQPPAASVVPPLATGFHNHSLYPAIGGTNGGGGGNNGMSMPGTFFS
ALGPPQQHMGNNAASPYAALGGAGTCKDFRYTAYGIRSLADEQSOLMTEAMNTSVENPWRLPSPSQTTTFFP
LSSYSPLGATSDLGQNNSSNNNSGVKAEGQQQQQLSPFGCGDFGSGDSAKQENQTLRPFDEWPKTRDS
WSDLTDDNSNVASFATQLSISIPMTSSDFSASSQSPNGMLFAGEMY

SEQ ID NO: 6

TaGIF1 amino acid sequence

MQQQHLQMNMQSMGGYASSTATTDLIQYLDENKQLILAILDNQNNKGVEECARNQAKLQONLMYLA
ATADSQPPQATSLSQYPNLMQSGPRYMQQSAQMMSPQSLMAARSSMMYAQQAMSPQQQQQQQ
QHQAHAHQQLGMSSGATTGFLNLHGEASMGGGGATGNSMMNAVSFSDYGRGGSGAKEGSTSLSADARG
ANSGAHSGDGEYLLKTEEGS

SEQ ID NO: 7 GmGRF5 amino acid sequence

MGELFGVGKRRNIISSNNNNSSSSSVLGLDVKVQSPALFHNRRMMMAHNNHHRPLSSPFDNNGDG
DGFTTYMSFTNHINLVSGASSVLGPAIDAGCGAAPPAPVRTLQPFDISSYTSSTPTTTTTTFNFKPPSAGVMAASL
GFPFTSAQWRELERQAMIKYMMASVVPVPHDLLTPSSRSSCMDGGFNRLRLANS TDPEPGRCCRDTGKKWRCS
RDVAPNHKYCERHMRHGRPRSRKPEVNTNSTTTPTSVNNNNHQIKKARHECNNNPFATPDVTAIISNPTSR
KNGSPHFLGSTTQPYLDSLSLDNFGFLKAASFDSVASVANSKEPRGLEWMLNGDPI SLGASDSQWQSLMH
NKDGMTSVSSCNTTESQYLNLSLALYNSGLEQQNRRHPLFLNPLVPMENLOPEKPRGFI DAWSNASNANTN
TTNKNSAASIGKLSLSSLDLSMGGAADVNDVGNVNMGLGLMEPNGKTHGTGKI SLSNQNPAPWVASSLGG
PLAEVLRSSVTATTTTTNEATSNTPSPATTTHAESPSGVQLKTLVLSLSDSSNNSSPRVASSRANSEMAALLRFQSN

SEQ ID NO: 8 GmGRF6 amino acid sequence

MMSASARNRSPFTQTQWQLEHQALVFKYMTGTPIPPDLIYSIKRSLDTSISSRLFPHHPIGWGCFEMGFGRK
VDPEPGRCCRDTGKKWRCSKEAYPDSKYCERHMRHGRNRSRKPVEVSSAISTATNTSQTIPSSYTRNLSLTNPN
MTPSSFPFPLPSSMPIESQPFSSQYQNSLNPFFYSQTSRPPDADFPQDATTQFLPMDSGSYSHDEKNY
RHVHGIREVDVERAFPEASGSARSYTESYQQLSMSYKSYSNSNFQININDATTNPRQEQQQQHCFVLGT
DFKSTRPTKEKAETATGQRPLHRFFGEWPPKNTTDSWLDLASNSRIQTDE

SEQ ID NO: 9 GmGRF11 amino acid sequence

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Sequences

SEQ ID NO: 10 GmGRF18 amino acid sequence
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SEQ ID NO: 11 GmGIF1 amino acid sequence
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SEQ ID NO: 12 linkerA
AAAA

SEQ ID NO: 13 linkers
SGGS

SEQ ID NO: 14 A3A base editor amino acid sequence
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SEQ ID NO: 15 SpCas9 amino acid sequence
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SEQ ID NO: 16 sgRNA scaffold sequence
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SEQ ID NO: 17 GmGRF5-GmGIF1
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Sequences

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SEQ ID NO: 18 GmGRF6-GmGIF1

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Sequences

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SEQ ID NO: 20 GmGRF18-GmGIF1

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SEQ ID NO: 21 TaGRF4-TaGIF1

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Sequences

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SEQ ID NO: 22 mTaGRF4-TaGIF1

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GGCGCGCACAGCGGCGCACGGGAGTACTCAAGGGCACGAGGAGGAAGGAAGCTAA

SEQ ID NO: 23 GmGRF5-GmGIF1

MGELGVGVGKKRNISSSSNNSSSSSVGLGLDVVKVQQSPALFHNRMMMMAHHNHHRPLSPFDNNNGD
DGPTTSMQSFNTNHLINLVSGASSVGLPAIDAGCGAAPVPTVLQFPDIISSYTSPTTTTTFNFKPKPSAGVMAASL
GPFPTSAQWRELQYAGIKYKMMASVVPDHLLTTPSSRSCMDGGFNRLRANSTDPEGRCCRRTDGKWKRC
RDVAPHNHYK CERHMHGRPRSRKPVENVNTSTTPTSVNNNNHGIKKARHECNMNPFPDPTDVTAAISNPTSR
KNGPSPHFLGSTTTQPYLDLSSLSLNDNFGLKAAFSVASVANKEPRGLEWMLNGDPTISLGADSSQWQS
LMH NKDGMTSVSSCNTTESQYLNSLALVNSGLEQGNRRHPLFLNPLVPMENLQPEKPRGFIADWSANESNANTN
TTNNKSAASI GKLSLSLSDLMGGAAVNDEGVNVMGLGLMEPNKGKTHTGKISLNNWQNPAPWASSLGG
PLAEVLRSSVTVTATTTEATSNTPSTPATTTHAESPSGVGLQKTLVLSLSSNNPVRASSRANSEMALLRPQSN
AAAAPGSGSLFHPHCLLHVHQSFYDACFSQRYLRSHLFTQITMSEYALPGDHKPSFYQNYLDEKSLILKIV
ESQNSGKLSCEAENQARLQRLNMLYLAADASQOPQPTMPGQYPPSGMQAGAHYMAQQQTQMSFPQ
QLMAERSLSLLYAQQPYSAQQQAMHSLGSSGLHMLQSEGSNNVNGSSGSVGGGFPDLVRGGGGGG
GSGTEGLHSGGRIGMSGLQEGTGGSGRGGGSGEGENLYKIIADDGN

SEQ ID NO: 24 GmGRF6-GmGIF1

MMSASARNRNSPFTQTQWQBEHQALVFKYVMVTGTPIDPDLIYSIKRSLDTSISRLLPPHHPITGSGCFEMGFGRK
VDEPEGRKRLPTDGKKVKCSKEAYPDSKCYERHHMRHGRNRSRKRPVEVSSAISTATNTSQTIPSSYTRNLSLTNPN
MTPESSFPSPRLPSSMPBIESPFSQSYQNSLNPFFYSQTSRRPDDFPQDATHQLFMSYSGHDEKYN
RHVHGIRVEDVERAFATPEASGSARSYTESYQQLSMSSYKSYNSNFQINIDATTNPRQEQEQQQQHCFVLGT
DFKSTRPTKEKEAETATGQRPRLHRFFGWEPPKNTDLSWDLANSRNIQTDEAAAAAMFGLSHFFHCHLLHLVHQ
SYDFACFSQRYLRHSLPTQITMSEYALPGDKHPSYQNYLNDENKSLIKTIVESQNGSKLSECAENQARLQRNL
MYLAAIADSQPQPTMPGQYPPSGNMVQQAHYMQAQQTQQMSPQQLMAARSSLYLAAQQFYSALQQ
QQAMHSAALGSSGLHMLQSESGSNVQSGSGSVGGGFPDLVRGGGGGGSGEGGLHSGGRGINMGSSKQEI
GGSGEGRGSGSGEGGELNLXIADDGN

SEQ ID NO: 25 GmGRF11-GmGIF1

MNNSGGGGRGRTLMGLSNYICGRSPPTVSQWQLEHQALIKYMLAGLPVPLDVLVPIQNSFHSTISLSHAFF
 HPTPLSYCSFYGKGVDPGRCRTRDGGKWCSCKAYPDKCYCERHMRGRNRSRKPESQTMTHSSSTVSL
 TVGGSGASKGTVNFQNLSTFNLGQDSTGDTHNYHLSIPIAYIPKERYVQGLKSGEGGHECFLPNSSEAS
 NKVLQMESQLENTFPMMLSTRVASFSTSKSSNDLHSHDYPRHSLFSGEYVSGEHVKEBGQPLRPFNEWPKSRE
 CSWGLBEDERSNQTFATSTLSIIPMSSNFATSQSPGHEDEIQFRAAAAMFGSLFHPHCHLVLHVHQSFYDA
 FSGQRYLRSHLFTFQITMSEYALPGDHHKPSYQNYLDEKSLILKTVESQNGKLECAENAGRCRNLMYLA
 AIADSQPQPTMPGQYPPSGMMQGAHYMQAQQQTQOMSPQOLMAARSSLLYAQQPYALQQQQAM
 HSLGSSGLHMLQLSEGSINNVNNGSGSGSVGGGGPDLVRGGGGGGSTGELHSGGGRGTMGSSKQIEIGSSE
 GRGGSGEGENGLIADDDN

SEQ ID NO: 26 GmGRF18-GmGIF1

MDLGVVGLGEGVVGSESGCVFGSSLSVDPETKHKWYSGLLKQERSAIATEDDEWIRISKVAKTDHDMSSASKA
MLFPQQRNNLSLLRSNNATLFSGDHHQSQMLSFSPKSDSLIDKASSNATLPFSHQLSYSTRNTGYNSGSI SMH
GALASVPGPRTPSQWMEHLQHAIYKYITANVPVPTLLIPIRKALDSVGFCNFSAGLRLPNSLGLGVGLGFST
NTDPEGRCCRTDDGKKWCRSDAVDDQVCKCYERHNNRGRHRSRKVPQEGSGHALTTSTNTNPASSNNGVFLP
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Sequences

SDEYSSRNNEFGLVTSDDLNPSEKRSFTSSQKNDSSSQQHSRLRHFIDDSKPKQSNHHHRSSSIWPELDN
 MQSDRTQLSISIPISDDHMSFTTSLPSNEKLTLSPLRLSRELDPIQMGLGVGSAPNEANTRQANWIPITWESSM
 GGPLGEVLNLSNNNSNASDQCGKNNNNNTSALNLMKDGWDNNPPSGSSPTGVLQKSAFGSLSNSAGSSP
 RGAENNKEGATLCNALAAAMFGSLFHFPHCLLHVHQSFYDACFSQRYLRSHFLTQITMSESEYALFGDHKP
 SPYQNYLDENKSLILKIVESQNSGKLSACAENQARLQRNLMYLAIAIDSQPPPTMPGQYPPSGMMQQGAH
 YMQAQOQTQOMSPQQLMMAARSLLYAQQPYSALQQQQAMHSALGSSSLHMLQSEGSNVNVGSGSGSV
 GGGFPDLVRGGGGGGSTGEGLHSGGRGIMGSSKQEIGGSGEGRGGGSEGGENLYLKIADDGN

SEQ ID NO: 27 TaGRF4-TaGIF1

MAMPYASLSPAGDRSSPAATATASLLPFCRSSPFSAGNGMGEEARMGRWMARVPVFTAAQYEELEHQ
 ALIYKYLIVAGVSVPPDLVLPIRRGIESLAARFYHNPLAIGYGSYLGKKVDPEPGRCRRTDGKKWRCACEAASDSKY
 CERHMHGRNRSRKPVETQLVSHSQPPAASVVPPLATGFHNHSLYPAIGGTNGGGGGGNNGMSMPGTFSS
 ALGPPQQHMGNNASPYAALGGAGTCKDFRYTAYGIRSLADEQSQMLTEAMNTSVENPWRLPPSSQTTFP
 LSSYSPQLGATSDLGQNNSSNNNSGVKAEGQQQQPLSFPGCCDFGSGDSAKQENQTLRPFDEWPKTRDS
 WSDLTDDNSNVASFATQLSISIPMTSSDFSAAASQSPNGMLFAGEMYAAAAMQQQHLQMNMQSMGG
 YASSTTATDQLIQQYLDENKQLILAILDNQNNNGKVEECARNQAKLQQNLMYLAIAIDSQPPQTASLSQYPSNL
 MMQSGPRYMQQQSAQMMSPOSLMAARSSMMYAQQAMSPLOQQQQQQHQAHAHGLGMSSGATT
 GFNLLHGEASMGGGGATGNSMMNASVFSYGRGGSGAKEGSTLSADARGANSGAHSGDGEYLGKTEEE
 GS

SEQ ID NO: 28 mTaGRF4-TaGIF1

MAMPYASLSPAGDRSSPAATATASLLPFCRSSPFSAGNGMGEEARMGRWMARVPVFTAAQYEEEL
 EHQALYKYLIVAGVSVPPDLVLPIRRGIESLAARFYHNPLAIGYGSYLGKKVDPEPGRCRRTDGKKWRCACEAAS
 DSKYCERHMHGRNRSRKPVETQLVSHSQPPAASVVPPLATGFHNHSLYPAIGGTNGGGGGGNNGMSMPG
 TFSSALGPPQQHMGNNASPYAALGGAGTCKDFRYTAYGIRSLADEQSQMLTEAMNTSVENPWRLPPSSQT
 TTFPLSSYSPQLGATSDLGQNNSSNNNSGVKAEGQQQQPLSFPGCCDFGSGDSAKQENQTLRPFDEWPK
 TRDSWDLTDDNSNVASFATQLSISIPMTSSDFSAAASQSPNGMLFAGEMYAAAAMQQQHLQMNMQSM
 MGGYASSTTATDQLIQQYLDENKQLILAILDNQNNNGKVEECARNQAKLQQNLMYLAIAIDSQPPQTASLSQY
 PSNLMMQSGPRYMQQQSAQMMSPOSLMAARSSMMYAQQAMSPLOQQQQQQHQAHAHGLGMSS
 GATTGFNLLHGEASMGGGGATGNSMMNASVFSYGRGGSGAKEGSTLSADARGANSGAHSGDGEYLGK
 TEEEGS

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 68

<210> SEQ ID NO 1

<211> LENGTH: 302

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<400> SEQUENCE: 1

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Gly	Ser	Val	Ala	Ala	Pro	Ala	Val	Cys	Arg	Pro	Ser	Gly	Ser	Arg	Trp
		20					25					30			
Thr	Pro	Thr	Pro	Glu	Gln	Ile	Arg	Met	Leu	Lys	Glu	Leu	Tyr	Tyr	Gly
		35				40						45			
Cys	Gly	Ile	Arg	Ser	Pro	Ser	Ser	Glu	Gln	Ile	Gln	Arg	Ile	Thr	Ala
	50				55					60					
Met	Leu	Arg	Gln	His	Gly	Lys	Ile	Glu	Gly	Lys	Asn	Val	Phe	Tyr	Trp
65			70						75					80	
Phe	Gln	Asn	His	Lys	Ala	Arg	Glu	Arg	Gln	Lys	Arg	Arg	Leu	Thr	Ser
			85					90						95	
Leu	Asp	Val	Asn	Val	Pro	Ala	Ala	Gly	Ala	Ala	Asp	Ala	Thr	Thr	Ser
		100					105							110	
Gln	Leu	Gly	Val	Leu	Ser	Leu	Ser	Pro	Pro	Pro	Ser	Gly	Ala	Ala	
		115				120					125				
Pro	Pro	Ser	Pro	Thr	Leu	Gly	Phe	Tyr	Ala	Ala	Gly	Asn	Gly	Gly	Gly
		130				135					140				
Ser	Ala	Val	Leu	Leu	Asp	Thr	Ser	Ser	Asp	Trp	Gly	Ser	Ser	Gly	Ala
145				150						155					160

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Ala	Met	Ala	Thr	Glu	Thr	Cys	Phe	Leu	Gln	Asp	Tyr	Met	Gly	Val	Thr
				165					170					175	
Asp	Thr	Gly	Ser	Ser	Ser	Gln	Trp	Pro	Arg	Phe	Ser	Ser	Ser	Asp	Thr
			180					185					190		
Ile	Met	Ala	Ala	Ala	Ala	Ala	Arg	Ala	Ala	Thr	Thr	Arg	Ala	Pro	Glu
	195						200					205			
Thr	Leu	Pro	Leu	Phe	Pro	Thr	Cys	Gly	Asp	Asp	Gly	Gly	Ser	Gly	Ser
	210					215					220				
Ser	Ser	Tyr	Leu	Pro	Phe	Trp	Gly	Ala	Ala	Ser	Thr	Thr	Ala	Gly	Ala
225					230					235				240	
Thr	Ser	Ser	Val	Ala	Ile	Gln	Gln	Gln	His	Gln	Leu	Gln	Glu	Gln	Tyr
			245						250					255	
Ser	Phe	Tyr	Ser	Asn	Ser	Asn	Ser	Thr	Gln	Leu	Ala	Gly	Thr	Gly	Asn
			260					265					270		
Gln	Asp	Val	Ser	Ala	Thr	Ala	Ala	Ala	Ala	Ala	Ala	Leu	Glu	Leu	Ser
		275					280					285			
Leu	Ser	Ser	Trp	Cys	Ser	Pro	Tyr	Pro	Ala	Ala	Gly	Ser	Met		
	290					295					300				

<210> SEQ ID NO 2
 <211> LENGTH: 710
 <212> TYPE: PRT
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 2

Met	Ala	Thr	Val	Asn	Asn	Trp	Leu	Ala	Phe	Ser	Leu	Ser	Pro	Gln	Glu
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Leu	Pro	Pro	Ser	Gln	Thr	Thr	Asp	Ser	Thr	Leu	Ile	Ser	Ala	Ala	Thr
			20				25						30		
Ala	Asp	His	Val	Ser	Gly	Asp	Val	Cys	Phe	Asn	Ile	Pro	Gln	Asp	Trp
	35					40						45			
Ser	Met	Arg	Gly	Ser	Glu	Leu	Ser	Ala	Leu	Val	Ala	Glu	Pro	Lys	Leu
	50				55					60					
Glu	Asp	Phe	Leu	Gly	Gly	Ile	Ser	Phe	Ser	Glu	Gln	His	His	Lys	Ala
65				70					75					80	
Asn	Cys	Asn	Met	Ile	Pro	Ser	Thr	Ser	Ser	Thr	Val	Cys	Tyr	Ala	Ser
		85						90						95	
Ser	Gly	Ala	Ser	Thr	Gly	Tyr	His	His	Gln	Leu	Tyr	His	Gln	Pro	Thr
		100					105						110		
Ser	Ser	Ala	Leu	His	Phe	Ala	Asp	Ser	Val	Met	Val	Ala	Ser	Ser	Ala
		115					120					125			
Gly	Val	His	Asp	Gly	Gly	Ala	Met	Leu	Ser	Ala	Ala	Ala	Asn	Gly	
	130				135						140				
Val	Ala	Gly	Ala	Ala	Ser	Ala	Asn	Gly	Gly	Gly	Ile	Gly	Leu	Ser	Met
145				150					155					160	
Ile	Lys	Asn	Trp	Leu	Arg	Ser	Gln	Pro	Ala	Pro	Met	Gln	Pro	Arg	Val
			165					170						175	
Ala	Ala	Ala	Glu	Gly	Ala	Gln	Gly	Leu	Ser	Leu	Ser	Met	Asn	Met	Ala
			180					185					190		
Gly	Thr	Thr	Gln	Gly	Ala	Ala	Gly	Met	Pro	Leu	Leu	Ala	Gly	Glu	Arg
		195					200						205		
Ala	Arg	Ala	Pro	Glu	Ser	Val	Ser	Thr	Ser	Ala	Gln	Gly	Gly	Ala	Val
	210					215						220			
Val	Val	Thr	Ala	Pro	Lys	Glu	Asp	Ser	Gly	Gly	Ser	Gly	Val	Ala	Gly
225					230					235				240	

Ala	Leu	Val	Ala	Val	Ser	Thr	Asp	Thr	Gly	Gly	Ser	Gly	Gly	Ala	Ser	
				245					250							
Ala	Asp	Asn	Thr	Ala	Arg	Lys	Thr	Val	Asp	Thr	Phe	Gly	Gln	Arg	Thr	
				260					265							
Ser	Ile	Tyr	Arg	Gly	Val	Thr	Arg	His	Arg	Trp	Thr	Gly	Arg	Tyr	Glu	
				275					280							
Ala	His	Leu	Trp	Asp	Asn	Ser	Cys	Arg	Arg	Glu	Gly	Gln	Thr	Arg	Lys	
				290					295							
Gly	Arg	Gln	Val	Tyr	Leu	Gly	Gly	Tyr	Asp	Lys	Glu	Glu	Lys	Ala	Ala	
				305					310							
Arg	Ala	Tyr	Asp	Leu	Ala	Ala	Leu	Lys	Tyr	Trp	Gly	Ala	Thr	Thr	Thr	
				325					330							
Thr	Asn	Phe	Pro	Val	Ser	Asn	Tyr	Glu	Lys	Glu	Leu	Glu	Asp	Met	Lys	
				340					345							
His	Met	Thr	Arg	Gln	Glu	Phe	Val	Ala	Ser	Leu	Arg	Arg	Lys	Ser	Ser	
				355					360							
Gly	Phe	Ser	Arg	Gly	Ala	Ser	Ile	Tyr	Arg	Gly	Val	Thr	Arg	His	His	
				370					375							
Gln	His	Gly	Arg	Trp	Gln	Ala	Arg	Ile	Gly	Arg	Val	Ala	Gly	Asn	Lys	
				385					390							
Asp	Leu	Tyr	Leu	Gly	Thr	Phe	Ser	Thr	Gln	Glu	Glu	Ala	Ala	Glu	Ala	
				405					410							
Tyr	Asp	Ile	Ala	Ala	Ile	Lys	Phe	Arg	Gly	Leu	Asn	Ala	Val	Thr	Asn	
				420					425							
Phe	Asp	Met	Ser	Arg	Tyr	Asp	Val	Lys	Ser	Ile	Leu	Asp	Ser	Ser	Ala	
				435					440							
Leu	Pro	Ile	Gly	Ser	Ala	Ala	Lys	Arg	Leu	Lys	Glu	Ala	Glu	Ala	Ala	
				450					455							
Ala	Ser	Ala	Gln	His	His	His	Ala	Gly	Val	Val	Ser	Tyr	Asp	Val	Gly	
				465					470							
Arg	Ile	Ala	Ser	Gln	Leu	Gly	Asp	Gly	Gly	Ala	Leu	Ala	Ala	Ala	Tyr	
				485					490							
Gly	Ala	His	Tyr	His	Gly	Ala	Ala	Trp	Pro	Thr	Ile	Ala	Phe	Gln	Pro	
				500					505							
Gly	Ala	Ala	Ser	Thr	Gly	Leu	Tyr	His	Pro	Tyr	Ala	Gln	Gln	Pro	Met	
				515					520							
Arg	Gly	Gly	Gly	Trp	Cys	Lys	Gln	Glu	Gln	Asp	His	Ala	Val	Ile	Ala	
				530					535							
Ala	Ala	His	Ser	Leu	Gln	Asp	Leu	His	His	Leu	Asn	Leu	Gly	Ala	Ala	
				545					550							
Gly	Ala	His	Asp	Phe	Phe	Ser	Ala	Gly	Gln	Gln	Ala	Ala	Ala	Ala	Ala	
				565					570							
Met	His	Gly	Leu	Gly	Ser	Ile	Asp	Ser	Ala	Ser	Leu	Glu	His	Ser	Thr	
				580					585							
Gly	Ser	Asn	Ser	Val	Val	Tyr	Asn	Gly	Gly	Val	Gly	Asp	Ser	Asn	Gly	
				595					600							
Ala	Ser	Ala	Val	Gly	Gly	Ser	Gly	Gly	Gly	Tyr	Met	Met	Pro	Met	Ser	
				610					615							
Ala	Ala	Gly	Ala	Thr	Thr	Thr	Ser	Ala	Met	Val	Ser	His	Glu	Gln	Val	
				625					630							
His	Ala	Arg	Ala	Tyr	Asp	Glu	Ala	Lys	Gln	Ala	Ala	Gln	Met	Gly	Tyr	
				645					650							

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Glu Ser Tyr Leu Val Asn Ala Glu Asn Asn Gly Gly Gly Arg Met Ser
660 665 670

Ala Trp Gly Thr Val Val Ser Ala Ala Ala Ala Ala Ala Ser Ser
675 680 685

Asn Asp Asn Met Ala Ala Asp Val Gly His Gly Gly Ala Gln Leu Phe
690 695 700

Ser Val Trp Asn Asp Thr
705 710

<210> SEQ ID NO 3
 <211> LENGTH: 564
 <212> TYPE: PRT
 <213> ORGANISM: Zea mays

<400> SEQUENCE: 3

Met Ala Ala Ala Glu Ala Arg Arg Arg Ser Ala Val Trp Ala Leu Leu
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Pro Leu Leu Leu Arg Leu Leu His Pro Ala Ala Leu Val Leu Ala Asn
20 25 30

Thr Glu Gly Asp Ala Leu His Ser Leu Arg Thr Asn Leu Asn Asp Pro
35 40 45

Asn Asn Val Leu Gln Ser Trp Asp Pro Thr Leu Val Asn Pro Cys Thr
50 55 60

Trp Phe His Val Thr Cys Asn Asn Asp Asn Ser Val Ile Arg Val Asp
65 70 75 80

Leu Gly Asn Ala Ala Leu Ser Gly Thr Leu Val Pro Gln Leu Gly Gln
85 90 95

Leu Lys Asn Leu Gln Tyr Leu Glu Leu Tyr Ser Asn Asn Ile Ser Gly
100 105 110

Thr Ile Pro Ser Glu Leu Gly Asn Leu Thr Asn Leu Val Ser Leu Asp
115 120 125

Leu Tyr Leu Asn Asn Phe Thr Gly Pro Ile Pro Asp Ser Leu Gly Lys
130 135 140

Leu Leu Lys Leu Arg Phe Leu Arg Leu Asn Asn Ser Leu Ser Gly
145 150 155 160

Ser Ile Pro Lys Ser Leu Thr Ala Ile Thr Ala Leu Gln Val Leu Asp
165 170 175

Leu Ser Asn Asn Asn Leu Ser Gly Glu Val Pro Ser Thr Gly Ser Phe
180 185 190

Ser Leu Phe Thr Pro Ile Ser Phe Gly Asn Asn Pro Asn Leu Cys Gly
195 200 205

Pro Gly Thr Thr Lys Pro Cys Pro Gly Ala Pro Pro Phe Ser Pro Pro
210 215 220

Pro Pro Tyr Asn Pro Thr Thr Pro Val Gln Ser Pro Gly Ser Ser Ser
225 230 235 240

Ser Ser Thr Gly Ala Ile Ala Gly Gly Val Ala Ala Gly Ala Ala Leu
245 250 255

Leu Phe Ala Ile Pro Ala Ile Ser Phe Ala Tyr Trp Arg Arg Arg Lys
260 265 270

Pro Gln Glu His Phe Phe Asp Val Pro Ala Glu Glu Asp Pro Glu Val
275 280 285

His Leu Gly Gln Leu Lys Arg Phe Ser Leu Arg Glu Leu Gln Val Ala
290 295 300

Thr Asp Gly Phe Ser Asn Lys Asn Ile Leu Gly Arg Gly Gly Phe Gly
305 310 315 320

Lys	Val	Tyr	Lys	Gly	Arg	Leu	Ala	Asp	Gly	Ser	Leu	Val	Ala	Val	Lys	
				325					330					335		
Arg	Leu	Lys	Glu	Glu	Arg	Thr	Pro	Gly	Gly	Glu	Leu	Gln	Phe	Gln	Thr	
				340					345					350		
Glu	Val	Glu	Met	Ile	Ser	Met	Ala	Val	His	Arg	Asn	Leu	Leu	Arg	Leu	
				355					360					365		
Arg	Gly	Phe	Cys	Met	Thr	Pro	Thr	Glu	Arg	Leu	Leu	Val	Tyr	Pro	Tyr	
				370					375					380		
Met	Ala	Asn	Gly	Ser	Val	Ala	Ser	Arg	Leu	Arg	Asp	Arg	Pro	Pro	Ala	
				385					390					395		
Glu	Pro	Pro	Leu	Asp	Trp	Gln	Thr	Arg	Gln	Arg	Ile	Ala	Leu	Gly	Ser	
				405					410					415		
Ala	Arg	Gly	Leu	Ser	Tyr	Leu	His	Asp	His	Cys	Asp	Pro	Lys	Ile	Ile	
				420					425					430		
His	Arg	Asp	Val	Lys	Ala	Ala	Asn	Ile	Leu	Leu	Asp	Glu	Asp	Phe	Glu	
				435					440					445		
Ala	Val	Val	Gly	Asp	Phe	Gly	Leu	Ala	Lys	Leu	Met	Asp	Tyr	Lys	Asp	
				450					455					460		
Thr	His	Val	Thr	Thr	Ala	Val	Arg	Gly	Thr	Ile	Gly	His	Ile	Ala	Pro	
				465					470					475		
Glu	Tyr	Leu	Ser	Thr	Gly	Lys	Ser	Ser	Glu	Lys	Thr	Asp	Val	Phe	Gly	
				485					490					495		
Tyr	Gly	Ile	Thr	Leu	Leu	Glu	Leu	Ile	Thr	Gly	Gln	Arg	Ala	Phe	Asp	
				500					505					510		
Leu	Ala	Arg	Leu	Ala	Asn	Asp	Asp	Asp	Val	Met	Leu	Leu	Asp	Trp	Val	
				515					520					525		
Val	Asn	Cys	Ala	Ile	Ser	Ser	Val	Gln	Ile	Val	Pro	Gln	Val	Phe	Met	
				530					535					540		
Pro	Phe	Leu	Trp	Leu	Pro	Tyr	Cys	Ala	Gly	Ala	Gly	Ala	His	Leu	Phe	
				545					550					555		
Leu	Cys	Asn	Cys													

<400> SEQUENCE: 4

Met	Ala	Met	Pro	Tyr	Ala	Ser	Leu	Ser	Pro	Ala	Gly	Asp	Arg	Arg	Ser
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Ser	Pro	Ala	Ala	Thr	Ala	Thr	Ala	Ser	Leu	Leu	Pro	Phe	Cys	Arg	Ser
			20					25					30		
Ser	Pro	Phe	Ser	Ala	Gly	Gly	Asn	Gly	Gly	Met	Gly	Glu	Glu	Ala	Arg
		35					40					45			
Met	Asp	Gly	Arg	Trp	Met	Ala	Arg	Pro	Val	Pro	Phe	Thr	Ala	Ala	Gln
	50					55					60				
Tyr	Glu	Glu	Leu	Glu	His	Gln	Ala	Leu	Ile	Tyr	Lys	Tyr	Leu	Val	Ala
65					70				75						80
Gly	Val	Ser	Val	Pro	Pro	Asp	Leu	Val	Leu	Pro	Ile	Arg	Arg	Gly	Ile
			85						90					95	
Glu	Ser	Leu	Ala	Ala	Arg	Phe	Tyr	His	Asn	Pro	Leu	Ala	Ile	Gly	Tyr
			100					105					110		
Gly	Ser	Tyr	Leu	Gly	Lys	Lys	Val	Asp	Pro	Glu	Pro	Gly	Arg	Cys	Arg
		115					120					125			

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Arg Thr Asp Gly Lys Lys Trp Arg Cys Ala Lys Glu Ala Ala Ser Asp
 130                               135               140

Ser Lys Tyr Cys Glu Arg His Met His Arg Gly Arg Asn Arg Ser Arg
 145                               150               155               160

Lys Pro Val Glu Thr Gln Leu Val Ser His Ser Gln Pro Pro Ala Ala
                165                               170               175

Ser Val Val Pro Pro Leu Ala Thr Gly Phe His Asn His Ser Leu Tyr
                180                               185               190

Pro Ala Ile Gly Gly Thr Asn Gly Gly Gly Gly Gly Gly Asn Asn Gly
                195                               200               205

Met Ser Met Pro Gly Thr Phe Ser Ser Ala Leu Gly Pro Pro Gln Gln
 210                               215               220

His Met Gly Asn Asn Ala Ala Ser Pro Tyr Ala Ala Leu Gly Gly Ala
 225                               230               235               240

Gly Thr Cys Lys Asp Phe Arg Tyr Thr Ala Tyr Gly Ile Arg Ser Leu
                245                               250               255

Ala Asp Glu Gln Ser Gln Leu Met Thr Glu Ala Met Asn Thr Ser Val
                260                               265               270

Glu Asn Pro Trp Arg Leu Pro Pro Ser Ser Gln Thr Thr Thr Phe Pro
 275                               280               285

Leu Ser Ser Tyr Ser Pro Gln Leu Gly Ala Thr Ser Asp Leu Gly Gln
 290                               295               300

Asn Asn Ser Ser Asn Asn Asn Ser Gly Val Lys Ala Glu Gly Gln Gln
 305                               310               315               320

Gln Gln Gln Pro Leu Ser Phe Pro Gly Cys Gly Asp Phe Gly Ser Gly
                325                               330               335

Asp Ser Ala Lys Gln Glu Asn Gln Thr Leu Arg Pro Phe Phe Asp Glu
                340                               345               350

Trp Pro Lys Thr Arg Asp Ser Trp Ser Asp Leu Thr Asp Asp Asn Ser
 355                               360               365

Asn Val Ala Ser Phe Ser Ala Thr Gln Leu Ser Ile Ser Ile Pro Met
 370                               375               380

Thr Ser Ser Asp Phe Ser Ala Ala Ser Ser Gln Ser Pro Asn Gly Met
 385                               390               395               400

Leu Phe Ala Gly Glu Met Tyr
                405

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<210> SEQ ID NO 5

<211> LENGTH: 407

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mir396 binding site mutated TaGRF4 amino acid sequence

<400> SEQUENCE: 5

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Met Ala Met Pro Tyr Ala Ser Leu Ser Pro Ala Gly Asp Arg Arg Ser
 1                               5               10               15

Ser Pro Ala Ala Thr Ala Thr Ala Ser Leu Leu Pro Phe Cys Arg Ser
                20                               25               30

Ser Pro Phe Ser Ala Gly Gly Asn Gly Gly Met Gly Glu Glu Ala Arg
 35                               40               45

Met Asp Gly Arg Trp Met Ala Arg Pro Val Pro Phe Thr Ala Ala Gln
 50                               55               60

Tyr Glu Glu Leu Glu His Gln Ala Leu Ile Tyr Lys Tyr Leu Val Ala

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65	70	75	80
Gly Val Ser Val Pro Pro Asp Leu Val Leu Pro Ile Arg Arg Gly Ile	85	90	95
Glu Ser Leu Ala Ala Arg Phe Tyr His Asn Pro Leu Ala Ile Gly Tyr	100	105	110
Gly Ser Tyr Leu Gly Lys Lys Val Asp Pro Glu Pro Gly Arg Cys Arg	115	120	125
Arg Thr Asp Gly Lys Lys Trp Arg Cys Ala Lys Glu Ala Ala Ser Asp	130	135	140
Ser Lys Tyr Cys Glu Arg His Met His Arg Gly Arg Asn Arg Ser Arg	145	150	155
Lys Pro Val Glu Thr Gln Leu Val Ser His Ser Gln Pro Pro Ala Ala	165	170	175
Ser Val Val Pro Pro Leu Ala Thr Gly Phe His Asn His Ser Leu Tyr	180	185	190
Pro Ala Ile Gly Gly Thr Asn Gly Gly Gly Gly Gly Gly Asn Asn Gly	195	200	205
Met Ser Met Pro Gly Thr Phe Ser Ser Ala Leu Gly Pro Pro Gln Gln	210	215	220
His Met Gly Asn Asn Ala Ala Ser Pro Tyr Ala Ala Leu Gly Gly Ala	225	230	235
Gly Thr Cys Lys Asp Phe Arg Tyr Thr Ala Tyr Gly Ile Arg Ser Leu	245	250	255
Ala Asp Glu Gln Ser Gln Leu Met Thr Glu Ala Met Asn Thr Ser Val	260	265	270
Glu Asn Pro Trp Arg Leu Pro Pro Ser Ser Gln Thr Thr Thr Phe Pro	275	280	285
Leu Ser Ser Tyr Ser Pro Gln Leu Gly Ala Thr Ser Asp Leu Gly Gln	290	295	300
Asn Asn Ser Ser Asn Asn Asn Ser Gly Val Lys Ala Glu Gly Gln Gln	305	310	315
Gln Gln Gln Pro Leu Ser Phe Pro Gly Cys Gly Asp Phe Gly Ser Gly	325	330	335
Asp Ser Ala Lys Gln Glu Asn Gln Thr Leu Arg Pro Phe Phe Asp Glu	340	345	350
Trp Pro Lys Thr Arg Asp Ser Trp Ser Asp Leu Thr Asp Asp Asn Ser	355	360	365
Asn Val Ala Ser Phe Ser Ala Thr Gln Leu Ser Ile Ser Ile Pro Met	370	375	380
Thr Ser Ser Asp Phe Ser Ala Ala Ser Ser Gln Ser Pro Asn Gly Met	385	390	395
Leu Phe Ala Gly Glu Met Tyr	405		

<210> SEQ ID NO 6

<211> LENGTH: 225

<212> TYPE: PRT

<213> ORGANISM: Triticum aestivum

<400> SEQUENCE: 6

Met Gln Gln Gln His Leu Met Gln Met Asn Gln Ser Met Met Gly Gly	1	5	10	15
Tyr Ala Ser Ser Thr Thr Ala Thr Thr Asp Leu Ile Gln Gln Tyr Leu	20	25	30	

Leu	Thr	Pro	Ser	Ser	Arg	Ser	Ser	Cys	Met	Asp	Gly	Gly	Phe	Asn	Leu
			180					185					190		
Arg	Leu	Ala	Asn	Ser	Thr	Asp	Pro	Glu	Pro	Gly	Arg	Cys	Arg	Arg	Thr
		195					200					205			
Asp	Gly	Lys	Lys	Trp	Arg	Cys	Ser	Arg	Asp	Val	Ala	Pro	Asn	His	Lys
	210					215					220				
Tyr	Cys	Glu	Arg	His	Met	His	Arg	Gly	Arg	Pro	Arg	Ser	Arg	Lys	Pro
225					230					235					240
Val	Glu	Val	Asn	Thr	Asn	Ser	Thr	Thr	Thr	Pro	Thr	Ser	Val	Asn	Asn
				245					250					255	
Asn	Asn	His	Gln	Ile	Lys	Lys	Ala	Arg	His	Glu	Cys	Asn	Asn	Asn	Pro
			260					265					270		
Phe	Ala	Thr	Pro	Asp	Val	Thr	Ala	Ala	Ile	Ser	Asn	Pro	Thr	Ser	Arg
		275					280					285			
Lys	Asn	Gly	Ser	Ser	Pro	His	Phe	Leu	Gly	Ser	Thr	Thr	Thr	Gln	Pro
	290					295					300				
Tyr	Leu	Asp	Ser	Ser	Leu	Ser	Leu	Asp	Asn	Phe	Gly	Leu	Lys	Ala	Ala
305					310					315					320
Ser	Phe	Asp	Ser	Val	Ala	Ser	Val	Ser	Ala	Asn	Lys	Glu	Pro	Arg	Gly
				325					330					335	
Leu	Glu	Trp	Met	Leu	Asn	Gly	Asp	Pro	Ile	Ser	Leu	Gly	Ala	Ser	Asp
			340					345					350		
Ser	Gln	Trp	Gln	Ser	Leu	Met	His	Asn	Lys	Asp	Gly	Met	Thr	Ser	Val
		355					360					365			
Ser	Ser	Cys	Asn	Thr	Thr	Glu	Ser	Gln	Tyr	Leu	Asn	Ser	Leu	Ala	Leu
		370				375					380				
Tyr	Asn	Ser	Gly	Leu	Glu	Gln	Gln	Asn	Arg	Arg	His	Pro	Leu	Phe	Leu
385					390					395					400
Asn	Pro	Leu	Val	Val	Pro	Met	Glu	Asn	Leu	Gln	Pro	Glu	Lys	Pro	Arg
			405						410					415	
Gly	Phe	Ile	Asp	Ala	Trp	Ser	Asn	Ala	Glu	Ser	Asn	Ala	Asn	Thr	Asn
			420					425					430		
Thr	Thr	Asn	Lys	Asn	Ser	Ala	Ala	Ser	Ile	Gly	Lys	Leu	Ser	Leu	Ser
		435					440					445			
Ser	Leu	Asp	Leu	Ser	Met	Gly	Gly	Ala	Ala	Val	Asn	Glu	Asp	Val	Gly
		450				455					460				
Asn	Val	Asn	Met	Gly	Leu	Gly	Leu	Met	Glu	Pro	Asn	Gly	Lys	Thr	His
465					470					475					480
Thr	Gly	Thr	Lys	Ile	Ser	Leu	Ser	Asn	Trp	Gln	Asn	Pro	Ala	Pro	Trp
			485						490					495	
Val	Ala	Ser	Ser	Leu	Gly	Gly	Pro	Leu	Ala	Glu	Val	Leu	Arg	Ser	Ser
			500					505					510		
Thr	Val	Thr	Ala	Thr	Thr	Thr	Thr	Asn	Glu	Ala	Thr	Ser	Asn	Thr	Pro
		515					520					525			
Ser</															

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<210> SEQ ID NO 8
<211> LENGTH: 345
<212> TYPE: PRT
<213> ORGANISM: Glycine max

<400> SEQUENCE: 8

Met Met Ser Ala Ser Ala Arg Asn Arg Ser Pro Phe Thr Gln Thr Gln
 1             5             10             15

Trp Gln Glu Leu Glu His Gln Ala Leu Val Phe Lys Tyr Met Val Thr
 20             25             30

Gly Thr Pro Ile Pro Pro Asp Leu Ile Tyr Ser Ile Lys Arg Ser Leu
 35             40             45

Asp Thr Ser Ile Ser Ser Arg Leu Phe Pro His His Pro Ile Gly Trp
 50             55             60

Gly Cys Phe Glu Met Gly Phe Gly Arg Lys Val Asp Pro Glu Pro Gly
 65             70             75             80

Arg Cys Arg Arg Thr Asp Gly Lys Lys Trp Arg Cys Ser Lys Glu Ala
 85             90             95

Tyr Pro Asp Ser Lys Tyr Cys Glu Arg His Met His Arg Gly Arg Asn
100            105            110

Arg Ser Arg Lys Pro Val Glu Val Ser Ser Ala Ile Ser Thr Ala Thr
115            120            125

Asn Thr Ser Gln Thr Ile Pro Ser Ser Tyr Thr Arg Asn Leu Ser Leu
130            135            140

Thr Asn Pro Asn Met Thr Pro Pro Ser Ser Phe Pro Phe Ser Pro Leu
145            150            155            160

Pro Ser Ser Met Pro Ile Glu Ser Gln Pro Phe Ser Gln Ser Tyr Gln
165            170            175

Asn Ser Ser Leu Asn Pro Phe Phe Tyr Ser Gln Ser Thr Ser Ser Arg
180            185            190

Pro Pro Asp Ala Asp Phe Pro Pro Gln Asp Ala Thr Thr His Gln Leu
195            200            205

Phe Met Asp Ser Gly Ser Tyr Ser His Asp Glu Lys Asn Tyr Arg His
210            215            220

Val His Gly Ile Arg Glu Asp Val Asp Glu Arg Ala Phe Phe Pro Glu
225            230            235            240

Ala Ser Gly Ser Ala Arg Ser Tyr Thr Glu Ser Tyr Gln Gln Leu Ser
245            250            255

Met Ser Ser Tyr Lys Ser Tyr Ser Asn Ser Asn Phe Gln Asn Ile Asn
260            265            270

Asp Ala Thr Thr Asn Pro Arg Gln Gln Glu Gln Gln Gln Gln His
275            280            285

Cys Phe Val Leu Gly Thr Asp Phe Lys Ser Thr Arg Pro Thr Lys Glu
290            295            300

Lys Glu Ala Glu Thr Ala Thr Gly Gln Arg Pro Leu His Arg Phe Phe
305            310            315            320

Gly Glu Trp Pro Pro Lys Asn Thr Thr Asp Ser Trp Leu Asp Leu Ala
325            330            335

Ser Asn Ser Arg Ile Gln Thr Asp Glu
340            345

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<210> SEQ ID NO 9
<211> LENGTH: 341
<212> TYPE: PRT
<213> ORGANISM: Glycine max

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<400> SEQUENCE: 9

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Met Asn Asn Ser Ser Gly Gly Gly Gly Arg Gly Thr Leu Met Gly Leu
1      5      10      15
Ser Asn Gly Tyr Cys Gly Arg Ser Pro Phe Thr Val Ser Gln Trp Gln
20     25     30
Glu Leu Glu His Gln Ala Leu Ile Phe Lys Tyr Met Leu Ala Gly Leu
35     40     45
Pro Val Pro Leu Asp Leu Val Phe Pro Ile Gln Asn Ser Phe His Ser
50     55     60
Thr Ile Ser Leu Ser His Ala Phe Phe His His Pro Thr Leu Ser Tyr
65     70     75     80
Cys Ser Phe Tyr Gly Lys Lys Val Asp Pro Glu Pro Gly Arg Cys Arg
85     90     95
Arg Thr Asp Gly Lys Lys Trp Arg Cys Ser Lys Glu Ala Tyr Pro Asp
100    105    110
Ser Lys Tyr Cys Glu Arg His Met His Arg Gly Arg Asn Arg Ser Arg
115    120    125
Lys Pro Val Glu Ser Gln Thr Met Thr His Ser Ser Ser Thr Val Thr
130    135    140
Ser Leu Thr Val Thr Gly Gly Ser Gly Ala Ser Lys Gly Thr Val Asn
145    150    155    160
Phe Gln Asn Leu Ser Thr Asn Thr Phe Gly Asn Leu Gln Gly Thr Asp
165    170    175
Ser Gly Thr Asp His Thr Asn Tyr His Leu Asp Ser Ile Pro Tyr Ala
180    185    190
Ile Pro Ser Lys Glu Tyr Arg Tyr Val Gln Gly Leu Lys Ser Glu Gly
195    200    205
Gly Glu His Cys Phe Phe Ser Glu Ala Ser Gly Ser Asn Lys Val Leu
210    215    220
Gln Met Glu Ser Gln Leu Glu Asn Thr Trp Pro Leu Met Ser Thr Arg
225    230    235    240
Val Ala Ser Phe Ser Thr Ser Lys Ser Ser Asn Asp Ser Leu Leu His
245    250    255
Ser Asp Tyr Pro Arg His Ser Phe Leu Ser Gly Glu Tyr Val Ser Gly
260    265    270
Glu His Val Lys Glu Glu Gly Gln Pro Leu Arg Pro Phe Phe Asn Glu
275    280    285
Trp Pro Lys Ser Arg Glu Ser Trp Ser Gly Leu Glu Asp Glu Arg Ser
290    295    300
Asn Gln Thr Ala Phe Ser Thr Thr Gln Leu Ser Ile Ser Ile Pro Met
305    310    315    320
Ser Ser Asn Phe Ser Ala Thr Ser Ser Gln Ser Pro His Gly Glu Asp
325    330    335
Glu Ile Gln Phe Arg
340

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<210> SEQ ID NO 10

<211> LENGTH: 600

<212> TYPE: PRT

<213> ORGANISM: Glycine max

<400> SEQUENCE: 10

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Met Asp Leu Gly Val Val Gly Leu Glu Gly Val Val Gly Ser Glu Ser
1      5      10      15

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Gly	Cys	Val	Phe	Gly	Ser	Ser	Leu	Val	Ser	Asp	Pro	Glu	Thr	Lys	His	
			20					25					30			
Lys	Trp	Tyr	Gly	Ser	Gly	Leu	Leu	Lys	Gln	Glu	Arg	Ser	Ala	Ile	Ala	
		35					40					45				
Thr	Glu	Asp	Asp	Glu	Trp	Arg	Ile	Ser	Lys	Val	Ala	Lys	Thr	Asp	His	
		50					55					60				
Asp	Met	Ser	Ser	Ala	Ser	Lys	Ala	Met	Leu	Phe	Gln	Gln	Arg	Asn	Asn	
		65					70					75		80		
Ser	Leu	Leu	Arg	Ser	Asn	Asn	Ala	Thr	Leu	Phe	Ser	Asp	Gly	His	His	
				85					90					95		
Gln	Ser	Gln	Met	Leu	Ser	Phe	Ser	Ser	Pro	Lys	Ser	Asp	Ser	Leu	Leu	
		100					105					110				
Ile	Asp	Lys	Ala	Ser	Ser	Asn	Ala	Thr	Leu	Pro	Phe	Ser	Ser	His	Gln	
		115					120					125				
Leu	Ser	Ser	Tyr	Thr	Arg	Asn	Thr	Gly	Tyr	Asn	Ser	Gly	Ser	Ile	Ser	
		130					135					140				
Met	His	Gly	Ala	Leu	Ala	Ser	Val	Arg	Gly	Pro	Phe	Thr	Pro	Ser	Gln	
		145					150					155		160		
Trp	Met	Glu	Leu	Glu	His	Gln	Ala	Leu	Ile	Tyr	Lys	Tyr	Ile	Thr	Ala	
		165					170					175				
Asn	Val	Pro	Val	Pro	Thr	His	Leu	Leu	Ile	Pro	Ile	Arg	Lys	Ala	Leu	
		180					185					190				
Asp	Ser	Val	Gly	Phe	Cys	Asn	Phe	Ser	Ala	Gly	Leu	Leu	Arg	Pro	Asn	
		195					200					205				
Ser	Leu	Gly	Trp	Gly	Gly	Phe	His	Leu	Gly	Phe	Ser	Asn	Asn	Thr	Asp	
		210					215					220				
Pro	Glu	Pro	Gly	Arg	Cys	Arg	Arg	Thr	Asp	Gly	Lys	Lys	Trp	Arg	Cys	
		225					230					235		240		
Ser	Arg	Asp	Ala	Val	Val	Asp	Gln	Lys	Tyr	Cys	Glu	Arg	His	Met	Asn	
		245					250					255				
Arg	Gly	Arg	His	Arg	Ser	Arg	Lys	Pro	Val	Glu	Gly	Gln	Ser	Gly	His	
		260					265					270				
Ala	Leu	Thr	Thr	Thr	Thr	Ser	Asn	Thr	Pro	Asn	Ala	Ser	Ser	Asn	Ser	
		275					280					285				
Val	Val	Pro	Gly	Asn	Asn	Asn	Asn	Thr	Phe	Ala	His	Asn	Asn	Val	His	
		290					295					300				
His	Pro	Ile	Pro	Pro	His	Ser	Ser	Pro	Val	Asn	Thr	Ile	Thr	Arg	Met	
		305					310					315		320		
Phe	Thr	Ser	Asn	Lys	Glu	Asn	Asn	Asn	Ser	Thr	Ser	Glu	Arg	Met	Gln	
		325					330					335				
Asp	Pro	Ala	Leu	Pro	Met	Leu	Pro	Pro	Thr	Leu	Glu	Leu	Lys	Pro	Lys	
		340					345					350				
Glu	Asn	Asn	Pro	Phe	Met	Ile	His	Lys	His	Gln	Ile	Pro	Ser	Asp	Glu	
		355					360					365				
Tyr	Ser	Ser	Arg	Asn	Asn	Asn	Glu	Phe	Gly	Leu	Val	Thr	Ser	Asp	Ser	
		370					375					380				
Leu	Leu	Asn	Pro	Ser	Glu	Lys	Arg	Ser	Phe	Thr	Ser	Ser	Gln	Lys	Asn	
		385					390					395		400		
Asp	Ser	Ser	Glu													

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435	440	445
Ser Ile Pro Ile Ser Ser	Ser Asp His Phe Met	Ser Phe Thr Thr Ser
450	455	460
Leu Pro Ser Asn Glu Lys	Leu Thr Leu Ser Pro	Leu Arg Leu Ser Arg
465	470	475 480
Glu Leu Asp Pro Ile Gln Met	Gly Leu Gly Val Gly	Ser Ala Pro Asn
485	490	495
Glu Ala Asn Thr Arg Gln Ala	Asn Trp Ile Pro Ile	Thr Trp Glu Ser
500	505	510
Ser Met Gly Gly Pro Leu Gly	Glu Val Leu Asn Leu	Ser Asn Asn Asn
515	520	525
Asn Ser Asn Ala Ser Asp	Gln Cys Gly Lys Asn	Asn Asn Thr Ser
530	535	540
Ala Leu Asn Leu Met Lys Asp	Gly Trp Asp Asn Asn	Pro Pro Ser Gly
545	550	555 560
Ser Ser Pro Thr Gly Val Leu	Gln Lys Ser Ala Phe	Gly Ser Leu Ser
565	570	575
Asn Ser Ser Ala Gly Ser Ser	Pro Arg Gly Ala Glu	Asn Asn Lys Glu
580	585	590
Gly Ala Thr Leu Cys Asn Ala	Leu	
595	600	

<210> SEQ ID NO 11

<211> LENGTH: 257

<212> TYPE: PRT

<213> ORGANISM: Glycine max

<400> SEQUENCE: 11

Met Phe Gly Ser Leu Phe His	Phe Pro His Cys Leu Leu His	Val Leu
1	5 10	15
His Gln Ser Phe Tyr Asp Ala	Cys Phe Ser Gln Arg Tyr Leu	Arg Ser
20	25	30
His Phe Leu Thr Phe Gln Ile	Thr Met Ser Glu Ser Glu Tyr	Ala Leu
35	40	45
Phe Gly Asp His Lys Pro Ser	Pro Tyr Gln Asn Tyr Leu Asp	Glu Asn
50	55	60
Lys Ser Leu Ile Leu Lys Ile	Val Glu Ser Gln Asn Ser Gly	Lys Leu
65	70	75 80
Ser Glu Cys Ala Glu Asn Gln	Ala Arg Leu Gln Arg Asn Leu	Met Tyr
85	90	95
Leu Ala Ala Ile Ala Asp Ser	Gln Pro Gln Pro Pro Thr	Met Pro Gly
100	105	110
Gln Tyr Pro Pro Ser Gly Met	Met Gln Gln Gly Ala His	Tyr Met Gln
115	120	125
Ala Gln Gln Gln Thr Gln Gln	Met Ser Pro Gln Gln Leu	Met Ala Ala
130	135	140
Arg Ser Ser Leu Leu Tyr Ala	Gln Gln Pro Tyr Ser Ala	Leu Gln Gln
145	150	155 160
Gln Gln Ala Met His Ser Ala	Leu Gly Ser Ser Ser Gly	Leu His Met
165	170	175
Leu Gln Ser Glu Gly Ser Asn	Val Asn Val Gly Ser Gly	Ser Gly Ser
180	185	190
Val Gly Gly Gly Phe Pro Asp	Leu Val Arg Gly Gly Gly	Gly Gly Gly
195	200	205

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Gly Ser Thr Gly Glu Gly Leu His Ser Gly Gly Arg Gly Ile Met Gly
 210 215 220

Ser Ser Lys Gln Glu Ile Gly Gly Ser Ser Glu Gly Arg Gly Gly Gly
 225 230 235 240

Ser Ser Glu Gly Gly Glu Asn Leu Tyr Leu Lys Ile Ala Asp Asp Gly
 245 250 255

Asn

<210> SEQ ID NO 12
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: linkerA

<400> SEQUENCE: 12

Ala Ala Ala Ala
 1

<210> SEQ ID NO 13
 <211> LENGTH: 4
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: linkerS

<400> SEQUENCE: 13

Ser Gly Gly Ser
 1

<210> SEQ ID NO 14
 <211> LENGTH: 1701
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: A3A base editor amino acid sequence

<400> SEQUENCE: 14

Met Glu Ala Ser Pro Ala Ser Gly Pro Arg His Leu Met Asp Pro His
 1 5 10 15

Ile Phe Thr Ser Asn Phe Asn Asn Gly Ile Gly Arg His Lys Thr Tyr
 20 25 30

Leu Cys Tyr Glu Val Glu Arg Leu Asp Asn Gly Thr Ser Val Lys Met
 35 40 45

Asp Gln His Arg Gly Phe Leu His Asn Gln Ala Lys Asn Leu Leu Cys
 50 55 60

Gly Phe Tyr Gly Arg His Ala Glu Leu Arg Phe Leu Asp Leu Val Pro
 65 70 75 80

Ser Leu Gln Leu Asp Pro Ala Gln Ile Tyr Arg Val Thr Trp Phe Ile
 85 90 95

Ser Trp Ser Pro Cys Phe Ser Trp Gly Cys Ala Gly Glu Val Arg Ala
 100 105 110

Phe Leu Gln Glu Asn Thr His Val Arg Leu Arg Ile Phe Ala Ala Arg
 115 120 125

Ile Tyr Asp Tyr Asp Pro Leu Tyr Lys Glu Ala Leu Gln Met Leu Arg
 130 135 140

Asp Ala Gly Ala Gln Val Ser Ile Met Thr Tyr Asp Glu Phe Lys His
 145 150 155 160

Cys Trp Asp Thr Phe Val Asp His Gln Gly Cys Pro Phe Gln Pro Trp
 165 170 175

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Lys	Pro	Ile	Leu	Glu	Lys	Met	Asp	Gly	Thr	Glu	Glu	Leu	Leu	Val	Lys	
		595					600					605				
Leu	Asn	Arg	Glu	Asp	Leu	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	
	610					615					620					
Ser	Ile	Pro	His	Gln	Ile	His	Leu	Gly	Glu	Leu	His	Ala	Ile	Leu	Arg	
	625				630					635					640	
Arg	Gln	Glu	Asp	Phe	Tyr	Pro	Phe	Leu	Lys	Asp	Asn	Arg	Glu	Lys	Ile	
			645						650					655		
Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile	Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	
			660					665					670			
Arg	Gly	Asn	Ser	Arg	Phe	Ala	Trp	Met	Thr	Arg	Lys	Ser	Glu	Glu	Thr	
		675					680					685				
Ile	Thr	Pro	Trp	Asn	Phe	Glu	Glu	Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	
	690					695					700					
Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr	Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	
	705				710					715					720	
Glu	Lys	Val	Leu	Pro	Lys	His	Ser	Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	
			725						730					735		
Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys	Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	
		740						745					750			
Pro	Ala	Phe	Leu	Ser	Gly	Glu	Gln	Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	
		755					760					765				
Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr	Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	
	770					775					780					
Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp	Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	
	785				790					795					800	
Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly	Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	
			805						810					815		
Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp	Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	
		820						825					830			
Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr	Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	
		835					840					845				
Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala	His	Leu	Phe	Asp	Asp	Lys	Val	Met	
	850					855					860					
Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr	Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	
	865				870					875					880	
Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp	Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	
			885						890					895		
Asp	Phe	Leu	Lys	Ser	Asp	Gly	Phe	Ala	Asn	Arg	Asn	Phe	Met	Gln	Leu	
		900						905					910			
Ile	His	Asp	Asp	Ser	Leu	Thr	Phe	Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	
		915					920					925				
Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu	His	Glu	His	Ile	Ala	Asn	Leu	Ala	
	930					935					940					
Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly	Ile	Leu	Gln	Thr	Val	Lys	Val	Val	
	945				950					955					960	
Asp	Glu	Leu	Val	Lys	Val	Met	Gly	Arg	His	Lys	Pro	Glu	Asn	Ile	Val	
			965					970						975		
Ile	Glu	Met	Ala	Arg	Glu	Asn	Gln	Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	
		980					985						990			
Ser	Arg	Glu	Arg	Met	Lys	Arg	Ile	Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	
		995					1000					1005				
Ser	Gln	Ile	Leu	Lys	Glu	His	Pro	Val	Glu	Asn	Thr	Gln	Leu	Gln		

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1010	1015	1020
Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu Gln Asn Gly Arg Asp Met		
1025	1030	1035
Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg Leu Ser Asp Tyr Asp		
1040	1045	1050
Val Asp His Ile Val Pro Gln Ser Phe Leu Lys Asp Asp Ser Ile		
1055	1060	1065
Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg Gly Lys Ser		
1070	1075	1080
Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys Asn Tyr		
1085	1090	1095
Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys Phe		
1100	1105	1110
Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp		
1115	1120	1125
Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile		
1130	1135	1140
Thr Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys		
1145	1150	1155
Tyr Asp Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr		
1160	1165	1170
Leu Lys Ser Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe		
1175	1180	1185
Tyr Lys Val Arg Glu Ile Asn Asn Tyr His His Ala His Asp Ala		
1190	1195	1200
Tyr Leu Asn Ala Val Val Gly Thr Ala Leu Ile Lys Lys Tyr Pro		
1205	1210	1215
Lys Leu Glu Ser Glu Phe Val Tyr Gly Asp Tyr Lys Val Tyr Asp		
1220	1225	1230
Val Arg Lys Met Ile Ala Lys Ser Glu Gln Glu Ile Gly Lys Ala		
1235	1240	1245
Thr Ala Lys Tyr Phe Phe Tyr Ser Asn Ile Met Asn Phe Phe Lys		
1250	1255	1260
Thr Glu Ile Thr Leu Ala Asn Gly Glu Ile Arg Lys Arg Pro Leu		
1265	1270	1275
Ile Glu Thr Asn Gly Glu Thr Gly Glu Ile Val Trp Asp Lys Gly		
1280	1285	1290
Arg Asp Phe Ala Thr Val Arg Lys Val Leu Ser Met Pro Gln Val		
1295	1300	1305
Asn Ile Val Lys Lys Thr Glu Val Gln Thr Gly Gly Phe Ser Lys		
1310	1315	1320
Glu Ser Ile Leu Pro Lys Arg Asn Ser Asp Lys Leu Ile Ala Arg		
1325	1330	1335
Lys Lys Asp Trp Asp Pro Lys Lys Tyr Gly Gly Phe Asp Ser Pro		
1340	1345	1350
Thr Val Ala Tyr Ser Val Leu Val Val Ala Lys Val Glu Lys Gly		
1355	1360	1365
Lys Ser Lys Lys Leu Lys Ser Val Lys Glu Leu Leu Gly Ile Thr		
1370	1375	1380
Ile Met Glu Arg Ser Ser Phe Glu Lys Asn Pro Ile Asp Phe Leu		
1385	1390	1395
Glu Ala Lys Gly Tyr Lys Glu Val Lys Lys Asp Leu Ile Ile Lys		
1400	1405	1410

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Leu Pro Lys Tyr Ser Leu Phe Glu Leu Glu Asn Gly Arg Lys Arg
 1415 1420 1425
 Met Leu Ala Ser Ala Gly Glu Leu Gln Lys Gly Asn Glu Leu Ala
 1430 1435 1440
 Leu Pro Ser Lys Tyr Val Asn Phe Leu Tyr Leu Ala Ser His Tyr
 1445 1450 1455
 Glu Lys Leu Lys Gly Ser Pro Glu Asp Asn Glu Gln Lys Gln Leu
 1460 1465 1470
 Phe Val Glu Gln His Lys His Tyr Leu Asp Glu Ile Ile Glu Gln
 1475 1480 1485
 Ile Ser Glu Phe Ser Lys Arg Val Ile Leu Ala Asp Ala Asn Leu
 1490 1495 1500
 Asp Lys Val Leu Ser Ala Tyr Asn Lys His Arg Asp Lys Pro Ile
 1505 1510 1515
 Arg Glu Gln Ala Glu Asn Ile Ile His Leu Phe Thr Leu Thr Asn
 1520 1525 1530
 Leu Gly Ala Pro Ala Ala Phe Lys Tyr Phe Asp Thr Thr Ile Asp
 1535 1540 1545
 Arg Lys Arg Tyr Thr Ser Thr Lys Glu Val Leu Asp Ala Thr Leu
 1550 1555 1560
 Ile His Gln Ser Ile Thr Gly Leu Tyr Glu Thr Arg Ile Asp Leu
 1565 1570 1575
 Ser Gln Leu Gly Gly Asp Lys Arg Pro Ala Ala Thr Lys Lys Ala
 1580 1585 1590
 Gly Gln Ala Lys Lys Lys Lys Thr Arg Asp Ser Gly Gly Ser Thr
 1595 1600 1605
 Asn Leu Ser Asp Ile Ile Glu Lys Glu Thr Gly Lys Gln Leu Val
 1610 1615 1620
 Ile Gln Glu Ser Ile Leu Met Leu Pro Glu Glu Val Glu Glu Val
 1625 1630 1635
 Ile Gly Asn Lys Pro Glu Ser Asp Ile Leu Val His Thr Ala Tyr
 1640 1645 1650
 Asp Glu Ser Thr Asp Glu Asn Val Met Leu Leu Thr Ser Asp Ala
 1655 1660 1665
 Pro Glu Tyr Lys Pro Trp Ala Leu Val Ile Gln Asp Ser Asn Gly
 1670 1675 1680
 Glu Asn Lys Ile Lys Met Leu Ser Gly Gly Ser Pro Lys Lys Lys
 1685 1690 1695
 Arg Lys Val
 1700

<210> SEQ ID NO 15
 <211> LENGTH: 1384
 <212> TYPE: PRT
 <213> ORGANISM: Streptococcus pyogenes

<400> SEQUENCE: 15

Met Asp Lys Lys Tyr Ser Ile Gly Leu Asp Ile Gly Thr Asn Ser Val
 1 5 10 15

Gly Trp Ala Val Ile Thr Asp Glu Tyr Lys Val Pro Ser Lys Lys Phe
 20 25 30

Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile
 35 40 45

Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu

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50	55	60
Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys		
65	70	75 80
Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser		
	85	90 95
Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys		
	100	105 110
His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr		
	115	120 125
His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp		
	130	135 140
Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His		
	145	150 155 160
Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro		
	165	170 175
Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr		
	180	185 190
Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala		
	195	200 205
Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn		
	210	215 220
Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn		
	225	230 235 240
Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe		
	245	250 255
Asp Leu Ala Glu Asp Ala Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp		
	260	265 270
Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp		
	275	280 285
Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp		
	290	295 300
Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser		
	305	310 315 320
Met Ile Lys Arg Tyr Asp Glu His His Gln Asp Leu Thr Leu Leu Lys		
	325	330 335
Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe		
	340	345 350
Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser		
	355	360 365
Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile Leu Glu Lys Met Asp		
	370	375 380
Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg Glu Asp Leu Leu Arg		
	385	390 395 400
Lys Gln Arg Thr Phe Asp Asn Gly Ser Ile Pro His Gln Ile His Leu		
	405	410 415
Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu Asp Phe Tyr Pro Phe		
	420	425 430
Leu Lys Asp Asn Arg Glu Lys Ile Glu Lys Ile Leu Thr Phe Arg Ile		
	435	440 445
Pro Tyr Tyr Val Gly Pro Leu Ala Arg Gly Asn Ser Arg Phe Ala Trp		
	450	455 460
Met Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro Trp Asn Phe Glu Glu		
	465	470 475 480

Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr		
				485					490					495			
Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser		
				500					505					510			
Leu	Leu	Tyr	Glu	Tyr	Phe	Thr	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Lys		
				515					520					525			
Tyr	Val	Thr	Glu	Gly	Met	Arg	Lys	Pro	Ala	Phe	Leu	Ser	Gly	Glu	Gln		
				530					535					540			
Lys	Lys	Ala	Ile	Val	Asp	Leu	Leu	Phe	Lys	Thr	Asn	Arg	Lys	Val	Thr		
				545					550					555			
Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp		
				565					570					575			
Ser	Val	Glu	Ile	Ser	Gly	Val	Glu	Asp	Arg	Phe	Asn	Ala	Ser	Leu	Gly		
				580					585					590			
Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp		
				595					600					605			
Asn	Glu	Glu	Asn	Glu	Asp	Ile	Leu	Glu	Asp	Ile	Val	Leu	Thr	Leu	Thr		
				610					615					620			
Leu	Phe	Glu	Asp	Arg	Glu	Met	Ile	Glu	Glu	Arg	Leu	Lys	Thr	Tyr	Ala		
				625					630					635			
His	Leu	Phe	Asp	Asp	Lys	Val	Met	Lys	Gln	Leu	Lys	Arg	Arg	Arg	Tyr		
				645					650					655			
Thr	Gly	Trp	Gly	Arg	Leu	Ser	Arg	Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp		
				660					665					670			
Lys	Gln	Ser	Gly	Lys	Thr	Ile	Leu	Asp	Phe	Leu	Lys	Ser	Asp	Gly	Phe		
				675					680					685			
Ala	Asn	Arg	Asn	Phe	Met	Gln	Leu	Ile	His	Asp	Asp	Ser	Leu	Thr	Phe		
				690					695					700			
Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu		
				705					710					715			
His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly		
				725					730					735			
Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu	Val	Lys	Val	Met	Gly		
				740					745					750			
Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu	Met	Ala	Arg	Glu	Asn	Gln		
				755					760					765			
Thr	Thr	Gln	Lys	Gly	Gln	Lys	Asn	Ser	Arg	Glu	Arg	Met	Lys	Arg	Ile		
				770					775					780			
Glu	Glu	Gly	Ile	Lys	Glu	Leu	Gly	Ser	Gln	Ile	Leu	Lys	Glu	His	Pro		
				785					790					795			
Val	Glu	Asn	Thr	Gln	Leu	Gln	Asn	Glu	Lys	Leu	Tyr	Leu	Tyr	Tyr	Leu		
				805					810					815			
Gln	Asn	Gly	Arg	Asp	Met	Tyr	Val	Asp	Gln	Glu	Leu	Asp	Ile	Asn	Arg		
				820					825					830			
Leu	Ser	Asp	Tyr	Asp	Val	Asp	His	Ile	Val	Pro	Gln	Ser	Phe	Leu	Lys		
				835					840					845			
Asp	Asp	Ser	Ile	Asp	Asn	Lys	Val	Leu	Thr	Arg	Ser	Asp	Lys	Asn	Arg		
				850					855					860			
Gly	Lys	Ser	Asp	Asn	Val	Pro	Ser	Glu	Glu	Val	Val	Lys	Lys	Met	Lys		
				865					870								

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1295	1300	1305
Ile Ile His Leu Phe Thr Leu Thr Asn Leu Gly Ala Pro Ala Ala		
1310	1315	1320
Phe Lys Tyr Phe Asp Thr Thr Ile Asp Arg Lys Arg Tyr Thr Ser		
1325	1330	1335
Thr Lys Glu Val Leu Asp Ala Thr Leu Ile His Gln Ser Ile Thr		
1340	1345	1350
Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ser Gln Leu Gly Gly Asp		
1355	1360	1365
Lys Arg Pro Ala Ala Thr Lys Lys Ala Gly Gln Ala Lys Lys Lys		
1370	1375	1380

Lys

<210> SEQ ID NO 16
 <211> LENGTH: 76
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: sgRNA scaffold

<400> SEQUENCE: 16

gttttagagc tagaaatagc aagttaaaat aaggctagtc cgttatcaac ttgaaaaagt	60
ggcaccgagt cggtgc	76

<210> SEQ ID NO 17
 <211> LENGTH: 2517
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: GmGRF5-GmGIF1

<400> SEQUENCE: 17

atgggtgaat tatttggcgt tggaaaaaga aggaacatca gcagcagcaa caacaacaac	60
aacagttcta gtagtagcgt tcttgggttg gatgtgaagg tgcaacagag ccctgaagca	120
ttattccata ataggatgat gatgatggcg catcataatc atcaccaccg tccattgtca	180
tcacccttgg ataataatgg tgatggcgat ggtcccacga cttacatgtc ttttactaat	240
catataaacc ttgttagtgg tgcttcttct gttcttggtc ctgctattga tgctggctgt	300
ggtgtctgct ctcctgcacc tgtaagaact ttgcagcctt ttgacatttc ttcttatact	360
tcttctccca ccaccactac cacaaccttc aacttcaaac ccccttcagc aggtgtgatg	420
gcggcttcgt tgggggttcc ttccacaagt gcacaatgga gggagcttga aagacaagct	480
atgatataca agtacatgat ggcttctggt cctgttccac atgatctcct cacaccctct	540
tctcgtctct cctgcattga tgggtggttc aatctgaggt tggcaaatag cactgaccct	600
gagccaggta ggtgtagaag aacagatggt aaaaaatgga gatgttcaag agatgtggct	660
cctaaccaca agtactgtga gcgcccatatg catagaggcc gtccccgttc aagaaagcct	720
gtggaagtta acaccaacag caccaccact cccactagcg tcaacaataa caaccatcaa	780
atcaaaaaagg ctgcocatga gtgtaataat aatccttttg ctacacctga cgttactgcg	840
gctattttcca accccacatc cagaaaaaat ggatcttctc cccattttct tgggtctact	900
accactcagc cataccttga ttcttccctc tcccttgata actttggtct aaaagctgct	960
agttttgact ccgtggcttc tgtttctgct aataaggaac ccaggggttt agagtggatg	1020
ctgaatggag atcctatttc cctgggtgct tctgactcac aatggcagtc tctgatgcac	1080

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aataaagatg gaatgaccag tgttagttcc tgtaacacca ccgagtctca gtatctgaat	1140
tcattagcac tatataactc tggactagaa caacagaata gacgccatcc tttgttccctg	1200
aaccctcttg ttgttcccat ggaaaatctc caaccggaga aaccaagggg ttttattgat	1260
gcttggtcta acgctgaaag caatgccaac accaacacca ccaacaagaa ctctgctgca	1320
tcaattggta aattatccct ttcttctctt gatctatcaa tgggggggtgc tgctgtgaat	1380
gaagatgtgg gtaatgttaa catgggtttg ggctaattgg agcctaattg aaaaacgcac	1440
actggtacta aaatttctct ctccaattgg caaaaccag cacttgggt ggcttcac	1500
cttgggggtc cactagctga agttctaagg tcaagcacag tcaatgccac caccaccacc	1560
aatgaagcaa cctccaacac accctcgcca gccaccacta cacatgctga atctccatct	1620
ggggtgttgc agaaaacgct tgtttcattg tctgatagca gtaacaatag cagccaagg	1680
gttgcacat caagggccaa ttctgagatg gccttgctaa ggtttcaatc aaatgcggcc	1740
gtgcccattg tcgggtcttt gtccatttt ccacactgtc tactacatgt gctgcatcag	1800
agcttttatg acgcgtgctt ttctcagcgc tacctcagat cacattttct cactttcag	1860
atcacaatgt ctgaatccga atatgctttg ttggagatc acaaaccttc tccctatcaa	1920
aactatctgg atgagaacaa gtccttaatt ctgaagattg ttgaaagcca gaattcaggc	1980
aagcttagcg agtgtgccga gaaccaagca aggttcaga gaaatctcat gtacttagct	2040
gcaatagctg attctcaacc ccaaccaccc accatgcctg gtcagtaccc tccgagtggg	2100
atgatgcagc agggagcaca ctacatgcag gctcaacaac agacacagca gatgtcacca	2160
caacaactaa tggcggcacg ctgctccctt ttgtacgcac agcagccata ctacgactt	2220
caacagcagc aagccatgca cagtgcactc gggtcgagtt caggactcca catgctgcaa	2280
agtgaaggca gcaatgtgaa tgtgggatca ggcagtggct ctgtaggagg agggtttct	2340
gacttggtgc gcggtggtgg tgggtggcgt ggctcgacag gggagggttt gcacagtgg	2400
ggaaggggta tcattgggaag tagcaagcag gaaattggg gtccaagtga aggccgaggt	2460
gggggaagct cagaggtggt tgaaaacctt tacctcaaaa ttgccgacga tggaaac	2517

<210> SEQ ID NO 18

<211> LENGTH: 1818

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF6-GmGIF1

<400> SEQUENCE: 18

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gagcatcaag ctcttgtttt taagtacatg gttacaggaa caccatccc accagatctc	120
atctactcta ttaaaagaag tctagacact tcaatttctt caaggctctt cccacatcat	180
ccaattgggt ggggatgttt tgaatggga ttggcagaa aagtagacc agagccagg	240
agggtgcagaa gaacagatgg caagaaatgg agatgctcaa aggaggcata tccagactcc	300
aagtactgtg aaagacacat gcacagaggc agaaaccgtt caagaaagcc tgtggaagtt	360
tcttcagcaa taagcaccgc cacaacacc tccaaacaa tccatcttc ttataaccga	420
aaccttctct tgaccaaccc caacatgaca ccacctctt ccttcccttt ctctcctttg	480
ccctcttcta tgctattga gtcccaaccc ttttccaat cctacaaaa ctctctctc	540
aatccctctt tctactccca atcaacctcc tctagacccc cagatgctga ttttccccc	600
caagatgccca ccaccacca gctattcatg gactctgggt cttattcgca tgatgaaaag	660

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aattataggc atgttcattg aataagagaa gatgtggatg agagagcttt cttcccagaa 720
gcatcaggat cagctaggag ctacactgaa tcataccagc aactatcaat gagctcctac 780
aagtcctatt caaactccaa ctttcagaac atcaatgatg ccaccaccaa cccaagacag 840
caagagcagc aacaacaaca aactgcttt gttttgggga cagacttcaa atcaacaaga 900
ccaactaaag agaaagaagc tgagacagct acgggtcaga gaccccttca cgttttcttt 960
ggggagtggc caccaaagaa cacaacagat tcatggctag atcttgcttc caactccaga 1020
atccaaaccg atgaagcggc cgtgcccagc ttgggtctt tgttcattt tccacactgt 1080
ctactacatg tgctgcatca gagcttttat gacgcgtgct tttctcagcg ctacctcaga 1140
tcacatttct tcacctttca gatcacaatg tctgaatccg aatatgcttt gtttgagat 1200
cacaacacct ctccctatca aaactatctg gatgagaaca agtccttaat tctgaagatt 1260
gttgaaagcc agaattcagg caagcttagc gagtgtgccg agaaccaagc aaggcttcag 1320
agaaatctca tgtacttagc tgcaatagct gattctcaac cccaaccacc caccatgcct 1380
ggtcagtacc ctccgagtgg gatgatgcag caggaggcac actacatgca ggctcaacaa 1440
cagacacagc agatgtcacc acaacaacta atggcggcac gctcgtccct tttgtacgca 1500
cagcagccat actcagcact tcaacagcag caagccatgc acagtgcact cgggtcgagt 1560
tcaggactcc acatgtgca aagtgaagc agcaatgtga atgtgggac aggcagtggc 1620
tctgtaggag gagggtttcc tgacttggtg cgcggtggtg gtggtggcgg tggtcgcaca 1680
ggggagggtt tgcacagtgg tggaaagggc atcatgggaa gtagcaagca ggaattggg 1740
ggttcaagtg aaggccgagg tgggggaagc tcagagggtg gtgaaaacct ttacctcaaa 1800
attgccgacg atggaaac 1818

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<210> SEQ ID NO 19
<211> LENGTH: 1806
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: GmGRF11-GmGIF1

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<400> SEQUENCE: 19

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tgtgggaggt cgccattcac agtgtctcag tggcaggaa tggagacca agctttgatc 120
ttcaagtaca tgcttgccgg tcttctgtt cctctcgatc tcgtgttccc cattcagaac 180
agcttcact ctactatctc gctctcgac gctttcttc accatcccac gttgagttac 240
tgttccttct atgggaagaa ggtggaccct gagccaggac gatgcaggag gactgatgga 300
aaaaagtgga ggtgctccaa ggaagcatac ccagactcca agtactcgca gcgccacatg 360
caccgtggcc gcaaccgttc aagaaagcct gtggaatcac aaactatgac tactcatct 420
tcaactgtca catcactcac tgtcactggg gtagtggtg ccagcaaagg aactgtaaat 480
ttccaaaacc tttctacaaa tacctttggt aatctccagg gtaccgatc tggaaactgac 540
cacaccaatt atcatctaga ttccattccc tatgcgattc caagtaaaga atacaggat 600
gttcaaggac ttaaatctga ggtggtgag cactgctttt tttctgaagc ttctggaagc 660
aacaaggctc tccaaatgga gtcacagctg gaaaacacat ggcctttgat gtcaaccaga 720
gttgctctt tttctacgtc aaaatcaagt aatgattccc tgttgcatag tgattatccc 780
cggcattcgt ttttatctgg tgaatatgtg tcgggagaac acgtaaagga ggagggccag 840

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cctcttcgac ctttttttaa tgaatggcct aaaagcaggg agtcatgggc tggctctagaa	900
gatgagagat ccaaccaaac agccttctcc acaactcaac totcaatc ctttcctatg	960
tcttccaatt tctctgcaac gagctctcag tccccacatg gtgaagatga gattcaattt	1020
agggcgcccg ctgccatgtt cgggtctttg ttccatttcc cactctgtct actacatgtg	1080
ctgcatcaga gcttttatga cgcgtgcttt tctcagcgc acctcagatc acattttctc	1140
acctttcaga tcacaatgtc tgaatccgaa tatgctttgt ttggagatca caaaccttct	1200
ccctatcaaa actatctgga tgagaacaag tccttaattc tgaagattgt tgaaagccag	1260
aattcaggca agcttagcga gtgtgccgag aaccaagcaa ggcttcagag aaatctcatg	1320
tacttagctg caatagctga ttctcaaccc caaccacca ccatgcctgg tcagtaccct	1380
ccgagtggga tgatgcagca gggagcacac tacatgcagg ctcaacaaca gacacagcag	1440
atgtcaccac aacaactaat ggcggcacgc tcgtcccttt tgtacgcaca gcagccatac	1500
tcagcacttc aacagcagca agccatgcac agtgcactcg ggtcgagtcc aggactccac	1560
atgttgcaaa gtgaaggcag caatgtgaat gtgggatcag gcagtggctc tgtaggagga	1620
gggtttcctg acttggtgcg cgggtggtgt ggtggcgggt gctcgacagg ggagggtttg	1680
cacagtgggt gaaggggtat catgggaagt agcaagcagg aaattggggg ttcaagtga	1740
ggccgaggtg ggggaagctc agaggggtgt gaaaaccttt acctcaaat tgccgacgat	1800
ggaaac	1806

<210> SEQ ID NO 20

<211> LENGTH: 2583

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF18-GmGIF1

<400> SEQUENCE: 20

atggatcttg ggggtggtggg tttggagggg gtggtgggtt cagaaagtgg ttgtgtgttt	60
ggttctttctc ttgtttcaga tcttgagaca aagcacaagt ggtacggatc tggtttgctc	120
aagcaagaga gatctgcat agctactgaa gatgatgagt ggagaatttc caaagttgct	180
aaaactgatc atgacatgtc ttcagcctcc aaagcaatgc tctttcagca aagaaacaac	240
tctttgttga gatctaataa tgcaactctc ttctctgatg gtcacacca atcacaatg	300
ttgagcttct cttctccaaa gtcagattct ttgttgatag ataaggcttc ctcaaatgcc	360
acattgcctt tttcttccca ccaattgtct agctacacca gaaatacagg ttacaattca	420
ggaagcataa gcattgcattg ggttttggtt agtgtgagag ggccattcac tccatcacag	480
tggatggagc ttgaacacca agccttgatc tacaagtaca tcacagcaaa tgttcctgtg	540
ccaactcatc ttctcattcc catcagaaaa gcacttgatt ctgttggtct ctgcaacttc	600
tcagccggac tctcagacc caactcattg ggatggggag gtttccatct aggattctcg	660
aacaatacag acctgagcc agggagggtgt aggagaacag atggaaagaa atggcgatgt	720
tcaagagatg ccgtagtaga tcagaagtat tgcgagcggc acatgaaccg aggacgccat	780
cgttcaagaa agcctgtgga aggccaatca ggccatgccc tcaccaccac caccagtaat	840
acacctaattg cttctcccaa ctctgtggtg cctggcaaca acaacaacac ctttgcacac	900
aacaatgtgc accaccttat tctcctcat tctctccgg tcaacaccat cactaggatg	960
tttacaagca acaagagaa taataacagt accagtgaga ggatgcagga ccctgcactt	1020
cccatgcttc ctcccactct tgagctgaaa ccaaaggaga acaatccttt catgattcat	1080

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aaacacaaaa tcccatctga tgaatactca agtaggaaca acaatgagtt tgggcttgtc 1140
acttctgatt ctttgettaa cccctcagag aaaagaagct ttactttctc acaaaagaat 1200
gattcttctg agtcccaaca acaacattcc ctcaggcact tcattgatga ctctcccaaa 1260
ccacagtcta atcatcatca tcgttcgtcg tctatatggc ctgaacttga caacatgcag 1320
tcagacagga ctcagttatc aatctccata ccaatatctt cctcagatca ctctatgtca 1380
ttcactactt ccttgccctc gaacgagaaa ctcacgttgt caccacttag gctttcaagg 1440
gagtttagacc ccattcaaat ggggttggga gtgggaagtg cccccaatga agcaaact 1500
aggcaagcca attggattcc aatcacttgg gagagttcaa tgggtggtcc tcttgagag 1560
gttttgaacc ttagtaacaa taacaacagc aatgctagtg atcaatgtgg caagaacaac 1620
aacaacactt cagctctcaa cctcatgaaa gatggatggg acaataatcc tccatcaggg 1680
tcatcccaaa ctggggtgct tcaaaaatct gcatttggat cactttccaa tagcagtgtc 1740
gggagcagtc caaggggggc agagaacaac aaagaagggtg ccaccttgtg caatgccttg 1800
gcgcccgctg ccattgttcg gtctttgttc cattttccac actgtctact acatgtgctg 1860
catcagagct tttatgacgc gtgcttttct cagcgtacc tcagatcaca tttctcacc 1920
tttcagatca caatgtctga atccgaatat gctttgtttg gagatcaca accttctccc 1980
tatcaaaact atctggatga gaacaagtcc ttaattctga agattgttga aagccagaat 2040
tcaggcaagc ttagcgagtg tgccgagAAC caagcaaggc ttcagagaaa tctcatgtac 2100
ttagctgcaa tagctgattc tcaaccccaa ccaccacca tgcttggtca gtaccctccg 2160
agtgggatga tgcagcaggg agcacactac atgcaggctc aacaacagac acagcagatg 2220
tcaccacaac aactaatggc ggcacgctcg tcccttttgt acgcacagca gccatactca 2280
gcacttcaac agcagcaagc catgcacagt gcactcgggt cgagttcagg actccacatg 2340
ctgcaaatgA aaggcagcaa tgtgaatgtg ggatcaggca gtggctctgt aggaggaggg 2400
tttctgact tgggtgcgcg tggtggtggt ggcggtggct cgacagggga gggtttgac 2460
agtgttgaa ggggtatcat gggaaagtagc aagcaggaaa ttgggggttc aagtgaaggc 2520
cgaggtgggg gaagctcaga ggggtgtgaa aacctttacc tcaaaattgc cgacgatgga 2580
aac

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<210> SEQ ID NO 21
<211> LENGTH: 1911
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TaGRF4-TaGIF1

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<400> SEQUENCE: 21

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atggcgatgc cgtatgcctc tctttccccc gcaggcgacc gccgtcctc cccggccgcc 60
accgccaccg cctccctcct ccccttctgc cgctcctccc cctctccgc cgccgccaat 120
ggcgcatgg gggaggaggc gcggatggac gggagggtga tggcgaggcc ggtgcccttc 180
acggcggcgc agtacgagga gctggagcac caggcgctca tatacaagta cctggtggcc 240
ggcgtgtccg tcccgccgga tctcgtgtc cccatccgcc gcggcatcga gtccctcgcc 300
gcccgttctt accacaaccc cctcgccatc gggtaaggat cgtacctggg caagaagggtg 360
gatccggagc cgggccgggt cgggcgcacg gacggcaaga agtggcggtg cgccaaggag 420
gcgcctccg actccaagta ttgcgagcg cacatgcacc gcggccgcaa ccgttcaaga 480

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aagcctgtgg aaacgcagct cgtctcgcac tcccagccgc cggccgcctc cgtcgtgccg 540
cccccgcca ccggttcca caaactctcc ctctacccc ccatcgccgg caccaacggt 600
ggtggaggcg gggggaacaa cggcatgtcc atgcccgca cgttctctc cgcgtgggg 660
ccgctcagc agcacatggg caacaatgcc gcctctcct acgcggctct cggcgccgc 720
ggaacatgca aagatttcag gtataccgca tatggaataa gatctttggc agacgagcag 780
agtcagctca tgacagaagc catgaacacc tccgtggaga acccatggcg ctcgcccga 840
tcttctcaaa cgactacatt ccgctctca agctactctc ctacgttgg agcaacgagt 900
gacctgggtc agaacaacag cagcaacaac aacagccggc tcaaggccga gggacagcag 960
cagcagcagc cgtctctctt ccgggggtgc ggcgacttcg gcagcgccga ctccgcaag 1020
caggagaacc agacgctgcg gccgttcttc gacgagtggc cgaagacgag ggactcgtgg 1080
tcggacctga ccgacgaca ctgcaacgct gcctcctct cggccacca gctgtcgatc 1140
tcgataccca tgacgtctc cgactctctc gccgccagct ccagtcgcc caacggcatg 1200
ctgttcgccg gcgaaatgta cgcggccgct gccatgcagc agcaaacct gatgcagatg 1260
aaccagagca tgatgggggg ctacgcttcc tctaccactg ccaccactga tctcattcag 1320
cagtacctgg atgagaacaa gcagctgatc ctggccatcc tcgacaacca gaacaacggc 1380
aaggtggagg agtgccgacg gaaccaagct aagctccagc agaacctcat gtacctgcc 1440
gccatcgccg acagccagcc tccgcagacg gcacgctgt ctcagtacc gtccaacctg 1500
atgatgcagt ccgggcccgc gtacatgcag cagcagtcgg cgcagatgat gtcgcccag 1560
tcgtgatgg cggcgccgct gtcgatgatg tacgcgcagc aggccatgtc gccgctccag 1620
cagcagcagc agcagcagca gcaccaggcg gccgcgcagc gccagctggg gatgtcctcc 1680
ggcgcgacca ccgggttcaa cctcctgcac ggtgaggcca gcattggcgg cggcgccggc 1740
gccactggca acagcatgat gaacgccagc gtcttctcgg actatggccg cggcgccagc 1800
ggcgccaagg aggggtcgac ctgcgtgtcg gccgacgctc gcggcgccaa ctctggcgcg 1860
cacagcggcg acggggagta cctcaagggc accgaggagg aaggagcta a 1911

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<210> SEQ ID NO 22

<211> LENGTH: 1911

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mTaGRF4-TaGIF1

<400> SEQUENCE: 22

```

atggcgatgc cgtatgcctc tctttccccc gcaggcgacc gccgtcctc ccgggcccgc 60
accgccaccg cctccctcct cccttctctc cgtctctccc cttctctcgc cggcgccaat 120
ggcgcatgg gggaggaggc gcggatggac gggagggtga tggcgaggcc ggtgcccttc 180
acggcgccgc agtacagga gctggagcac caggcgctca tatacaagta cctggtggcc 240
ggcgtgtccg tcccgcggga tctcgtgtc cccatccgcc gcggcatcga gtccctcgcc 300
gcccgtctt accacaaccc cctcgccatc gggtaaggat cgtacctggg caagaagggtg 360
gatccggagc cgggcccgtg ccggcgccag gacggcaaga agtggcggtg cgccaaggag 420
gccgcctccg actccaagta ttgcgagcgc cacatgcacc gcggccgcaa ccgttctaga 480
aaaccagtag agacgcagct cgtctcgcac tcccagccgc cggccgcctc cgtcgtgccg 540
cccccgcca ccggttcca caaactctcc ctctacccc ccatcgccgg caccaacggt 600
ggtggaggcg gggggaacaa cggcatgtcc atgcccgca cgttctctc cgcgtgggg 660

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ccgcctcagc agcacatggg caacaatgcc gcctctccct acgcggctct cggcggcgcc 720
ggaacatgca aagatttcag gtataccgca tatggaataa gatctttggc agacgagcag 780
agtcagctca tgacagaagc catgaacacc tccgtggaga acccatggcg cctgccgcca 840
tcttctcaaa cgactacatt cccgctctca agctactctc ctcagcttgg agcaacgagt 900
gacctgggtc agaacaacag cagcaacaac aacagcggcg tcaaggccga gggacagcag 960
cagcagcagc cgctctcctt cccgggggtgc ggcgacttcg gcagcggcga ctccgcgaag 1020
caggagaacc agacgctgcg gccgttcttc gacgagtggc cgaagacgag ggactcgtgg 1080
tcggacctga ccgacgacaa ctcgaacgtc gcctccttct cggccacca gctgtcgatc 1140
tcgatacca tgacgtcttc cgacttctcc gccgccagct ccagtcgcc caacggcatg 1200
ctgttcgccg gcgaaatgta cgcggccgct gccatgcagc agcaacacct gatgcagatg 1260
aaccagagca tgatgggggg ctacgcttcc tctaccactg ccaccactga tctcattcag 1320
cagtacctgg atgagaacaa gcagctgatc ctggccatcc tcgacaacca gaacaacggc 1380
aaggtggagg agtgcgcacg gaaccaagct aagctccagc agaacctcat gtacctgcc 1440
gccatcgccg acagccagcc tccgcagacg gcacgctgt ctcagtacc gtccaacctg 1500
atgatgcagt ccgggcccgc gtacatgcag cagcagtcgg cgcagatgat gtcgccgcag 1560
tcgctgatgg cggcgcggtc gtcgatgatg tacgcgcagc aggccatgct gccgctccag 1620
cagcagcagc agcagcagca gcaccaggcg gccgcgcagc gccagctggg gatgtcctcc 1680
ggcgcgacca ccgggttcaa cctcctgcac ggtgaggcca gcatgggcgg cggcggcggc 1740
gccactggca acagcatgat gaacgccagc gtcttctcgg actatggccg cggcggcagc 1800
ggcgccaagg aggggtcgac ctcgtgtcg gccgacgctc gcggcgccaa ctctggcgcg 1860
cacagcggcg acggggagta cctcaagggc accgaggagg aaggaagcta a 1911

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<210> SEQ ID NO 23

<211> LENGTH: 839

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF5-GmGIF1

<400> SEQUENCE: 23

```

Met Gly Glu Leu Phe Gly Val Gly Lys Arg Arg Asn Ile Ser Ser Ser
1           5           10           15
Asn Asn Asn Asn Asn Ser Ser Ser Ser Val Leu Gly Leu Asp Val
20          25          30
Lys Val Gln Gln Ser Pro Glu Ala Leu Phe His Asn Arg Met Met Met
35          40          45
Met Ala His His Asn His His His Arg Pro Leu Ser Ser Pro Phe Asp
50          55          60
Asn Asn Gly Asp Gly Asp Gly Pro Thr Thr Tyr Met Ser Phe Thr Asn
65          70          75          80
His Ile Asn Leu Val Ser Gly Ala Ser Ser Val Leu Gly Pro Ala Ile
85          90          95
Asp Ala Gly Cys Gly Ala Ala Pro Pro Ala Pro Val Arg Thr Leu Gln
100         105         110
Pro Phe Asp Ile Ser Ser Tyr Thr Ser Ser Pro Thr Thr Thr Thr Thr
115         120         125
Thr Phe Asn Phe Lys Pro Pro Ser Ala Gly Val Met Ala Ala Ser Leu
130         135         140

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Gly Phe Pro Phe Thr Ser Ala Gln Trp Arg Glu Leu Glu Arg Gln Ala															
145					150				155					160	
Met Ile Tyr Lys Tyr Met Met Ala Ser Val Pro Val Pro His Asp Leu					165			170					175		
Leu Thr Pro Ser Ser Arg Ser Ser Cys Met Asp Gly Gly Phe Asn Leu					180			185					190		
Arg Leu Ala Asn Ser Thr Asp Pro Glu Pro Gly Arg Cys Arg Arg Thr					195			200				205			
Asp Gly Lys Lys Trp Arg Cys Ser Arg Asp Val Ala Pro Asn His Lys					210			215			220				
Tyr Cys Glu Arg His Met His Arg Gly Arg Pro Arg Ser Arg Lys Pro					225			230			235			240	
Val Glu Val Asn Thr Asn Ser Thr Thr Thr Pro Thr Ser Val Asn Asn					245				250					255	
Asn Asn His Gln Ile Lys Lys Ala Arg His Glu Cys Asn Asn Asn Pro					260			265					270		
Phe Ala Thr Pro Asp Val Thr Ala Ala Ile Ser Asn Pro Thr Ser Arg					275			280				285			
Lys Asn Gly Ser Ser Pro His Phe Leu Gly Ser Thr Thr Thr Gln Pro					290			295			300				
Tyr Leu Asp Ser Ser Leu Ser Leu Asp Asn Phe Gly Leu Lys Ala Ala					305			310			315			320	
Ser Phe Asp Ser Val Ala Ser Val Ser Ala Asn Lys Glu Pro Arg Gly					325			330					335		
Leu Glu Trp Met Leu Asn Gly Asp Pro Ile Ser Leu Gly Ala Ser Asp					340			345					350		
Ser Gln Trp Gln Ser Leu Met His Asn Lys Asp Gly Met Thr Ser Val					355			360				365			
Ser Ser Cys Asn Thr Thr Glu Ser Gln Tyr Leu Asn Ser Leu Ala Leu					370			375			380				
Tyr Asn Ser Gly Leu Glu Gln Gln Asn Arg Arg His Pro Leu Phe Leu					385			390			395			400	
Asn Pro Leu Val Val Pro Met Glu Asn Leu Gln Pro Glu Lys Pro Arg					405				410				415		
Gly Phe Ile Asp Ala Trp Ser Asn Ala Glu Ser Asn Ala Asn Thr Asn					420			425					430		
Thr Thr Asn Lys Asn Ser Ala Ala Ser Ile Gly Lys Leu Ser Leu Ser					435			440					445		
Ser Leu Asp Leu Ser Met Gly Gly Ala Ala Val Asn Glu Asp Val Gly					450			455			460				
Asn Val Asn Met Gly Leu Gly Leu Met Glu Pro Asn Gly Lys Thr His					465			470			475			480	
Thr Gly Thr Lys Ile Ser Leu Ser Asn Trp Gln Asn Pro Ala Pro Trp					485				490					495	
Val Ala Ser Ser Leu Gly Gly Pro Leu Ala Glu Val Leu Arg Ser Ser					500				505				510		
Thr Val Thr Ala Thr Thr Thr Thr Asn Glu Ala Thr Ser Asn Thr Pro					515				520				525		
Ser Pro Ala Thr Thr Thr His Ala Glu Ser Pro Ser Gly Val Leu Gln					530			535			540				
Lys Thr Leu Val Ser Leu Ser Asp Ser Ser Asn Asn Ser Ser Pro Arg					545				550			555			560

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Val	Ala	Ser	Ser	Arg	Ala	Asn	Ser	Glu	Met	Ala	Leu	Leu	Arg	Phe	Gln
				565					570					575	
Ser	Asn	Ala	Ala	Ala	Ala	Met	Phe	Gly	Ser	Leu	Phe	His	Phe	Pro	His
				580				585					590		
Cys	Leu	Leu	His	Val	Leu	His	Gln	Ser	Phe	Tyr	Asp	Ala	Cys	Phe	Ser
		595					600					605			
Gln	Arg	Tyr	Leu	Arg	Ser	His	Phe	Leu	Thr	Phe	Gln	Ile	Thr	Met	Ser
	610					615					620				
Glu	Ser	Glu	Tyr	Ala	Leu	Phe	Gly	Asp	His	Lys	Pro	Ser	Pro	Tyr	Gln
625					630					635					640
Asn	Tyr	Leu	Asp	Glu	Asn	Lys	Ser	Leu	Ile	Leu	Lys	Ile	Val	Glu	Ser
				645				650						655	
Gln	Asn	Ser	Gly	Lys	Leu	Ser	Glu	Cys	Ala	Glu	Asn	Gln	Ala	Arg	Leu
			660					665					670		
Gln	Arg	Asn	Leu	Met	Tyr	Leu	Ala	Ala	Ile	Ala	Asp	Ser	Gln	Pro	Gln
		675					680					685			
Pro	Pro	Thr	Met	Pro	Gly	Gln	Tyr	Pro	Pro	Ser	Gly	Met	Met	Gln	Gln
	690					695					700				
Gly	Ala	His	Tyr	Met	Gln	Ala	Gln	Gln	Gln	Thr	Gln	Gln	Met	Ser	Pro
705					710					715					720
Gln	Gln	Leu	Met	Ala	Ala	Arg	Ser	Ser	Leu	Leu	Tyr	Ala	Gln	Gln	Pro
				725					730					735	
Tyr	Ser	Ala	Leu	Gln	Gln	Gln	Gln	Ala	Met	His	Ser	Ala	Leu	Gly	Ser
			740					745					750		
Ser	Ser	Gly	Leu	His	Met	Leu	Gln	Ser	Glu	Gly	Ser	Asn	Val	Asn	Val
		755					760					765			
Gly	Ser	Gly	Ser	Gly	Ser	Val	Gly	Gly	Gly	Phe	Pro	Asp	Leu	Val	Arg
	770					775					780				
Gly	Gly	Gly	Gly	Gly	Gly	Gly	Ser	Thr	Gly	Glu	Gly	Leu	His	Ser	Gly
785					790					795					800
Gly	Arg	Gly	Ile	Met	Gly	Ser	Ser	Lys	Gln	Glu	Ile	Gly	Gly	Ser	Ser
				805				810						815	
Glu	Gly	Arg	Gly	Gly	Gly	Ser	Ser	Glu	Gly	Gly	Glu	Asn	Leu	Tyr	Leu
			820					825				830			
Lys	Ile	Ala	Asp	Asp	Gly	Asn									
			835												

<210> SEQ ID NO 24

<211> LENGTH: 606

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF6-GmGIF1

<400> SEQUENCE: 24

Met	Met	Ser	Ala	Ser	Ala	Arg	Asn	Arg	Ser	Pro	Phe	Thr	Gln	Thr	Gln
1				5					10					15	
Trp	Gln	Glu	Leu	Glu	His	Gln	Ala	Leu	Val	Phe	Lys	Tyr	Met	Val	Thr
			20					25					30		
Gly	Thr	Pro	Ile	Pro	Pro	Asp	Leu	Ile	Tyr	Ser	Ile	Lys	Arg	Ser	Leu
		35					40					45			
Asp	Thr	Ser	Ile	Ser	Ser	Arg	Leu	Phe	Pro	His	His	Pro	Ile	Gly	Trp
	50					55					60				
Gly	Cys	Phe	Glu	Met	Gly	Phe	Gly	Arg	Lys	Val	Asp	Pro	Glu	Pro	Gly
65					70					75					80

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Arg	Cys	Arg	Arg	Thr	Asp	Gly	Lys	Lys	Trp	Arg	Cys	Ser	Lys	Glu	Ala	
				85					90					95		
Tyr	Pro	Asp	Ser	Lys	Tyr	Cys	Glu	Arg	His	Met	His	Arg	Gly	Arg	Asn	
			100					105					110			
Arg	Ser	Arg	Lys	Pro	Val	Glu	Val	Ser	Ser	Ala	Ile	Ser	Thr	Ala	Thr	
		115					120						125			
Asn	Thr	Ser	Gln	Thr	Ile	Pro	Ser	Ser	Tyr	Thr	Arg	Asn	Leu	Ser	Leu	
	130					135					140					
Thr	Asn	Pro	Asn	Met	Thr	Pro	Pro	Ser	Ser	Phe	Pro	Phe	Ser	Pro	Leu	
145					150					155					160	
Pro	Ser	Ser	Met	Pro	Ile	Glu	Ser	Gln	Pro	Phe	Ser	Gln	Ser	Tyr	Gln	
			165						170					175		
Asn	Ser	Ser	Leu	Asn	Pro	Phe	Phe	Tyr	Ser	Gln	Ser	Thr	Ser	Ser	Arg	
			180					185					190			
Pro	Pro	Asp	Ala	Asp	Phe	Pro	Pro	Gln	Asp	Ala	Thr	Thr	His	Gln	Leu	
		195					200						205			
Phe	Met	Asp	Ser	Gly	Ser	Tyr	Ser	His	Asp	Glu	Lys	Asn	Tyr	Arg	His	
	210					215					220					
Val	His	Gly	Ile	Arg	Glu	Asp	Val	Asp	Glu	Arg	Ala	Phe	Phe	Pro	Glu	
225					230					235					240	
Ala	Ser	Gly	Ser	Ala	Arg	Ser	Tyr	Thr	Glu	Ser	Tyr	Gln	Gln	Leu	Ser	
				245					250					255		
Met	Ser	Ser	Tyr	Lys	Ser	Tyr	Ser	Asn	Ser	Asn	Phe	Gln	Asn	Ile	Asn	
			260					265					270			
Asp	Ala	Thr	Thr	Asn	Pro	Arg	Gln	Gln	Glu	Gln	Gln	Gln	Gln	Gln	His	
		275					280					285				
Cys	Phe	Val	Leu	Gly	Thr	Asp	Phe	Lys	Ser	Thr	Arg	Pro	Thr	Lys	Glu	
	290					295					300					
Lys	Glu	Ala	Glu	Thr	Ala	Thr	Gly	Gln	Arg	Pro	Leu	His	Arg	Phe	Phe	
305					310					315					320	
Gly	Glu	Trp	Pro	Pro	Lys	Asn	Thr	Thr	Asp	Ser	Trp	Leu	Asp	Leu	Ala	
			325						330					335		
Ser	Asn	Ser	Arg	Ile	Gln	Thr	Asp	Glu	Ala	Ala	Ala	Ala	Met	Phe	Gly	
			340				345						350			
Ser	Leu	Phe	His	Phe	Pro	His	Cys	Leu	Leu	His	Val	Leu	His	Gln	Ser	
		355					360					365				
Phe	Tyr	Asp	Ala	Cys	Phe	Ser	Gln	Arg	Tyr	Leu	Arg	Ser	His	Phe	Leu	
	370					375					380					
Thr	Phe	Gln	Ile	Thr	Met	Ser	Glu	Ser	Glu	Tyr	Ala	Leu	Phe	Gly	Asp	
385					390					395					400	
His	Lys	Pro	Ser	Pro	Tyr	Gln	Asn	Tyr	Leu	Asp	Glu	Asn	Lys	Ser	Leu	
			405						410					415		
Ile	Leu	Lys	Ile	Val	Glu	Ser	Gln	Asn	Ser	Gly	Lys	Leu	Ser	Glu	Cys	
			420					425					430			
Ala	Glu	Asn	Gln	Ala	Arg	Leu	Gln	Arg	Asn	Leu	Met	Tyr	Leu	Ala	Ala	
		435					440					445				
Ile	Ala	Asp	Ser	Gln	Pro	Gln	Pro	Pro	Thr	Met	Pro	Gly	Gln	Tyr	Pro	
	450					455					460					
Pro	Ser	Gly	Met	Met	Gln	Gly	Ala	His	Tyr	Met	Gln	Ala	Gln	Gln		
465					470				475					480		
Gln	Thr	Gln	Gln	Met	Ser	Pro	Gln	Gln	Leu	Met	Ala	Ala	Arg	Ser	Ser	
				485					490					495		
Leu	Leu	Tyr	Ala	Gln	Gln	Pro	Tyr	Ser	Ala	Leu	Gln	Gln	Gln	Gln	Ala	

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500					505					510					
Met	His	Ser	Ala	Leu	Gly	Ser	Ser	Ser	Gly	Leu	His	Met	Leu	Gln	Ser
	515					520					525				
Glu	Gly	Ser	Asn	Val	Asn	Val	Gly	Ser	Gly	Ser	Gly	Ser	Val	Gly	Gly
	530					535					540				
Gly	Phe	Pro	Asp	Leu	Val	Arg	Gly	Gly	Gly	Gly	Gly	Gly	Gly	Ser	Thr
	545					550					555				560
Gly	Glu	Gly	Leu	His	Ser	Gly	Gly	Arg	Gly	Ile	Met	Gly	Ser	Ser	Lys
			565					570						575	
Gln	Glu	Ile	Gly	Gly	Ser	Ser	Glu	Gly	Arg	Gly	Gly	Gly	Ser	Ser	Glu
			580					585					590		
Gly	Gly	Glu	Asn	Leu	Tyr	Leu	Lys	Ile	Ala	Asp	Asp	Gly	Asn		
		595					600					605			

<210> SEQ ID NO 25

<211> LENGTH: 602

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF11-GmGIF1

<400> SEQUENCE: 25

Met	Asn	Asn	Ser	Ser	Gly	Gly	Gly	Gly	Arg	Gly	Thr	Leu	Met	Gly	Leu
1				5					10					15	
Ser	Asn	Gly	Tyr	Cys	Gly	Arg	Ser	Pro	Phe	Thr	Val	Ser	Gln	Trp	Gln
			20					25					30		
Glu	Leu	Glu	His	Gln	Ala	Leu	Ile	Phe	Lys	Tyr	Met	Leu	Ala	Gly	Leu
		35					40					45			
Pro	Val	Pro	Leu	Asp	Leu	Val	Phe	Pro	Ile	Gln	Asn	Ser	Phe	His	Ser
		50				55					60				
Thr	Ile	Ser	Leu	Ser	His	Ala	Phe	Phe	His	His	Pro	Thr	Leu	Ser	Tyr
				70						75				80	
Cys	Ser	Phe	Tyr	Gly	Lys	Lys	Val	Asp	Pro	Glu	Pro	Gly	Arg	Cys	Arg
				85					90					95	
Arg	Thr	Asp	Gly	Lys	Lys	Trp	Arg	Cys	Ser	Lys	Glu	Ala	Tyr	Pro	Asp
			100					105					110		
Ser	Lys	Tyr	Cys	Glu	Arg	His	Met	His	Arg	Gly	Arg	Asn	Arg	Ser	Arg
			115				120					125			
Lys	Pro	Val	Glu	Ser	Gln	Thr	Met	Thr	His	Ser	Ser	Ser	Thr	Val	Thr
		130				135					140				
Ser	Leu	Thr	Val	Thr	Gly	Gly	Ser	Gly	Ala	Ser	Lys	Gly	Thr	Val	Asn
				150					155					160	
Phe	Gln	Asn	Leu	Ser	Thr	Asn	Thr	Phe	Gly	Asn	Leu	Gln	Gly	Thr	Asp
			165					170						175	
Ser	Gly	Thr	Asp	His	Thr	Asn	Tyr	His	Leu	Asp	Ser	Ile	Pro	Tyr	Ala
			180					185				190			
Ile	Pro	Ser	Lys	Glu	Tyr	Arg	Tyr	Val	Gln	Gly	Leu	Lys	Ser	Glu	Gly
		195				200						205			
Gly	Glu	His	Cys	Phe	Phe	Ser	Glu	Ala	Ser	Gly	Ser	Asn	Lys	Val	Leu
		210				215					220				
Gln	Met	Glu	Ser	Gln	Leu	Glu	Asn	Thr	Trp	Pro	Leu	Met	Ser	Thr	Arg
				230						235				240	
Val	Ala	Ser	Phe	Ser	Thr	Ser	Lys	Ser	Ser	Asn	Asp	Ser	Leu	Leu	His
				245				250						255	
Ser	Asp	Tyr	Pro	Arg	His	Ser	Phe	Leu	Ser	Gly	Glu	Tyr	Val	Ser	Gly

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260				265				270							
Glu	His	Val	Lys	Glu	Glu	Gly	Gln	Pro	Leu	Arg	Pro	Phe	Phe	Asn	Glu
		275					280							285	
Trp	Pro	Lys	Ser	Arg	Glu	Ser	Trp	Ser	Gly	Leu	Glu	Asp	Glu	Arg	Ser
	290					295					300				
Asn	Gln	Thr	Ala	Phe	Ser	Thr	Thr	Gln	Leu	Ser	Ile	Ser	Ile	Pro	Met
	305				310					315				320	
Ser	Ser	Asn	Phe	Ser	Ala	Thr	Ser	Ser	Gln	Ser	Pro	His	Gly	Glu	Asp
			325							330				335	
Glu	Ile	Gln	Phe	Arg	Ala	Ala	Ala	Ala	Met	Phe	Gly	Ser	Leu	Phe	His
			340						345					350	
Phe	Pro	His	Cys	Leu	Leu	His	Val	Leu	His	Gln	Ser	Phe	Tyr	Asp	Ala
		355					360						365		
Cys	Phe	Ser	Gln	Arg	Tyr	Leu	Arg	Ser	His	Phe	Leu	Thr	Phe	Gln	Ile
	370					375					380				
Thr	Met	Ser	Glu	Ser	Glu	Tyr	Ala	Leu	Phe	Gly	Asp	His	Lys	Pro	Ser
	385				390					395				400	
Pro	Tyr	Gln	Asn	Tyr	Leu	Asp	Glu	Asn	Lys	Ser	Leu	Ile	Leu	Lys	Ile
			405						410					415	
Val	Glu	Ser	Gln	Asn	Ser	Gly	Lys	Leu	Ser	Glu	Cys	Ala	Glu	Asn	Gln
			420						425					430	
Ala	Arg	Leu	Gln	Arg	Asn	Leu	Met	Tyr	Leu	Ala	Ala	Ile	Ala	Asp	Ser
		435				440							445		
Gln	Pro	Gln	Pro	Pro	Thr	Met	Pro	Gly	Gln	Tyr	Pro	Pro	Ser	Gly	Met
	450					455					460				
Met	Gln	Gln	Gly	Ala	His	Tyr	Met	Gln	Ala	Gln	Gln	Gln	Thr	Gln	Gln
	465				470					475				480	
Met	Ser	Pro	Gln	Gln	Leu	Met	Ala	Ala	Arg	Ser	Ser	Leu	Leu	Tyr	Ala
			485						490					495	
Gln	Gln	Pro	Tyr	Ser	Ala	Leu	Gln	Gln	Gln	Gln	Ala	Met	His	Ser	Ala
		500							505					510	
Leu	Gly	Ser	Ser	Ser	Gly	Leu	His	Met	Leu	Gln	Ser	Glu	Gly	Ser	Asn
	515					520							525		
Val	Asn	Val	Gly	Ser	Gly	Ser	Gly	Ser	Val	Gly	Gly	Gly	Phe	Pro	Asp
	530				535						540				
Leu	Val	Arg	Gly	Gly	Gly	Gly	Gly	Gly	Gly	Ser	Thr	Gly	Glu	Gly	Leu
	545				550					555				560	
His	Ser	Gly	Gly	Arg	Gly	Ile	Met	Gly	Ser	Ser	Lys	Gln	Glu	Ile	Gly
			565						570					575	
Gly	Ser	Ser	Glu	Gly	Arg	Gly	Gly	Gly	Ser	Ser	Glu	Gly	Gly	Glu	Asn
			580						585					590	
Leu	Tyr	Leu	Lys	Ile	Ala	Asp	Asp	Gly	Asn						
	595					600									

<210> SEQ ID NO 26

<211> LENGTH: 861

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: GmGRF18-GmGIF1

<400> SEQUENCE: 26

Met	Asp	Leu	Gly	Val	Val	Gly	Leu	Glu	Gly	Val	Val	Gly	Ser	Glu	Ser
1				5						10				15	

Gly Cys Val Phe Gly Ser Ser Leu Val Ser Asp Pro Glu Thr Lys His

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20							25					30				
Lys	Trp	Tyr	Gly	Ser	Gly	Leu	Leu	Lys	Gln	Glu	Arg	Ser	Ala	Ile	Ala	
		35					40					45				
Thr	Glu	Asp	Asp	Glu	Trp	Arg	Ile	Ser	Lys	Val	Ala	Lys	Thr	Asp	His	
		50					55					60				
Asp	Met	Ser	Ser	Ala	Ser	Lys	Ala	Met	Leu	Phe	Gln	Gln	Arg	Asn	Asn	
65					70					75					80	
Ser	Leu	Leu	Arg	Ser	Asn	Asn	Ala	Thr	Leu	Phe	Ser	Asp	Gly	His	His	
				85					90					95		
Gln	Ser	Gln	Met	Leu	Ser	Phe	Ser	Ser	Pro	Lys	Ser	Asp	Ser	Leu	Leu	
				100					105					110		
Ile	Asp	Lys	Ala	Ser	Ser	Asn	Ala	Thr	Leu	Pro	Phe	Ser	Ser	His	Gln	
				115			120					125				
Leu	Ser	Ser	Tyr	Thr	Arg	Asn	Thr	Gly	Tyr	Asn	Ser	Gly	Ser	Ile	Ser	
						135					140					
Met	His	Gly	Ala	Leu	Ala	Ser	Val	Arg	Gly	Pro	Phe	Thr	Pro	Ser	Gln	
145					150					155					160	
Trp	Met	Glu	Leu	Glu	His	Gln	Ala	Leu	Ile	Tyr	Lys	Tyr	Ile	Thr	Ala	
				165					170					175		
Asn	Val	Pro	Val	Pro	Thr	His	Leu	Leu	Ile	Pro	Ile	Arg	Lys	Ala	Leu	
				180					185					190		
Asp	Ser	Val	Gly	Phe	Cys	Asn	Phe	Ser	Ala	Gly	Leu	Leu	Arg	Pro	Asn	
				195			200					205				
Ser	Leu	Gly	Trp	Gly	Gly	Phe	His	Leu	Gly	Phe	Ser	Asn	Asn	Thr	Asp	
						215					220					
Pro	Glu	Pro	Gly	Arg	Cys	Arg	Arg	Thr	Asp	Gly	Lys	Lys	Trp	Arg	Cys	
225					230							235			240	
Ser	Arg	Asp	Ala	Val	Val	Asp	Gln	Lys	Tyr	Cys	Glu	Arg	His	Met	Asn	
				245					250					255		
Arg	Gly	Arg	His	Arg	Ser	Arg	Lys	Pro	Val	Glu	Gly	Gln	Ser	Gly	His	
				260					265					270		
Ala	Leu	Thr	Thr	Thr	Thr	Ser	Asn	Thr	Pro	Asn	Ala	Ser	Ser	Asn	Ser	
				275			280					285				
Val	Val	Pro	Gly	Asn	Asn	Asn	Asn	Thr	Phe	Ala	His	Asn	Asn	Val	His	
				290			295					300				
His	Pro	Ile	Pro	Pro	His	Ser	Ser	Pro	Val	Asn	Thr	Ile	Thr	Arg	Met	
305					310							315			320	
Phe	Thr	Ser	Asn	Lys	Glu	Asn	Asn	Asn	Ser	Thr	Ser	Glu	Arg	Met	Gln	
				325					330					335		
Asp	Pro	Ala	Leu	Pro	Met	Leu	Pro	Pro	Thr	Leu	Glu	Leu	Lys	Pro	Lys	
				340			345							350		
Glu	Asn	Asn	Pro	Phe	Met	Ile	His	Lys	His	Gln	Ile	Pro	Ser	Asp	Glu	
				355			360					365				
Tyr	Ser	Ser	Arg	Asn	Asn	Asn	Glu	Phe	Gly	Leu	Val	Thr	Ser	Asp	Ser	
						375					380					
Leu	Leu	Asn	Pro	Ser	Glu	Lys	Arg	Ser	Phe	Thr	Ser	Ser	Gln	Lys	Asn	
385					390							395			400	
Asp	Ser	Ser	Glu	Ser	Gln	Gln	Gln	His	Ser	Leu	Arg	His	Phe	Ile	Asp	
				405					410					415		
Asp	Ser	Pro	Lys	Pro	Gln	Ser	Asn	His	His	His	Arg	Ser	Ser	Ser	Ile	
				420			425							430		
Trp	Pro	Glu	Leu	Asp	Asn	Met	Gln	Ser	Asp	Arg	Thr	Gln	Leu	Ser	Ile	
				435			440							445		

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Ser	Ile	Pro	Ile	Ser	Ser	Ser	Asp	His	Phe	Met	Ser	Phe	Thr	Thr	Ser
450						455					460				
Leu	Pro	Ser	Asn	Glu	Lys	Leu	Thr	Leu	Ser	Pro	Leu	Arg	Leu	Ser	Arg
465				470						475					480
Glu	Leu	Asp	Pro	Ile	Gln	Met	Gly	Leu	Gly	Val	Gly	Ser	Ala	Pro	Asn
				485					490					495	
Glu	Ala	Asn	Thr	Arg	Gln	Ala	Asn	Trp	Ile	Pro	Ile	Thr	Trp	Glu	Ser
			500					505					510		
Ser	Met	Gly	Gly	Pro	Leu	Gly	Glu	Val	Leu	Asn	Leu	Ser	Asn	Asn	Asn
		515					520					525			
Asn	Ser	Asn	Ala	Ser	Asp	Gln	Cys	Gly	Lys	Asn	Asn	Asn	Asn	Thr	Ser
530						535					540				
Ala	Leu	Asn	Leu	Met	Lys	Asp	Gly	Trp	Asp	Asn	Asn	Pro	Pro	Ser	Gly
545					550					555					560
Ser	Ser	Pro	Thr	Gly	Val	Leu	Gln	Lys	Ser	Ala	Phe	Gly	Ser	Leu	Ser
			565						570					575	
Asn	Ser	Ser	Ala	Gly	Ser	Ser	Pro	Arg	Gly	Ala	Glu	Asn	Asn	Lys	Glu
			580					585						590	
Gly	Ala	Thr	Leu	Cys	Asn	Ala	Leu	Ala	Ala	Ala	Ala	Met	Phe	Gly	Ser
		595					600					605			
Leu	Phe	His	Phe	Pro	His	Cys	Leu	Leu	His	Val	Leu	His	Gln	Ser	Phe
610						615					620				
Tyr	Asp	Ala	Cys	Phe	Ser	Gln	Arg	Tyr	Leu	Arg	Ser	His	Phe	Leu	Thr
625					630					635					640
Phe	Gln	Ile	Thr	Met	Ser	Glu	Ser	Glu	Tyr	Ala	Leu	Phe	Gly	Asp	His
			645						650					655	
Lys	Pro	Ser	Pro	Tyr	Gln	Asn	Tyr	Leu	Asp	Glu	Asn	Lys	Ser	Leu	Ile
			660					665					670		
Leu	Lys	Ile	Val	Glu	Ser	Gln	Asn	Ser	Gly	Lys	Leu	Ser	Glu	Cys	Ala
		675					680					685			
Glu	Asn	Gln	Ala	Arg	Leu	Gln	Arg	Asn	Leu	Met	Tyr	Leu	Ala	Ala	Ile
690						695					700				
Ala	Asp	Ser	Gln	Pro	Gln	Pro	Pro	Thr	Met	Pro	Gly	Gln	Tyr	Pro	Pro
705				710						715				720	
Ser	Gly	Met	Met	Gln	Gln	Gly	Ala	His	Tyr	Met	Gln	Ala	Gln	Gln	Gln
			725						730					735	
Thr	Gln	Gln	Met	Ser	Pro	Gln	Gln	Leu	Met	Ala	Ala	Arg	Ser	Ser	Leu
			740					745					750		
Leu	Tyr	Ala	Gln	Gln	Pro	Tyr	Ser	Ala	Leu	Gln	Gln	Gln	Gln	Ala	Met
		755					760					765			
His	Ser	Ala	Leu	Gly	Ser	Ser	Ser	Gly	Leu	His	Met	Leu	Gln	Ser	Glu
770					775						780				
Gly	Ser	Asn	Val	Asn	Val	Gly	Ser	Gly	Ser	Gly	Ser	Val	Gly	Gly	Gly
785				790					795						800
Phe	Pro	Asp	Leu	Val	Arg	Gly	Gly	Gly	Gly	Gly	Gly	Gly	Ser	Thr	Gly
			805					810						815	
Glu	Gly	Leu	His	Ser	Gly	Gly	Arg	Gly	Ile	Met	Gly	Ser	Ser	Lys	Gln
			820					825					830		
Glu	Ile	Gly	Gly	Ser	Ser	Glu	Gly	Arg	Gly	Gly	Gly	Ser	Ser	Glu	Gly
	835						840					845			
Gly	Glu	Asn	Leu	Tyr	Leu	Lys	Ile	Ala	Asp	Asp	Gly	Asn			
850						855					860				

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<210> SEQ ID NO 27
<211> LENGTH: 636
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TaGRF4-TaGIF1

<400> SEQUENCE: 27

Met Ala Met Pro Tyr Ala Ser Leu Ser Pro Ala Gly Asp Arg Arg Ser
 1             5             10             15

Ser Pro Ala Ala Thr Ala Thr Ala Ser Leu Leu Pro Phe Cys Arg Ser
          20             25             30

Ser Pro Phe Ser Ala Gly Gly Asn Gly Gly Met Gly Glu Glu Ala Arg
          35             40             45

Met Asp Gly Arg Trp Met Ala Arg Pro Val Pro Phe Thr Ala Ala Gln
          50             55             60

Tyr Glu Glu Leu Glu His Gln Ala Leu Ile Tyr Lys Tyr Leu Val Ala
          65             70             75             80

Gly Val Ser Val Pro Pro Asp Leu Val Leu Pro Ile Arg Arg Gly Ile
          85             90             95

Glu Ser Leu Ala Ala Arg Phe Tyr His Asn Pro Leu Ala Ile Gly Tyr
          100            105            110

Gly Ser Tyr Leu Gly Lys Lys Val Asp Pro Glu Pro Gly Arg Cys Arg
          115            120            125

Arg Thr Asp Gly Lys Lys Trp Arg Cys Ala Lys Glu Ala Ala Ser Asp
          130            135            140

Ser Lys Tyr Cys Glu Arg His Met His Arg Gly Arg Asn Arg Ser Arg
          145            150            155            160

Lys Pro Val Glu Thr Gln Leu Val Ser His Ser Gln Pro Pro Ala Ala
          165            170            175

Ser Val Val Pro Pro Leu Ala Thr Gly Phe His Asn His Ser Leu Tyr
          180            185            190

Pro Ala Ile Gly Gly Thr Asn Gly Gly Gly Gly Gly Gly Asn Asn Gly
          195            200            205

Met Ser Met Pro Gly Thr Phe Ser Ser Ala Leu Gly Pro Pro Gln Gln
          210            215            220

His Met Gly Asn Asn Ala Ala Ser Pro Tyr Ala Ala Leu Gly Gly Ala
          225            230            235            240

Gly Thr Cys Lys Asp Phe Arg Tyr Thr Ala Tyr Gly Ile Arg Ser Leu
          245            250            255

Ala Asp Glu Gln Ser Gln Leu Met Thr Glu Ala Met Asn Thr Ser Val
          260            265            270

Glu Asn Pro Trp Arg Leu Pro Pro Ser Ser Gln Thr Thr Thr Phe Pro
          275            280            285

Leu Ser Ser Tyr Ser Pro Gln Leu Gly Ala Thr Ser Asp Leu Gly Gln
          290            295            300

Asn Asn Ser Ser Asn Asn Asn Ser Gly Val Lys Ala Glu Gly Gln Gln
          305            310            315            320

Gln Gln Gln Pro Leu Ser Phe Pro Gly Cys Gly Asp Phe Gly Ser Gly
          325            330            335

Asp Ser Ala Lys Gln Glu Asn Gln Thr Leu Arg Pro Phe Phe Asp Glu
          340            345            350

Trp Pro Lys Thr Arg Asp Ser Trp Ser Asp Leu Thr Asp Asp Asn Ser
          355            360            365

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Asn Val Ala Ser Phe Ser Ala Thr Gln Leu Ser Ile Ser Ile Pro Met
 370                375                380

Thr Ser Ser Asp Phe Ser Ala Ala Ser Ser Gln Ser Pro Asn Gly Met
385                390                395                400

Leu Phe Ala Gly Glu Met Tyr Ala Ala Ala Ala Met Gln Gln Gln His
                405                410                415

Leu Met Gln Met Asn Gln Ser Met Met Gly Gly Tyr Ala Ser Ser Thr
                420                425                430

Thr Ala Thr Thr Asp Leu Ile Gln Gln Tyr Leu Asp Glu Asn Lys Gln
 435                440                445

Leu Ile Leu Ala Ile Leu Asp Asn Gln Asn Asn Gly Lys Val Glu Glu
 450                455                460

Cys Ala Arg Asn Gln Ala Lys Leu Gln Gln Asn Leu Met Tyr Leu Ala
465                470                475                480

Ala Ile Ala Asp Ser Gln Pro Pro Gln Thr Ala Ser Leu Ser Gln Tyr
                485                490                495

Pro Ser Asn Leu Met Met Gln Ser Gly Pro Arg Tyr Met Gln Gln Gln
 500                505                510

Ser Ala Gln Met Met Ser Pro Gln Ser Leu Met Ala Ala Arg Ser Ser
 515                520                525

Met Met Tyr Ala Gln Gln Ala Met Ser Pro Leu Gln Gln Gln Gln
 530                535                540

Gln Gln Gln His Gln Ala Ala Ala His Gly Gln Leu Gly Met Ser Ser
545                550                555                560

Gly Ala Thr Thr Gly Phe Asn Leu Leu His Gly Glu Ala Ser Met Gly
 565                570                575

Gly Gly Gly Gly Ala Thr Gly Asn Ser Met Met Asn Ala Ser Val Phe
 580                585                590

Ser Asp Tyr Gly Arg Gly Gly Ser Gly Ala Lys Glu Gly Ser Thr Ser
 595                600                605

Leu Ser Ala Asp Ala Arg Gly Ala Asn Ser Gly Ala His Ser Gly Asp
 610                615                620

Gly Glu Tyr Leu Lys Gly Thr Glu Glu Glu Gly Ser
625                630                635

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<210> SEQ ID NO 28
<211> LENGTH: 636
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mTaGRF4-TaGIF1

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<400> SEQUENCE: 28

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Met Ala Met Pro Tyr Ala Ser Leu Ser Pro Ala Gly Asp Arg Arg Ser
 1                5                10                15

Ser Pro Ala Ala Thr Ala Thr Ala Ser Leu Leu Pro Phe Cys Arg Ser
 20                25                30

Ser Pro Phe Ser Ala Gly Gly Asn Gly Gly Met Gly Glu Glu Ala Arg
 35                40                45

Met Asp Gly Arg Trp Met Ala Arg Pro Val Pro Phe Thr Ala Ala Gln
 50                55                60

Tyr Glu Glu Leu Glu His Gln Ala Leu Ile Tyr Lys Tyr Leu Val Ala
 65                70                75                80

Gly Val Ser Val Pro Pro Asp Leu Val Leu Pro Ile Arg Arg Gly Ile
 85                90                95

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Glu 100	Ser 101	Leu 102	Ala 103	Ala 104	Arg 105	Phe 106	Tyr 107	His 108	Asn 109	Pro 110	Leu 111	Ala 112	Ile 113	Gly 114	Tyr 115
Gly 115	Ser 116	Tyr 117	Leu 118	Gly 119	Lys 120	Lys 121	Val 122	Asp 123	Pro 124	Glu 125	Pro 126	Gly 127	Arg 128	Cys 129	Arg 130
Arg 130	Thr 131	Asp 132	Gly 133	Lys 134	Lys 135	Trp 136	Arg 137	Cys 138	Ala 139	Lys 140	Glu 141	Ala 142	Ala 143	Ser 144	Asp 145
Ser 145	Lys 146	Tyr 147	Cys 148	Glu 149	Arg 150	His 151	Met 152	His 153	Arg 154	Gly 155	Arg 156	Asn 157	Arg 158	Ser 159	Arg 160
Lys 160	Pro 161	Val 162	Glu 163	Thr 164	Gln 165	Leu 166	Val 167	Ser 168	His 169	Ser 170	Gln 171	Pro 172	Pro 173	Ala 174	Ala 175
Ser 175	Val 176	Val 177	Pro 178	Pro 179	Leu 180	Ala 181	Thr 182	Gly 183	Phe 184	His 185	Asn 186	His 187	Ser 188	Leu 189	Tyr 190
Pro 190	Ala 191	Ile 192	Gly 193	Gly 194	Thr 195	Asn 196	Gly 197	Gly 198	Gly 199	Gly 200	Gly 201	Gly 202	Asn 203	Asn 204	Gly 205
Met 205	Ser 206	Met 207	Pro 208	Gly 209	Thr 210	Phe 211	Ser 212	Ser 213	Ala 214	Leu 215	Gly 216	Pro 217	Pro 218	Gln 219	Gln 220
His 220	Met 221	Gly 222	Asn 223	Asn 224	Ala 225	Ala 226	Ser 227	Pro 228	Tyr 229	Ala 230	Ala 231	Leu 232	Gly 233	Gly 234	Ala 235
Gly 235	Thr 236	Cys 237	Lys 238	Asp 239	Phe 240	Arg 241	Tyr 242	Thr 243	Ala 244	Tyr 245	Gly 246	Ile 247	Arg 248	Ser 249	Leu 250
Ala 250	Asp 251	Glu 252	Gln 253	Ser 254	Gln 255	Leu 256	Met 257	Thr 258	Glu 259	Ala 260	Met 261	Asn 262	Thr 263	Ser 264	Val 265
Glu 265	Asn 266	Pro 267	Trp 268	Arg 269	Leu 270	Pro 271	Pro 272	Ser 273	Ser 274	Gln 275	Thr 276	Thr 277	Thr 278	Phe 279	Pro 280
Leu 280	Ser 281	Ser 282	Tyr 283	Ser 284	Pro 285	Gln 286	Leu 287	Gly 288	Ala 289	Thr 290	Ser 291	Asp 292	Leu 293	Gly 294	Gln 295
Asn 295	Asn 296	Ser 297	Ser 298	Asn 299	Asn 300	Asn 301	Ser 302	Gly 303	Val 304	Lys 305	Ala 306	Glu 307	Gly 308	Gln 309	Gln 310
Gln 310	Gln 311	Gln 312	Pro 313	Leu 314	Ser 315	Phe 316	Pro 317	Gly 318	Cys 319	Gly 320	Asp 321	Phe 322	Gly 323	Ser 324	Gly 325
Asp 325	Ser 326	Ala 327	Lys 328	Gln 329	Glu 330	Asn 331	Gln 332	Thr 333	Leu 334	Arg 335	Pro 336	Phe 337	Phe 338	Asp 339	Glu 340
Trp 340	Pro 341	Lys 342	Thr 343	Arg 344	Asp 345	Ser 346	Trp 347	Ser 348	Asp 349	Leu 350	Thr 351	Asp 352	Asp 353	Asn 354	Ser 355
Asn 355	Val 356	Ala 357	Ser 358	Phe 359	Ser 360	Ala 361	Thr 362	Gln 363	Leu 364	Ser 365	Ile 366	Ser 367	Ile 368	Pro 369	Met 370
Thr 370	Ser 371	Ser 372	Asp 373	Phe 374	Ser 375	Ala 376	Ala 377	Ser 378	Ser 379	Gln 380	Ser 381	Pro 382	Asn 383	Gly 384	Met 385
Leu 385	Phe 386	Ala 387	Gly 388	Glu 389	Met 390	Tyr 391	Ala 392	Ala 393	Ala 394	Ala 395	Met 396	Gln 397	Gln 398	Gln 399	His 400
Leu 400	Met 401	Gln 402	Met 403	Asn 404	Gln 405	Ser 406	Met 407	Met 408	Gly 409	Gly 410	Tyr 411	Ala 412	Ser 413	Ser 414	Thr 415
Thr 415	Ala 416	Thr 417	Thr 418	Asp 419	Leu 420	Ile 421	Gln 422	Gln 423	Tyr 424	Leu 425	Asp 426	Glu 427	Asn 428	Lys 429	Gln 430
Leu 430	Ile 431	Leu 432	Ala 433	Ile 434	Leu 435	Asp 436	Asn 437	Gln 438	Asn 439	Asn 440	Gly 441	Lys 442	Val 443	Glu 444	Glu 445
Cys 445	Ala 446	Arg 447	Asn 448	Gln 449	Ala 450	Lys 451	Leu 452	Gln 453	Gln 454	Asn 455	Leu 456	Met 457	Tyr 458	Leu 459	Ala 460
Ala 460	Ile 461	Ala 462	Asp 463												

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515	520	525	
Met Met Tyr Ala Gln Gln Ala Met Ser Pro Leu Gln Gln Gln Gln Gln			
530	535	540	
Gln Gln Gln His Gln Ala Ala Ala His Gly Gln Leu Gly Met Ser Ser			
545	550	555	560
Gly Ala Thr Thr Gly Phe Asn Leu Leu His Gly Glu Ala Ser Met Gly			
	565	570	575
Gly Gly Gly Gly Ala Thr Gly Asn Ser Met Met Asn Ala Ser Val Phe			
	580	585	590
Ser Asp Tyr Gly Arg Gly Gly Ser Gly Ala Lys Glu Gly Ser Thr Ser			
	595	600	605
Leu Ser Ala Asp Ala Arg Gly Ala Asn Ser Gly Ala His Ser Gly Asp			
	610	615	620
Gly Glu Tyr Leu Lys Gly Thr Glu Glu Glu Gly Ser			
625	630	635	
<210> SEQ ID NO 29			
<211> LENGTH: 22			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: miRNA site			
<400> SEQUENCE: 29			
ccgttcaaga aagcctgtgg aa			22
<210> SEQ ID NO 30			
<211> LENGTH: 22			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: miRNA site			
<400> SEQUENCE: 30			
ccgttctaga aaaccagtag ag			22
<210> SEQ ID NO 31			
<211> LENGTH: 71			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: WT			
<400> SEQUENCE: 31			
tccttctctca acccttttcc ctcatgtcat ggccaatcta ttgggttctc caaggttgcc			60
ttctcactgg t			71
<210> SEQ ID NO 32			
<211> LENGTH: 61			
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<220> FEATURE:
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<400> SEQUENCE: 34

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<210> SEQ ID NO 35
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 35

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ctcaactggt                                           69

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 36

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 37

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 38

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<210> SEQ ID NO 39
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<212> TYPE: DNA
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<400> SEQUENCE: 39

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 41

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 42

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ggt                                                                    63

<210> SEQ ID NO 43
<211> LENGTH: 67
<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 43

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cactgggt                                                                    67

<210> SEQ ID NO 44
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<212> TYPE: DNA
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<400> SEQUENCE: 44

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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 45

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gt	62
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 46

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t	61
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<210> SEQ ID NO 47
<211> LENGTH: 66
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 47

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actggt	66
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<210> SEQ ID NO 48
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 48

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ctcactggt	69
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<210> SEQ ID NO 49
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 49

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ctggt	65
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<210> SEQ ID NO 50
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 50

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<400> SEQUENCE: 51

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<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 52

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cactggt 67

<210> SEQ ID NO 53

<211> LENGTH: 67

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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cactggt 67

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<211> LENGTH: 68

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mutant sequence

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 55

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<210> SEQ ID NO 56

<211> LENGTH: 68

<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 56

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tcactggt 68

<210> SEQ ID NO 57
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 57

tccttctctca accctttttcc ctcatgtcat ggccaatcta ttctccaagg ttgccttctc 60
actggt 66

<210> SEQ ID NO 58
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 58

tccttctctca accctttttcc ctcatgtcat ggccaatctcc aaggttgctt tctcactggt 60

<210> SEQ ID NO 59
<211> LENGTH: 69
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 59

tccttctctca accctttttcc ctcatgtcat ggccaatcta ttggtctcca aggttgctt 60
ctcactggt 69

<210> SEQ ID NO 60
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 60

tccttctctca accctttttcc ctcatgtcat ctccaaggtt gccttctcac tggt 54

<210> SEQ ID NO 61
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant sequence

<400> SEQUENCE: 61

tccttctctca accctttttcc ctcatgtcat ggccaatcta tctccaaggt tgccttctca 60
ctggt 65

<210> SEQ ID NO 62
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: target

<400> SEQUENCE: 62

gtatccgcaa cagcactgaa agg 23

<210> SEQ ID NO 63
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: WT

<400> SEQUENCE: 63

tccgcaacag cactgaaagg agaa 24

<210> SEQ ID NO 64
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Fa-6

<400> SEQUENCE: 64

tccgcaacaa cactgaaagg agaa 24

<210> SEQ ID NO 65
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Fa-21

<400> SEQUENCE: 65

tccccaaggc ccctgaaagg agaa 24

<210> SEQ ID NO 66
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Fa-34

<400> SEQUENCE: 66

tccccaacca cccttgaaag gagaa 25

<210> SEQ ID NO 67
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: TaALS target

<400> SEQUENCE: 67

caggtccccc gccgcattgat cgg 23

<210> SEQ ID NO 68
 <211> LENGTH: 23
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: GmFAD2 target

<400> SEQUENCE: 68

cattgcatgg ccaatctatt ggg

23

What we claimed is:

1. A method for improving plant cell regeneration efficiency, the method comprising:

(a) introducing into a cell of the plant

i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM, and an expression construct comprising a nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or

ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7 and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or

iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22;

(b) regenerating an intact plant from the plant cell.

2. A method of transforming at least one exogenous nucleic acid sequence of interest into a plant, the method comprising:

(a) introducing into a cell of the plant

i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or

ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7, and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or

iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22;

(b) introducing at least one expression construct comprising the at least one exogenous nucleic acid sequence of interest into the plant cell; and

(c) regenerating an intact plant from the plant cell.

3. A method for gene editing in a plant, the method comprising:

(a) introducing into a cell of the plant

i) an expression construct comprising a coding nucleic acid sequence of WUS, an expression construct comprising a coding nucleic acid sequence of BBM and an expression construct comprising a coding nucleic acid sequence of SERK, wherein the WUS comprises the amino acid sequence shown in SEQ ID NO: 1, the BBM comprises the amino acid sequence shown in SEQ ID NO: 2, or the SERK comprises the amino acid sequence shown in SEQ ID NO: 3; and/or

ii) an expression construct comprising a coding nucleic acid sequence of GRF and an expression construct comprising a coding nucleic acid sequence of GIF, wherein the GRF comprises the amino acid sequence of SEQ ID NO: 7, and the GIF1 comprises the amino acid sequence shown in SEQ ID NO: 11; and/or

iii) an expression construct comprising the nucleotide sequence of SEQ ID NO: 22;

(b) introducing at least one expression construct comprising the at least one exogenous nucleic acid sequence of interest into the plant cell, wherein the at least one exogenous nucleic acid sequence of interest encodes a component of a gene editing system; or introducing at least one component of the gene editing system into the plant cell; and

(c) regenerating an intact plant from the plant cell.

4. The method according to any one of claims 1-3, wherein the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM and the coding nucleic acid sequence of SERK are placed in a same expression construct.

5. The method according to claim 4, wherein the coding nucleic acid sequence of WUS, the coding nucleic acid sequence of BBM, the coding nucleic acid sequence of SERK, and the at least one exogenous nucleic acid sequence of interest, if present, are placed in a same expression construct.

6. The method according to any one of claims 1-3, wherein the coding nucleic acid sequence of GRF and the coding nucleic acid sequence of GIF are placed in a same expression construct.

7. The method according to claim 6, wherein the coding nucleic acid sequence of GRF the coding nucleic acid sequence of GIF and the at least one exogenous nucleic acid sequence of interest, if present, are placed in a same expression construct.

8. The method according to any one of claims 1-3, wherein the GRF comprises a mutated miRNA binding site, so as not to be regulated by the miRNA, wherein the miRNA is miR396.

9. The method according to any one of claims 1-3, wherein the GRF is fused to the GIF by a linker, and further wherein the linker comprises an amino acid sequence comprising SEQ ID NO: 12.

10. The method according to claim 3, wherein the gene editing system is selected from the group consisting of CRISPR system, TALEN, meganuclease and zinc finger nuclease.

11. The method according to claim 10, wherein the gene editing system is a base editing system.

12. The method according to claim 3, wherein the gene editing system is a base editor comprising an amino acid sequence comprising SEQ ID NO: 14.

* * * * *